

THE HORIZONTAL GEOMETRY OF PRODUCTION NETWORKS

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Abstract

Modern economies are organized as dense production networks, yet the transmission of shocks is conventionally understood through observed Input-Output linkages and vertical cascade effects. This paper argues that propagation additionally operates through horizontal complementarities: the extent to which sectors share upstream suppliers or downstream buyers. These interdependencies shape a latent network of connections, linking also otherwise unconnected sectors, and generate simultaneous responses to idiosyncratic shocks through structural demand and supply similarities embedded in the Input-Output system. Aggregate volatility then results from widespread correlations that prevent shocks from averaging out. This diffusion channel acts as a complement to vertical propagation, offering a unified view of comovement rooted in the horizontal geometry of common interactions rather than network trade intensities. Using Input-Output data, I show that sectoral responses vary systematically with horizontal relations, forging novel implications for the transmission of sectoral and aggregate shocks.

JEL: C67, D57, E32, J21

KEYWORDS: production networks, Input-Output economy, horizontal transmission, network distance, sectoral comovement, aggregate fluctuations

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1. INTRODUCTION

A hallmark of contemporary production systems is the comovement of economic activity across sectors. Recent advances in the production network literature have demonstrated how independent, idiosyncratic shocks propagate through observed Input-Output linkages, generating simultaneous adjustments in productions and contributing to aggregate fluctuations. Here, the presence of vertical “cascade effects” diffuses the shock along parallel supply chains via the tangle of sectoral ties.¹ Yet, a shock often travels through hidden network corridors: comovement may nonetheless emerge through common downstream buyers (*e.g.*, February 2, 2026: the HBM4 memory breakthrough reshuffles demand across competing chip providers) or common upstream suppliers (*e.g.*, the 2025 AI-driven surge in cloud demand for DRAM and HBM chips strained capacity for other users).

This paper argues that production networks connect sectors not only through existing trade linkages, but also by demand and supply similarities embedded in their common structure of inter-sectoral trade. I conceptualize these interdependencies as *network economic distances* (*e.g.*, Conley and Dapor 2003). Specifically, sectors exhibit “factor input demand” distance when they source intermediate inputs from the same upstream suppliers, and “factor input supply” distance when they sell to the same downstream buyers. Viewed through this lens, an Input-Output structure not only reflects *vertical* interdependencies, where transmitting shocks across sectors reflects their intensity of being inter-connected (*e.g.*, Shea 2002), but also *horizontal* complementarities, as network participants are additionally linked through structural demand and supply connections from a resembling Input-Output geometry.

Network-based comovement is therefore endogenous. Since sectors mutually adjust in response due to shared economic relationships, horizontal geometries become a source of systematic comovement across economically proximate sectors. The propagation of shocks depends not only on the sequentiality proper of observed linkages, but also on the exposure to the same Input-Output neighbourhoods: two sectors may display correlated responses even in the absence of strong vertical connections simply because they are economically connected through common buyers or suppliers. Hence, horizontal distances generate heterogeneous exposures to both idiosyncratic and aggregate disturbances. The question is therefore not only whether shocks propagate through production networks, but also whether the horizontal geometry of these networks matters for determining who is exposed to whom, and by how much.

Under this perspective, aggregate volatility reflects not only how idiosyncratic shocks are amplified, but also the way the network structure induces correlated behaviours. When a shock hits, it propagates vertically by cascading from suppliers to customers through input requirements, generating recursive feedback loops and sequential amplification along parallel supply chains (*e.g.*, Acemoglu, Carvalho, et al. 2012). At the same time, it propagates via horizontal

¹ Central insight is that a positive shock to a sector enhances its productive performance, triggering: (i) a *downstream effect*, as buyers have an incentive to expand their demand from that sector; (ii) an *upstream effect*, as higher production in the shocked sector raises its demand of intermediate inputs, benefiting its suppliers; and (iii) vertical *network spillovers*, where the initial effects ripple beyond immediate trading partners, benefiting indirectly connected sectors. Still through observed chains, the transmission of shocks proves less straightforward when non unitary elasticities of substitution limit the ability to adjust input combinations (*e.g.*, Baqaee and Farhi 2019). On the verticality of Input-Output economies refer to Pasinetti (1973).

complementarities, as sectors sharing buyers or suppliers adjust simultaneously to similar cost or supply conditions. These responses do not generate cascades but induce systematic comovement across economic activities, preventing shocks from averaging out and allowing aggregate fluctuations to emerge from the network’s horizontal geometry itself.

Indeed, building a coherent theoretical framework that emphasizes the role of demand- and supply-driven network economic distances on sectoral comovement and aggregate dynamics within an Input-Output economy, and validating its main predictions using sector-level data, constitute the central novelties of this paper.

Section 2 lays down the conceptual foundation for horizontal complementarities. On the input side, sectors buying from the same upstream suppliers reveal simultaneous adjustments as they rebalance labour and intermediate inputs. On the output side, sectors selling to the same downstream buyers face relative price adjustments, and thus revenue shifts when downstream demand changes. Theorem 1 establishes a “dual representation” of an Input-Output economy: the matrix of technical coefficients is a primitive object that embeds not only observed (direct and indirect) production linkages, but also horizontal relations from shared demand and supply structures. These horizontal interdependencies also bound otherwise disconnected nodes: as follows from Theorem 2, even if don’t directly trade, the existence of a common neighbour makes network participants economically connected.

Section 3 interprets the horizontal transmission of asymmetric shocks. Under demand linkages, opposite comovement arises only when intermediate inputs cannot be fully substituted, as a positive shock to one downstream user bids up the price of a shared supplier and scales other buyers’ demand. By contrast, in the supply-based case, opposite comovement is immediate and mechanical, as a sector capturing demand from a shared downstream buyer reduces activity among competing suppliers after a positive shock that effectively lowers its relative price. In this sense, demand linkages make comovement ambiguous and dependent on substitution patterns in downstream demand, whereas supply linkages make it transparent and directly tied to upstream competition for downstream markets. Crucially, the elasticity of substitution across intermediate inputs does not alter the direction of comovement.

In developing this perspective, Theorem 3 emerges: sectoral interdependencies in terms of common demand or supply relationships retain an independent force, distinct from the intensity of inter-sectoral trade flows. This separation strengthens the importance of horizontal complementarities, since these are not themselves weighted by the vertical Input-Output structure. In addition, Theorem 4 establishes how, for very large networks, horizontal effects from shared buyers and suppliers proliferate, making horizontal spillovers to dominate shock transmission over vertical chains.

Since horizontal complementarities operate through adjustments in factors of production, the analysis turns to employment comovement and its role on aggregate cycles. Dynamic patterns hinge on network distances: nearby sectors move in opposite directions due to shared Input-Output relations, causing diverging employment responses that dampen aggregate fluctuations, whereas distant sectors tend to comove as the vertical propagation prevails, generating broadly aligned employment dynamics. Theorem 5 then formalizes that horizontal interdependencies alone generate persistent aggregate fluctuations, thereby breaking the standard diversi-

fication argument by Lucas (1977): with many sectors sharing upstream or downstream connections, shocks propagate by the means of correlated behaviours, inducing mutual responses that accumulate at the aggregate level.

As a validation of theoretical insights, Section 4 uses U.S. sectoral and Input-Output data to simulate GDP dynamics. Sustained aggregate volatility stems from common upstream/-downstream connections, where the structural properties of the horizontal propagation induce aggregate fluctuations to mean revert at a slower rate of decay than the one predicted by the standard diversification. Delving into the mechanics of horizontal patterns reveals how they rely on diffuse interdependencies rather than a few vertical propagation hubs, and are strengthened as network sparsity increases, lending empirical support to Theorems 4 and 5 and their corollaries. A Bartik-style estimation further investigates whether horizontal geometries shape the transmission of aggregate disturbances when controlling for vertical Input-Output exposures. The relevant propagation channel depends on the nature of the disturbance: aggregate productivity shocks are primarily transmitted through supply-based horizontal relationships, whereas aggregate government spending shocks are amplified mainly through demand-based horizontal connections. Horizontal exposures therefore contain economically meaningful information about shock transmission not captured by conventional Leontief inverse linkages.

Section 5 implements the two-stage approach in Barattieri and Cacciatore (2023) to study sectoral comovement. The empirical results are threefold. Firstly, an increase in the set of circulating intermediate inputs of a sector leads to a comparatively smaller rise in its employment with minimal but alternate effects from similar changes in distanced sectors. Secondly, sectors closer under common supply distance exhibit opposite employment comovement: an increase in employment in nearby sectors tends to reduce employment in the sector; for closely demand-linked sectors, the responses are often ambiguous (increases for some, reductions for others), but mostly pointing to a dampening effect on the vertical shock’s transmission. Thirdly, positive comovement emerges among more distantly demand- and supply-related sectors, since sectoral employment rises in response to increases in farther ones.

To conclude, Section 6 examines further implications of the network’s horizontal geometry and how incorporating it can improve the design of policy interventions.

Overall, the discussion provides empirical support for the theoretical predictions: horizontal complementarities in terms of similar demand and supply relationships complement the canonical transmission of shocks, revealing the multiple channels through which vertical and horizontal structures of interdependencies shape the distribution and the propagation of economic activity across networked economies.

Literature review.— The objective of this paper is to advance both theoretically and empirically the understanding of how Input-Output economies work. On the theoretical side, it builds on the modern and rapidly expanding literature on production networks. In the wake of Gabaix (2011)’s “granular hypothesis”, the seminal observation of Long and Plosser (1983) – that comovements across sectors are not dictated by a shared disturbance but rather by sectoral interdependencies –, has been modernized through a series of papers of the early 2010s (*e.g.*, Acemoglu, Carvalho, et al. 2012; Carvalho and Gabaix 2013; Carvalho 2014; Barrot and Sauvagnat 2016). This renewed perspective gave rise to studies on production networks related

to efficient economies (*e.g.*, Baqaee and Farhi 2019; vom Lehn and Winberry 2022; Liu and Tsyvinski 2024), monetary policy and nominal rigidities (*e.g.*, La’O and Tahbaz-Salehi 2022; Rubbo 2023; Ghassibe and Nakov 2025), inefficiencies (*e.g.*, Jones 2011; Grassi 2017; Baqaee 2018; Baqaee and Farhi 2020), policy-oriented issues (*e.g.*, Liu 2019; Grassi and Sauvagnat 2019; Lane 2025), endogenous network formation (*e.g.*, Oberfield 2018; Acemoglu and Azar 2020; Ghassibe 2024; Taschereau-Dumouchel 2026), economic growth (*e.g.*, Hausmann and Hidalgo 2011; Gualdi and Mandel 2019; McNerney et al. 2022), and international contexts (*e.g.*, Caliendo et al. 2022; Qiu et al. 2026; Huo et al. 2025). A common mechanism explored in these works emphasizes the vertical propagation, where shocks originating in upstream sectors transmit downstream, amplifying their effects and contributing to aggregate outcomes.² My contribution broadens this perspective by introducing horizontal linkages from network “economic” distances, measured by common upstream or downstream Input-Output relationships. These novel interdependencies forge horizontal complementarities that bind even disconnected sectors, allowing asymmetric shocks to propagate across the network in ways beyond traditional and vertical supply chains.

On the empirical side, relatively few studies examine the role of sectoral production networks on macroeconomic outcomes (*e.g.*, Acemoglu, Akcigit, et al. 2015; Ghassibe 2021; Barattieri and Cacciatore 2023; Barattieri, Cacciatore, and Traum 2023; Monti and Van Keirsbilck 2025), with some illustrative spillovers for horizontal transmission (*e.g.*, Boehm and Oberfield 2018; Dragomirescu-Gaina and Elia 2021; Huremović et al. 2025). From this standpoint, the paper contributes novel empirical evidence on the role of Input-Output linkages in shaping interlinked dynamics, particularly in explaining the comovement of employment across sectors. Synchronized employment patterns are an important characteristic of business cycles, as most sectors tend to move together over time (*e.g.*, Christiano and Fitzgerald 1998). Several studies explore the nature and the sources of this sectoral comovement. Cooper and Haltiwanger (1990) find that sectoral employment levels are positively correlated, in ways not fully explained by aggregate shocks. Cassou and Vázquez (2014) attribute high employment comovement to similar shock transmission channels. In Yedid-Levi (2016) comovement is stronger along the extensive (number of workers) than the intensive (hours per worker) margin. Room for structural changes is in Kim (2020), where positive technology shocks in manufacturing raise employment in both manufacturing and services. While these studies provide valuable insights, they do not explicitly address how production networks shape employment comovement. By contrast, my paper focuses on short-run employment variations within a networked framework. Empirically, the paper contributes by: (i) advancing the comprehension of shocks’ transmission via network “economic” distances; and (ii) providing unprecedented evidence on employment comovement from a production network perspective.

² As noted by Shea (2002), a positive sectoral supply shock has a price (decrease in its price level, and reduction in its nominal expenditure for intermediate inputs) and a quantity (its supply to other sectors increase, sector’s production rises, thereby increasing its input demand) effect. These opposing forces affect upstream suppliers but, under Cobb-Douglas technology, they offset each other leaving intermediate inputs demand unchanged, thereby affecting only downstream suppliers. In general, only demand shocks can propagate upstream (*e.g.*, Acemoglu, Akcigit, et al. 2015; Barrot and Sauvagnat 2016; Ferrari 2024).

2. FOUNDATIONAL THEORY

The economy is populated by a finite set of sectors, $i, j, s = \{1, \dots, m\} \in \Phi$, with Φ_i denoting the set of all others not including sector- i . Each produces a single commodity using labour and a set of circulating intermediate inputs from other sectors. The whole economic system is thus represented by an $m \times 1$ vector of sectoral production outputs, $\mathbf{y} = [y_i > 0]$, an $m \times 1$ vector of sectoral employment levels, $\mathbf{n} = [n_i > 0]$, each associated to its output elasticity of labour $\alpha_i \in (0, 1)$, and by an $m \times m$ square matrix, $\mathbf{X} = [x_{ij} \geq 0]$, indicative of the inter-sectoral trade of intermediate input quantities, with $x_{ij} = 0$ stating that sector- i does not make use of the commodity produced by sector- j . Finally, an $m \times m$ square matrix, $\mathbf{A} = [\alpha_{ij} \geq 0]$, defines the intensity of the commodities produced by sectors as intermediate inputs in others; denote its Leontief inverse as $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1} = [\ell_{ij} \geq 0]$, with $\mathbf{I}$ being an $m \times m$ identity matrix.

Under these specifications, a perfectly competitive final good producer combines the production of different sectors, $Y = \prod_{i \in \Phi} y_i^{\omega_i} \propto \boldsymbol{\omega}' \mathbf{y}$, where $\boldsymbol{\omega}$ is a $1 \times m$ vector of $\omega \in (0, 1)$, weighting the relative importance of each sectoral commodity in the definition of aggregate output Y . Total consumption is an aggregator over final demands: let $\mathbf{c} = [c_i > 0]$ be an $m \times 1$ vector of sectoral consumption levels by households, it holds that $C = \prod_{i \in \Phi} c_i^{\omega_i} \propto \boldsymbol{\omega}' \mathbf{c}$. A representative household gets utility from consumption and leisure dis-utility from working: $\mathcal{U} := f^u[c_i, n_i]_{\forall i \in \Phi}$, with consumption of all sectoral commodities and labour supplied characterized by a price, p_i and w_i respectively, the latter chosen as the numeraire.

A representative firm characterizes each sector- i . Given an exogenous Hicks-neutral productivity z_i , the production function is

$$y_i = z_i f^y \left[n_i, \{x_{ij}\}_{\forall j \in \Phi} \right]$$

ASSUMPTION 1 (Production technology requirements) *As main characteristics of a production function: (i) it has constant returns to scale in either x_{ij} and n_i , so that production inputs shares sum to one, $\alpha_i + \sum_{j \in \Phi} \alpha_{ij} = 1$; (ii) it is differentiable, continuous, strictly quasi-concave, homogeneous of degree one, and increasing in z_i , x_{ij} , and n_i ; (iii) the case $f^y[0, \cdot]$ is ruled out as labour input is essential to production; (iv) at least two elements of x_{ij} have to be positive, thus $f^y[\cdot, 0]$ cannot exist; consequentially, (v) part of the sectoral output is directly produced by the sector, $x_{ii} > 0, \forall i \in \Phi$, that is a production plan with a bundle of intermediate inputs as $f^y[\cdot, x_{ij}]$, given any $j \in \Phi_i$, is precluded.*

The first two sub-assumptions are common in production technology (e.g., Acemoglu, Carvalho, et al. 2012). By way of *AI.iii*, workers cannot be fully substituted by any combination of intermediate inputs, and changes in the purchased quantity of intermediate inputs would determine subsequent variations in the labour force size. The assumptions characterizing the bundle of intermediates, *AI.iv* and *AI.v*, ensure the sector not only to participate in the network, but also that part of the production process is roundabout: for sector- i delivering its commodity y_i requires to be at least vertically connected (buying from one upstream sector), while imposing that a fraction of intermediate inputs must be directly produced within the sector.

Each perfectly competitive representative firm maximizes profits given input prices, w_i and $p_j, \forall j \in \Phi$. For any functional form of y_i , the following applies.

LEMMA 1 (Technological profile) Under Assumption 1, and being $\kappa_i(p, w)$ the unit cost function, the output shares of sector- i satisfy

$$\alpha_{ij} = \frac{\partial \log \kappa_i(p, w)}{\partial \log p_j} \quad \text{and} \quad \alpha_i = \frac{\partial \log \kappa_i(p, w)}{\partial \log w_i}$$

and the constraint $\alpha_i + \sum_{j \in \Phi} \alpha_{ij} = 1$ holds exactly for any $\{p, w\}$, so that the vectors of sectoral Input-Output profiles $\alpha_i^{\text{col}} := [\alpha_{ii}, \dots, \alpha_{ij}, \dots, \alpha_{im}]'$ and $\alpha_i^{\text{row}} := [\alpha_{ii}, \dots, \alpha_{ji}, \dots, \alpha_{mi}]$ belong to the unitary simplex Δ^{m-1} .

Proof: since $f^y[\cdot]$ is homogenous of degree one, total incurred costs are proportional to y_i , so that $\mathcal{K}_i^{\text{tot}} = \kappa_i(p, w) \frac{y_i}{z_i}$. By Sheppard's lemma, $x_{ij} = \frac{\partial \log \kappa_i(p, w)}{\partial \log p_j} \frac{y_i}{z_i}$ and $n_i = \frac{\partial \log \kappa_i(p, w)}{\partial \log w_i} \frac{y_i}{z_i}$. Since the α 's are, by definition, the input in total cost, multiplying each quantity by its price and dividing for total costs, the first part of the Lemma is delivered. Then, since $\kappa_i(p, w)$ is homogenous of degree one in p and w , its total derivative reads as $w_i \frac{\partial \kappa_i(p, w)}{\partial w_i} + p_j \frac{\partial \kappa_i(p, w)}{\partial p_j}$. Dividing by $\kappa_i(p, w)$ and applying the first part delivers the constraint. The last part is a collection of column-based (recording whenever a sector is buying from others) and row-based (recording whenever a sector is selling to others) Input-Output coefficients. ■

From the outlined economy, the sectoral Input-Output equilibrium is defined by final consumption and network-based allocations: $y_i = c_i + \sum_{j \in \Phi} x_{ji}$.

Distances.— In a networked economy, comovement of production inputs is endogenously induced. Each sector both relies on and supplies to others, so that changes in one sector affect not only its own optimal input composition but also the demand and supply conditions faced by other sectors. Sectors that trade with similar partners therefore face shared network exposure, providing a natural basis for a rigorous comparison. Paraphrasing Conley and Dupor (2003), two types of network “economic” distances between sectors, sketched in Panel 1a of Figure 1, can be defined: (a) *factor input demand*, when sectors are buying their intermediate inputs from similar upstream sectors; and (b) *factor input supply*, when sectors are selling their commodities as intermediate inputs to similar downstream sectors.

LEMMA 2 (Two Input-Output contractions) Consider two sectors, say $\{i, j\}$, related to a common one, say s . Then, for the pair:

- $[\mathcal{A}^2]_{ij} = \sum_{s \in \Phi} \alpha_{is} \alpha_{sj}$ denotes a recursive production chain;
- $[\mathcal{A}\mathcal{A}']_{ij} = \sum_{s \in \Phi} \alpha_{is} \alpha_{js}$ denotes the common upstream neighbourhood;
- $[\mathcal{A}'\mathcal{A}]_{ij} = \sum_{s \in \Phi} \alpha_{si} \alpha_{sj}$ denotes the common downstream neighbourhood;

Moreover, $\mathcal{A}\mathcal{A}'$ and $\mathcal{A}'\mathcal{A}$ are symmetric, positive semidefinite, and identify the Gram matrices of the Input-Output profile collection $\{\alpha_i\}_{i \in \Phi}$ across rows and columns.

Proof: by definition of matrix multiplication, the (i, j) -th entry of $\mathcal{A}^2$ is $[\mathcal{A}^2]_{ij} = \sum_{s=1}^m \alpha_{is} \alpha_{sj}$: a common node- s enters as the second argument in the first factor, and as a first argument in the second factor. Differently, the (i, j) -th entry of $\mathcal{A}\mathcal{A}'$ is given by $[\mathcal{A}\mathcal{A}']_{ij} = \sum_{s=1}^m \alpha_{is} \alpha_{js}$. Now observe that the right-hand side coincides with the standard inner product between the row vectors $\mathbf{a}_i$ and $\mathbf{a}_j$, namely $\langle \mathbf{a}_i, \mathbf{a}_j \rangle = \sum_{s=1}^m \alpha_{is} \alpha_{js}$. Therefore, $[\mathcal{A}\mathcal{A}']_{ij} = \langle \mathbf{a}_i, \mathbf{a}_j \rangle$. Hence, $\mathcal{A}\mathcal{A}'$ is the Gram matrix associated with $\{\mathbf{a}_i\}_{i=1}^m$. Such matrix is positive semidefinite since, for a given vector $\boldsymbol{\xi} \in \mathbb{R}^m$ then $\boldsymbol{\xi}' \mathcal{A}\mathcal{A}' \boldsymbol{\xi} = \|\mathcal{A}' \boldsymbol{\xi}\|_2^2$, while symmetric since $(\mathcal{A}\mathcal{A}')' = \mathcal{A}\mathcal{A}'$. Same for $\mathcal{A}'\mathcal{A}$. ■

Network connections extend beyond their flows of intermediate inputs. These additional interdependencies highlight the complexity of inter-sectoral dynamics and complement the *Leontief inverse*, a reminiscent result from Leontief (1936): it captures how sectors not directly

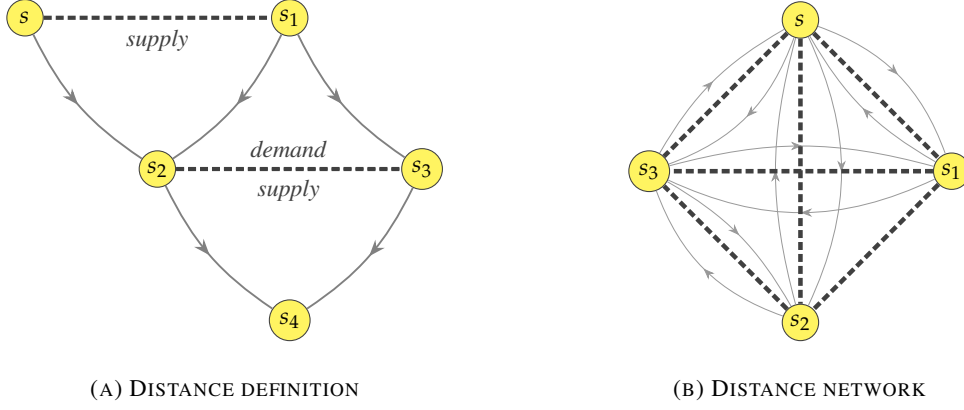


FIGURE 1: HORIZONTAL INPUT-OUTPUT GEOMETRIES

Note: the figure presents a stylized production network and its corresponding network of sectoral distances. Production network linkages are depicted with oriented thin lines, while dashed thick lines indicate the distance relationships.

trading are indirectly related – *e.g.*, a sector buying from an upstream one introjects such sector’s purchases as well the other of more upstream sectors –, emphasizing the depth of vertical linkages through layers of Input-Output relationships. Network-based economic distances, by contrast, consider how participants are mapped horizontally as well: sectors are additionally related because they are relying on the same suppliers or serving the same customers. These broader notions of linkages provide a richer understanding about systemic interdependencies and explain *horizontal complementarities* in demand and supply that classical Input-Output analysis overlook. They reveal how seemingly unrelated sectors can experience synchronized fluctuations, and how the structure of common upstream or downstream connections shapes the propagation of independent and idiosyncratic shocks. In other words, the notion of network interdependencies is not confined to “who trades with whom”, but also emerges from “who depends on whom”.

Panel 1b sketches a stylized economy associated to its distance-based network: distances effectively double Input-Output linkages and bound disconnected sectors.

EXAMPLE 1 (Distance in the network) Consider an economy populated by four sectors, $\{s, s_1, s_2, s_3\} \in \Phi$, where some trade with all others, while some do not. In Panel 1b, focus on the pair $\{s_1, s_2\}$: they do not mutually trade, so $\alpha_{s_1 s_2} = 0 = \alpha_{s_2 s_1}$. Nevertheless, their linkages with all the remaining sectors generate a common structure of inter-sectoral trade. Yet, while the underlying trade and distance structures are unique, the horizontal metrics vary under alternative specifications of network distance, as Examples 2-3 clarify.

Finally, I embed these notions of horizontal geometry into an Input-Output equilibrium. The same primitive structure governing the trade of commodities not only determines how connections spread through production chains, but also how sectors are positioned relative to each other in the space of interdependencies. In this sense, the network matrix induces both a dynamic vertical structure and a geometric delineation of sectoral similarities. The following result formalises this dual role.

THEOREM 1 (On the dual representation) Consider the equilibrium of an Input-Output economy, $\mathbf{y} = \mathbf{A}\mathbf{y} + \mathbf{c}$. The matrix $\mathbf{A}$ admits two distinct representations:

- vertical system, an equilibrium mapping governed by $\mathcal{L} = \sum_{k=0}^{\infty} \mathcal{A}^k$ encoding multiple production layers for circulating commodities;
- horizontal embedding, a collection of Input-Output profiles $\{\alpha_i^{col}, \alpha_i^{row}\}_{i \in \Phi}$ encoding a geometric representation of sectoral similarities;

Both representations are fully determined by the primitive Input-Output matrix.

Proof: let $G = (m, \Lambda)$ be a finite directed weighted graph, with $\Lambda = \{i, j \in m \times m : \alpha_{ij} \geq 0\}$, and let $\ell_{is} = \sum_{k=0}^{\infty} [\mathcal{A}^k]_{is}$ being the Neumann expansion of $\mathcal{L}$, where $[\mathcal{A}^k]_{is}$ is the total propagation weight of length- k paths from s to i , defined as the sum of the products of the corresponding fixed coefficients α_{ij} along those paths. From the market clearing condition, $y_i = \sum_j \alpha_{ij} y_j + c_i$, I obtain the propagation operator $\mathcal{P}_{is} \equiv \sum_j \alpha_{ij} \ell_{js} = [\mathcal{A}\mathcal{L}]_{is}$. It holds $\mathcal{P}_{is} = \sum_{k=0}^{\infty} \sum_j \alpha_{ij} [\mathcal{A}^k]_{js} = \alpha_{is} + \sum_j \alpha_{ij} \alpha_{js} + \sum_{k \geq 2} \sum_j \alpha_{ij} [\mathcal{A}^k]_{js}$, representing direct, order-1 indirect, and higher-order indirect linkages. The same applies to, say, path $i \leftarrow m$: $\mathcal{P}_{im} \equiv \sum_j \alpha_{ij} \ell_{jm} = [\mathcal{A}\mathcal{L}]_{im}$. Both propagations decomposes over $j \in \Lambda$ using identical Input-Output coefficients α_{ij} . Hence, $\mathcal{L}$ induces not only a propagation mechanism over k -stages, but also a geometric structure over simultaneous connections to a shared intermediate inputs space through the product of i 's and $\{s, m\}$'s trade shares with j , for $i, j, s, m \in \Lambda$. By Lemma 2, the definition of a matrix product is $[\mathcal{A}\mathcal{A}']_{is} = \langle \mathbf{a}_i, \mathbf{a}_s \rangle$, the inner product of rows $\{i, s\}$; symmetrically, for columns: $[\mathcal{A}'\mathcal{A}]_{is}$. So the same primitive $\mathcal{A}$ that builds the propagation operator $\mathcal{L}$ also builds these Gram matrices at one specific order of the same expansion. ■

The theorem states that a single Input-Output matrix generates two distinct but tightly connected objects. On one hand, its Leontief inverse captures vertical dynamics: how commodities are produced by the means of other commodities (Sraffa 1960) through successive layers of production from direct to increasingly indirect effects along length- k production chains. On the other hand, the same matrix defines a horizontal geometry: sectors are related insofar as they share similar demand and supply relationships, interacting with common segments of the network. This induces a set of sectoral profiles which embeds sectors according to their commonalities in the Input-Output space. Key is that these objects are not separated but different features of the same primitive structure: one capturing recursive chains, the other describing the overlap of network linkages.

Theorem 1 is stated in terms of the primitive coefficient matrix $\mathcal{A}$, but it applies also to its Leontief inverse representation. The entries of $\mathcal{A}$ are the direct Input-Output coefficients, while the powers $\mathcal{A}^k$ aggregate their propagation across production chains of length- k . The representation $\mathcal{L}$ carries over the primitive structure into its cumulative form rather than creating a new one.

REMARK 1 *The Leontief inverse inherits the dual representation, since $\mathcal{L} = \sum_{k \geq 0} \mathcal{A}^k$.*

So, the duality established by the theorem is intrinsic to any Input-Output system. The primitive matrix $\mathcal{A}$ and its Leontief inverse $\mathcal{L}$ are two representations of the same underlying system, so that constructing similarities from either objects is logically equivalent, although the two carry different economic interpretations.

2.1. STYLIZED ECONOMIES AND HORIZONTAL GEOMETRY

Before turning to technicalities, I illustrate the economic relevance of the network's horizontal geometry through highly stylized production structures, sketched in Figure 2. I consider three benchmark configurations: (i) *sequential economies*, where connections occurs purely along a vertical chain (sector-by-sector towards the final consumer) or, alternatively, purely horizontally

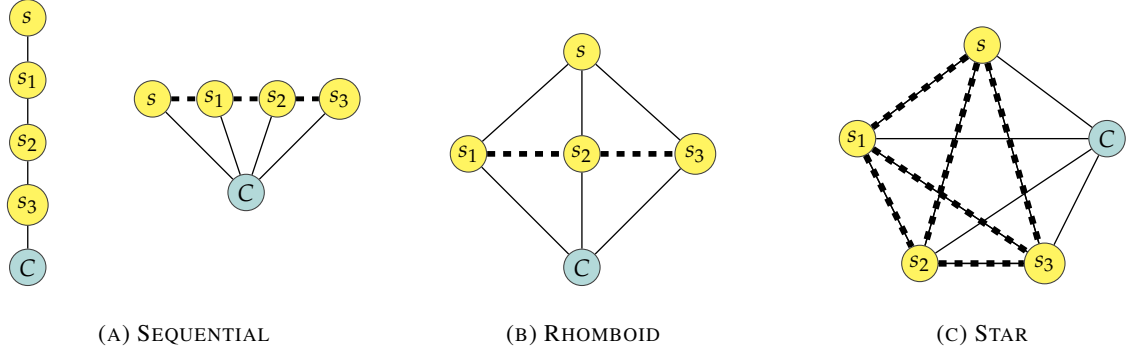


FIGURE 2: BENCHMARK ECONOMIES

Note: the figure illustrates some benchmark economic systems and their associated horizontal geometry (dashed) highlighting common Input-Output structures (solid). Sectors are in yellow, while the teal circles with C represent the final consumer.

(distinct sectors supplying only the final consumer); (ii) *rhomboid economies*, where sectors depend on a common upstream supplier while simultaneously selling to the final consumer; and (iii) *star economies*, with all sectors mutually connected while serving the final consumer.

Sequential economies.— Consider the “snake” in the left graph of Figure 2a. Connections are purely vertical, the upstream commodities flow sequentially along the supply chain, with no scope for horizontal interdependencies. By contrast, in the “half-spider” on the right graph, sectors sell directly to the final consumer without trading among themselves. This structure is a characteristic of standard multi-sector models with no inter-sectoral trade, and supply-based horizontal complementarities are central: even without explicit trade linkages, paired sectors are tied together by their shared exposure to final demand. Adjustments in one sector reshapes consumers’ expenditure across all others, moving horizontally through final demand reallocation and generating simultaneous responses absent any network structure.

Rhomboid economies.— A simple network in Figure 2b combines both sequential economies into a unique structure. This configuration blends vertical and horizontal propagation, as commodities travel vertically (via sector-specific supply chain) and horizontally across sectors simultaneously exposed to common supplier and final demand. Rhomboid economies thus epitomize both types of horizontal complementarities, as even sectors not directly trading with each other become tightly synchronized through their shared dependence on upstream and downstream markets.

Star economies.— Figure 2c classifies three archetypes. In fully symmetric and balanced star networks, where all sectors buy and sell with identical intensity, network distances collapse to zero: propagation is only vertical, but the network structure becomes irrelevant as sectoral shocks average out (e.g., Lucas 1977; Acemoglu, Carvalho, et al. 2012). In partially symmetric star networks, trade intensities in buying and selling are equal but sector-specific; vertical and horizontal dimensions coexist, though the two distance measures coincide. Finally, asymmetric star networks (i.e., different trade intensities across sectors) generate parallel vertical and horizontal channels, with comovement emerging endogenously from network distances: horizontal complementarities in production and shared exposure to final demand weave a dense web of horizontal interdependencies, and sectors with differing upstream and downstream connections experience unrelated effects.

These illustrative examples represent extreme cases. In more realistic, highly interconnected systems, the boundaries between these benchmark economies blur: vertical supply chains co-exist with horizontal interdependencies through shared buyers and suppliers, and sectors participate simultaneously in multiple propagation layers.

2.2. RECOVERING DEMAND AND SUPPLY DISTANCES

Consider the case in which two sectors, say $\{i, j\}$, are buying intermediate inputs from the same sector, say s . Let the price of the common input to vary by $d \log p_s$, holding all other prices fixed. By Lemma 1, the two sectors' costs respond as $d \log \kappa_i = \alpha_{is} \cdot d \log p_s$ and $d \log \kappa_j = \alpha_{js} \cdot d \log p_s$. The comovement induced by the common sector- s is therefore the covariance of the two responses:

$$b_{s \rightarrow}^{fd}[i, j] := \text{Cov}[d \log \kappa_i, d \log \kappa_j] = \alpha_{is} \alpha_{js} \geq 0 \quad (1)$$

Three properties follows. First, eq. (1) selects genuinely shared linkages, since $b_{s \rightarrow}^{fd}[i, j] > 0$ if and only if both sectors purchase from s – for strictly positive $\alpha_{is} \alpha_{js}$. Second, it weights by the network intensity of both: sharing a supplier from which each buys a large share of its inputs counts more than sharing one that is marginal for either. Third, since the α 's are cost shares, $b_{s \rightarrow}^{fd}[i, j]$ is the probability that one unit of expenditure drawn at random from sector- i and one drawn independently from sector- j end up at sector- s .

Iteratively, the above condition can be rewritten accounting for all sectors from which the pair is buying: $\mathbf{b}^{fd}[i, j] = [\alpha_{ii} \alpha_{ji}, \alpha_{ij} \alpha_{jj}, \dots, \alpha_{im} \alpha_{jm}]$, which is the $1 \times m$ vector of the pair's shared intermediate usage when sectors $\{i, j\}$ are buying from others. The vector carries two distinct pieces of information. Its support identifies the non-negative entries through which suppliers the pair is connected, and is the extensive margin; and its magnitudes identifying how much each channel weighs, and are the intensive margin. Summing across channels and stacking such condition across all sectors delivers the $m \times m$ demand-side proximity matrix $\mathbf{B}^{fd}$, whose generic element is $b^{fd}[i, j] = \sum_{s \in \Phi} \alpha_{is} \alpha_{js}$. The greater the value of each cell, the closer the sectors are within the production network.³ Using the demand composition of sector- i , $\mathbf{a}_i^{col} := [\alpha_{ii}, \alpha_{ij}, \dots, \alpha_{im}]'$ –, i.e., its column in $\mathbf{A}$ identified by Lemma 1 –, each entry is an inner product between profiles, $b^{fd}[i, j] = \langle \mathbf{a}_i^{col}, \mathbf{a}_j^{col} \rangle$ along the same row- s , $\forall s \in \Phi$, so that $\mathbf{B}^{fd} = \mathbf{A} \mathbf{A}'$. Since it is a transformation of the Input-Output matrix, by Theorem 1 the demand-profile proximity matrix is symmetric, positive semi-definite, and it is the Gram matrix of the collection $\{\mathbf{a}_i^{col}\}_{i \in \Phi}$.

Matrix $\mathbf{B}^{fd}$ measures *proximity*: large entries denote strongly co-exposed pairs. The corresponding *distance* follows from the standard Gram-to-metric conversion,

$$d^{fd}[i, j]^2 := b^{fd}[i, i] + b^{fd}[j, j] - 2 b^{fd}[i, j] = \|\mathbf{a}_i^{col} - \mathbf{a}_j^{col}\|_2^2 \quad (2)$$

where the second equality follows by expanding the squared norm through bi-linearity of

³ Co-exposure is increasing in both trade intensities: for $\alpha_{is} < \alpha_{js}$, a shock transmitted to the pair through s moves the two sectors' costs by unequal amounts and the induced comovement is weak, signaling low horizontal complementarity; conversely, when both intensities are large the same shock moves both strongly, and $b_{s \rightarrow}^{fd}[i, j]$ is large. The same logic applies to eq. (4).

the inner product. Explicitly,

$$d^{fd}[i, j] = \sqrt{[\alpha_{ii} - \alpha_{ji}]^2 + [\alpha_{ij} - \alpha_{jj}]^2 + \dots + [\alpha_{im} - \alpha_{jm}]^2} \quad (3)$$

thereby representing a certain network “economic” distance between two sectors, with larger values denoting how dissimilar they are in terms of shared Input-Output structure when buying from upstream sellers. Henceforth, the corresponding $m \times m$ matrix $\mathbf{D}^{fd}$ collects such distances, whose characterizing elements $d^{fd}[i, j] > 0, \forall i \neq j \in \Phi$, are symmetric outside the main diagonal (the distance from i to j is the same as from j to i), with $\text{Diag}[\mathbf{D}^{fd}] = 0$. Proximity and distance are thus two readings of one object: $\mathbf{B}^{fd}$ is foundational on the theory side, where larger entries mean closer sectors, while the conversion delivers the metric for $\mathbf{D}^{fd}$ used in the empirics of Sections 4-5, where larger entries mean the opposite.

EXAMPLE 2 (Factor input demand metrics) Consider the pair of sectors $\{s_1, s_2\}$ in an Input-Output matrix mirroring Example 1, where the common sectors from which both are purchasing are $\{s, s_3\}$, and assume fictional trade intensities for sector- s_1 to be $\alpha_{s_1s} = 0.2$ and $\alpha_{s_1s_3} = 0.1$, while that of sector- s_2 are $\alpha_{s_2s} = 0.1$ and $\alpha_{s_2s_3} = 0.2$. Accordingly, the co-exposure vector expressing their factor input demand relation is $\mathbf{b}^{fd}[s_1, s_2] = [.02, .02]$ when buying from sector- s and sector- s_3 , respectively. Then $d^{fd}[s_1, s_2] = \sqrt{.02}$. Demand-based proximity is positive even in the absence of a direct linkage between the two sectors in the pair.

Consider now the case in which two sectors, say $\{i, j\}$, are selling their own commodity as an intermediate input to the same sector, say s . By definition of cost share, the value of sector- i ’s sales to s is $p_i x_{si} = \alpha_{si} \cdot \mathcal{K}_s^{tot}$, with $\mathcal{K}_s^{tot}$ denoting the common buyer’s total cost. A variation in the buyer’s scale at constant shares therefore moves both suppliers’ sales in the same direction, and the covariance of the two responses per unit of shock variance delivers

$$b_{\rightarrow s}^{fs}[i, j] := \text{Cov} \left[d \log [p_i x_{si}], d \log [p_j x_{sj}] \right] = \alpha_{si} \alpha_{sj} \geq 0 \quad (4)$$

Collecting across all common buyers, $\mathbf{b}^{fs}[i, j] = [\alpha_{ii} \alpha_{ij}, \dots, \alpha_{mi} \alpha_{mj}]$, and stacking delivers the supply-side proximity matrix $\mathbf{B}^{fs}$, with generic element $b^{fs}[i, j] = \sum_{s \in \Phi} \alpha_{si} \alpha_{sj} = \langle \mathbf{a}_i^{row}, \mathbf{a}_j^{row} \rangle$ along the same column- s , $\forall s \in \Phi$, where $\mathbf{a}_i^{row} := [\alpha_{ii}, \alpha_{ji}, \dots, \alpha_{mi}]$ is the supply profile of sector- i – i.e., its row in $\mathbf{A}$. In matrix form $\mathbf{B}^{fs} = \mathbf{A}' \mathbf{A}$; the associated $m \times m$ distance matrix $\mathbf{D}^{fs}$ follows from eq. (2) applied to supply profiles.

EXAMPLE 3 (Factor input supply metrics) Consider the pair of sectors $\{s_1, s_2\}$ in an Input-Output matrix mirroring Example 1, where the common sectors to which both are selling are $\{s, s_3\}$, and assume fictional trade intensities for sector- s_1 to be $\alpha_{ss_1} = 0.4$ and $\alpha_{s_3s_1} = 0.3$, while that of sector- s_2 are $\alpha_{ss_2} = 0.6$ and $\alpha_{s_3s_2} = 0.6$. Accordingly, the co-exposure vector expressing their factor input supply relation is $\mathbf{b}^{fs}[s_1, s_2] = [.24, .18]$ when selling to sector- s and sector- s_3 , respectively. Then $d^{fs}[s_1, s_2] = \sqrt{.13}$. Supply-based proximity is positive even in the absence of a direct trade linkage between the two sectors in the pair.

An important aspect to consider when introducing economic distances in the production network is that, under the same configuration of the inter-sectoral trade structure, sectors can

be in a different “distance relation” depending on whether they are buying from or selling to other sectors. Moreover, all sectors are simultaneously connected by network distances due to common demand and supply. Asking when the two sides coincide amounts to asking when co-exposure fails to distinguish $\mathcal{A}$ from $\mathcal{A}'$, which is the content of the following result.

THEOREM 2 (On the configuration of network “economic” distances) *Let the set of sectors being Φ , and α_{ij}^q , $\forall i, j \in \Phi$ and $q = \{fd, fs\}$, the normalized Input-Output coefficients. Then, demand- and supply-based distance matrices satisfy: (i) $\mathcal{D}^{fd}$ and $\mathcal{D}^{fs}$ are non-negative, symmetric and positive semi-definite, with $d^q[i, j] > 0$ if and only if the pair shares at least one neighbour, so that both matrices are full whenever every pair does; (ii) a given Input-Output structure uniquely determines $\mathcal{D}^{fd}$ and $\mathcal{D}^{fs}$, which differ entry-wise unless the corresponding bilateral trade with the common neighbour coincides; and (iii) if α^q ’s are non-symmetric, then all entries of $\mathcal{D}^{fd}$ and $\mathcal{D}^{fs}$ generically differ; conversely, if $m \geq 4$, then both share at least one identical entry whenever at least $m - 2$ sectors trade symmetrically with a common sector and at least one additional symmetric trade relation elsewhere in the network exists.*

Proof: let α_{is}^{fd} being the i ’s expenditure share at s (row-sum normalization), α_{si}^{fs} the s ’s sale share at i (column-sum normalization), such that $\sum_{s \in \Phi} \alpha_{is}^{fd} = 1$ and $\sum_{s \in \Phi} \alpha_{si}^{fs} = 1$. Define $\Phi_i^{up} := \{s \in \Phi : \alpha_{si}^{fs} > 0\}$ and $\Phi_i^{dw} := \{s \in \Phi : \alpha_{is}^{fd} > 0\}$. For any $\{i, j\} \in \Phi$, let the set of common neighbours be $\Phi_{i,j}^{com} := \{s \in \Phi : s \in \Phi_i^{up} \cap \Phi_j^{up} \text{ or } s \in \Phi_i^{dw} \cap \Phi_j^{dw}\}$. Demand co-exposure is $b_{s \rightarrow}^{fd}[i, j] = \alpha_{is}^{fd} \alpha_{js}^{fd}$ and the supply one is $b_{\rightarrow s}^{fs}[i, j] = \alpha_{si}^{fs} \alpha_{sj}^{fs}$, both defining the vectors $\mathbf{b}^{fd}[i, j] := \{b_{s \rightarrow}^{fd}[i, j] : s \in \Phi_{i,j}^{com}\}$ and $\mathbf{b}^{fs}[i, j] := \{b_{\rightarrow s}^{fs}[i, j] : s \in \Phi_{i,j}^{com}\}$. *Existence.* Since each cell is a product of non-negative shares, $b^q[i, j] > 0$ if and only if $\Phi_{i,j}^{com} \neq \emptyset$. *Coincidence.* Cell-wise equality requires $\alpha_{is}^{fd} \alpha_{js}^{fd} = \alpha_{si}^{fs} \alpha_{sj}^{fs}$, which holds under bilateral symmetry at the common neighbour, $\alpha_{is}^{fd} = \alpha_{si}^{fs}$ and $\alpha_{js}^{fd} = \alpha_{sj}^{fs}$. *Counting:* if s is symmetric with $m' = m - 2$ sectors, every pair yields one coinciding entry, giving $\binom{m'}{2} = \frac{m'(m'-1)}{2}$. *Distances.* By eq. (2), the same applies to distance matrices. In relation to Corollary 1: $m = 3$ leaves no room for an independent extra symmetric pair, so $m \geq 4$ is the minimal non-degenerate case. ■

The first two properties clearly emerge in Examples 2-3 and Figure 1. The final clarifies when the relation between two sectors is uniquely determined: if two sectors buy and sell the same relative quantity of intermediate inputs with a common third sector, then their factor input demand and factor input supply co-exposures coincide. Their actual quantities may differ; what matters is that, for each sector individually, its purchases or sales with the common sector match in relative terms.⁴

COROLLARY 1 (Coinciding distances) *For $m \geq 4$, if a common sector trades symmetrically with $m' = m - 2$ others, then $\frac{m'(m'-1)}{2}$ sectors exhibit symmetric distances.*

3. AN INSPECTION INTO THE HORIZONTAL TRANSMISSION

Outlined horizontal geometry introduces novel implications for the propagation of asymmetric, independent and idiosyncratic shocks across the network. Crucially, the transmission depends not only on the size of existing connections among sectors, but also on their network “economic” distance. As shown in Figure 3, a shock originating in a given sector propagates

⁴ A “normalized” network refers to an Input-Output matrix once rows or columns are scaled by their sum; entries thus express relative and not absolute values, required to build $\mathcal{D}^{fd}$ and $\mathcal{D}^{fs}$.

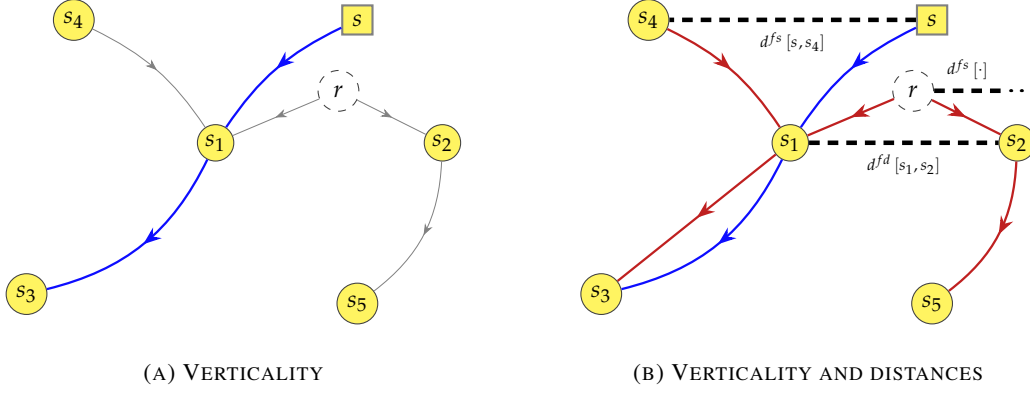


FIGURE 3: HORIZONTALITY OF A TRANSMISSION

Note: stylized example of how a sectoral shock propagates across the production network. Coloured and oriented bold-curves trace the propagation mechanism: vertical transmission is in blue, while additional transmission paths from the horizontal geometry are in red. Oriented thin-grey curves indicate further existing Input-Output (I-O) connections, and network economic distances through common nodes are represented by dashed lines. The underlying network structure is identical in both panels. Panel 3a shows a purely vertical propagation of a shock to sector- i , following standard sectoral supply-chain logics. Panel 3b depicts a richer mechanism, where the shock also spreads horizontally through sectors sharing common upstream suppliers or downstream buyers.

vertically through downstream buyers along the supply chain, while concurrently transmitting horizontally. Notably, this horizontal transmission materializes even between sectoral pairs that are not directly connected, reflecting their shared demand and/or supply relationships.⁵

EXAMPLE 4 (Propagating behaviour under horizontal distances) Consider an economy populated by $\{s, s_1, s_2, s_3, s_4, s_5, r\} \in \Phi$ sectors, with r representing the rest of the Input-Output network. Some trade with others, while some others do not: s_1 is buying from $\{s, s_4, r\}$ while selling to s_3 ; s_2 trades from r and to s_5 . Suppose a supply shock to s . In Panel 3a, the shock propagates downstream: it directly affects s_1 and indirectly reaches s_3 , yielding a purely vertical transmission with others unaffected.⁶ Unknown remain the potential effects of induced adjustments in s_1 , even though it shares linkages with others. Panel 3b illustrates both vertical and horizontal propagation: a shock to s not only affects its immediate trading connections but also alters its interaction with s_4 : changes in s_1 triggered by the initial shock generate an horizontal effect that links $\{s, s_4\}$ through their common supply-based connection with s_1 . By analogy, this reasoning falls under supply distances of $\{s, s_4\}$ with the rest of other sectors- r selling to s_1 , $d^{fs}[\cdot]$. With due demand-based scheme, analogous is the logic underlying $d^{fd}[s_1, s_2]$, simultaneously buying from r . Subsequent adjustments in $\{s_4, r\}$ feed back into s_1 , compounding the indirect propagation towards s_3 , and into s_2 : reverberating the initial shock on s , via $\{s_1, r\}$ to s_2 , an additional vertical effect further downstream, from s_2 to s_5 , is generated.

Example 4 illustrates how incorporating factor input demand and factor input supply net-

⁵ The example is presented as the more intuitive case of a sectoral supply shock propagating downstream, affecting sectors that depend on the initially shocked sector. Yet, the same underlying logic applies in the case of a demand shock, where the direction of a transmission is reversed, with the sectoral shock propagating upstream as sectors increase demand for intermediate inputs from their suppliers. In the presence of an already-existing Input-Output linkage between distanced sectors, the transmission is progressively amplified.

⁶ Upstream responses vanish because price and quantity effects offset each other (e.g., Acemoglu, Carvalho, et al. 2012). Yet, the horizontal mechanism uncovers these latent transmission channels via common network structures instead of tracking non-linearities in production (e.g., Baqaee and Farhi 2019).

work economic distances, and therefore horizontal complementarities, fundamentally reshapes the vertical transmission of micro-level shocks. The new propagation stems from similar demand/supply trade relationships between sectors sharing a comparable Input-Output structure (*i.e.*, common upstream suppliers or downstream buyers), and this paper is mostly on the conceptualisation of the $d[\cdot]$ -types of linkages. Stated differently, distance-based complementarities that emerge along the network's horizontal architecture alter the vertical transmission by either compounding or mitigating the propagating behaviours of a micro-originated shock.

3.1. REDUCED FORMS FOR DISTANCE-LEAD COMOVEMENT

A shadowed implication of the framework developed so far is that intermediate inputs generally function as substitutes, allowing sectors to freely adjust their input mix in response to shocks. However, the horizontal geometry of a network can embed two aspects: (i) vertical propagation through the supply chain may remain the dominant channel, and what may appear as a horizontal transmission can in fact be the by-product of vertical propagation; and (ii) intermediate inputs complementarity in production may alter the transmission of shocks. This tension between horizontal complementarities at the network level and substitutability at the input level sets the stage for a more precise characterization of how sectoral comovement occurs.

In the context of factor input supply, alleviating the tension is doable since competition for common downstream buyers yields a more direct and unambiguous form of horizontal interdependence. Conversely, the structure of upstream connections renders demand-based linkages ambiguous to interpret, since the coexistence of a dual transmission makes the direction and the strength of induced comovement subtle.

Consider a single-input (purchasing from only one upstream supplier, with no substitution possibilities), and a multiple-input (sourcing from multiple suppliers) cases, and a Constant Elasticity of Substitution (CES) among intermediate inputs.

PROPOSITION 1 (Demand-driven comovement) *In the single-input case, comovement in demand implies $\delta_{s \rightarrow [i, j]} \propto -\sigma \psi^{fd}$. In the multi-input CES case, then $\delta_{s \rightarrow [i, j]} \propto -\sigma [1 - e_{is}] \psi^{fd}$, with relative intermediate inputs changes from common supplier, $\delta_{s \rightarrow [i, j]} = \frac{\partial \log x_{is}}{\partial \log x_{js}}$, jointly determined by substitution elasticity $\sigma > 0$, downstream sectoral expenditure share e_{is} , and upstream pass-through ψ^{fd} . Opposite comovement occurs if $\psi^{fd} > 0$.*

Proof: let downstream buyer- i aggregate intermediates via $x_i = \left(\sum_{s \in \Phi} \alpha_{is}^{1/\sigma} x_{is}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}$. Differentiating Lemma 1 delivers own-price elasticity $\frac{\partial \log x_{is}}{\partial \log p_s} = -\sigma [1 - e_{is}]$ with $e_{is} = \frac{\alpha_{is} p_s^{1-\sigma}}{\sum_{s \in \Phi} \alpha_{is} p_s^{1-\sigma}}$. By the chain rule: $\frac{\partial \log x_{is}}{\partial \log x_{js}} = \frac{\partial \log x_{is}}{\partial \log p_s} \cdot \frac{\partial \log p_s}{\partial \log x_{js}} = -\sigma [1 - e_{is}] \psi^{fd}$. If purchasing one input, then $\frac{\partial \log x_{is}}{\partial \log p_s} = -\sigma$. ■

Opposite comovement between downstream sectors arises through upstream adjustments in supply. Intuitively, when j experiences a positive shock it might bids up the price of inputs from the common supplier- s , $\psi^{fd} = \frac{\partial \log p_s}{\partial \log x_{js}} > 0$, through increased demand. If this higher input cost makes production more expensive for i , its demand from s falls, generating opposite comovement. Whether this happens hinges on the complementarity between s 's input and the others purchased by i : when $\sigma > 0$, any price increase in s alters i 's input mix; in the limit $\sigma \rightarrow \infty$, inputs become perfect substitutes and i fully shifts away from s , generating no comovement with

sj . In the general multi-input CES case, the effect is further scaled by sector- i 's expenditure share on s , e_{is} , so that sectors more reliant on s are strongly affected. Thus, demand-driven comovement reflects not automatic crowding-out, but the interplay of upstream price pass-through (from vertical propagation), and downstream demand adjustments (indirectly generated by horizontal complementarities).

Considering instead supply-based distances, the reallocation of downstream demand across competing upstream suppliers is a clear channel for comovement.

PROPOSITION 2 (Supply-driven comovement) *In the single-input case no condition exists for comovement in supply due to the absence of competition for downstream markets. In the multi-input CES case it holds $\delta_{\rightarrow s}[i, j] \propto \sigma e_{sj} \psi^{fs}$, with relative intermediate inputs changes to common buyer, $\delta_{\rightarrow s}[i, j] = \frac{\partial \log x_{si}}{\partial \log x_{sj}}$, jointly determined by substitution elasticity σ , sectoral revenues for upstream supplier e_{sj} , and downstream pass-through ψ^{fs} . Opposite comovement occurs if $\psi^{fs} < 0$.*

Proof: let upstream supplier- i sell to a downstream sector- s with aggregator $x_s = \left(\sum_{i \in \Phi} \alpha_{si}^{1/\sigma} x_{si}^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}$. Differentiating Lemma 1 delivers cross-price elasticity $\frac{\partial \log x_{si}}{\partial \log p_j} = \sigma e_{sj}$, with $e_{sj} = \frac{\alpha_{sj} p_j^{1-\sigma}}{\sum_{i \in \Phi} \alpha_{si} p_i^{1-\sigma}}$. By the chain rule: $\frac{\partial \log x_{si}}{\partial \log x_{sj}} = \frac{\partial \log x_{si}}{\partial \log p_j} \cdot \frac{\partial \log p_j}{\partial \log x_{sj}} = \sigma e_{sj} \psi^{fs}$. If selling a single input, $\frac{\partial \log x_{si}}{\partial \log p_j} = 0$. ■

Opposite comovement between upstream suppliers mechanically arises from competition for downstream buyers. Intuitively, when j receives a positive shock it lowers its relative price, capturing a larger share of common buyer- s 's demand from adjustments in upstream price, $\psi^{fs} = \frac{\partial \log p_j}{\partial \log x_{sj}} < 0$. This revenue reallocation necessarily comes at the expense of the competing supplier- i , generating opposite comovement even when $\sigma \rightarrow \infty$ (shifting away of s 's demand from i , increased towards j due to relative price effect). The magnitude depends on two forces: the elasticity of substitution σ across sector- i 's inputs controls how easily s reallocates spending, while the expenditure share e_{sj} scales the importance of j for the common sector- s .

Discussion.— A networked economy inherently embeds complementarities in intermediate inputs – scaling the magnitude but not the direction of comovement –, as sectors depend on overlapping sets of buyers and suppliers. In the demand-distance case, comovement is ambiguous: it reflects a combination of vertical propagation along the supply chain and horizontal complementarities from shared suppliers. In the supply-distance case, comovement is transparent: the horizontal transmission directly arises from the reallocation of inputs to adjustments in relative prices, and the interaction between the two dimensions is mechanically determined.⁷

Importantly, the discrepancy between the ambiguous effects in demand and the clear responses in supply is in accordance with non-linear network theories (e.g., Atalay 2017; Baqaee and Farhi 2019), where downstream complementarities – and hence complementarities between purchased intermediate inputs – matter the most. This dimension is embodied into the supply-based distances through the downstream pass-through effect, which create horizontal interdependencies among sectors selling to the same buyers: when one supplier experiences a positive

⁷ Baqaee and Farhi (2019) develop a framework that emphasizes non-linear amplification; mine summarizes similar complementarity forces that drive non-linear dynamics by the means of horizontal distances.

shock, downstream complementarities of common buyers cause other suppliers to adjust their production in response, and the propagation operates directly through the interdependence of upstream inputs used together in downstream production. Such amplification is largely absent under demand-based distances, since the upstream pass-through reflects only substitution patterns across common upstream suppliers.

3.2. VERTICAL AND HORIZONTAL NON-DEPENDENCE

A key step in the analysis is to establish a dimensional separation.

THEOREM 3 (On the weighting of network “economic” distances) *The horizontal dimension of an Input-Output economy (defined by demand- and supply-based distance matrices) is not weighted by its vertical dimension (defined by the directed or Leontief inverse matrix).*

Proof: provided in [Appendix](#). ■

The theorem postulates that, within an Input-Output economy, the horizontal geometry of the network – captured by demand- and supply-driven distances – remains unweighted and independent of its trade intensities, encoded in the Input-Output matrix. This distinction is crucial for understanding the economics of production networks: while the vertical structure reflects the strength of sectoral relationships, the horizontal structure identifies demand/supply complementarities across sectors related through common upstream suppliers or downstream buyers. In other words, by separating the horizontal from the vertical geometry, the theorem clarifies that whatever distance matrix is not merely a transformation of the Input-Output matrix, but a complementary object that reveals otherwise hidden propagation channels.

Nevertheless, as linkages densify and the network expands, horizontal complementarities become increasingly important (with more sectors sharing common suppliers and buyers, interdependencies proliferate).

THEOREM 4 (On the importance of network “economic” distances) *Consider a network of size- m with Input-Output matrix $\mathbf{A}_m := \mathbf{V}_m \cup \mathbf{D}_m$, where $\mathbf{V}_m$ encodes vertical buyer-supplier ties and $\mathbf{D}_m$ encoding horizontal complementarities. Let $\boldsymbol{\epsilon}_m$ be a vector of independent shocks, and let $\mathcal{P}_m^{(\mathcal{V})}$ and $\mathcal{P}_m^{(\mathcal{D})}$ denote the associated matrices for propagation channels. Then, as the network size grows, the vertical propagation vanishes in spectral norm:*

$$\left\| \mathcal{P}_m^{(\mathcal{V})} \boldsymbol{\epsilon}_m \right\|_2 / \left\| \mathcal{P}_m^{(\mathcal{D})} \boldsymbol{\epsilon}_m \right\|_2 \xrightarrow{m \rightarrow \infty} 0.$$

Proof: let $G = (m, \Lambda)$ be a finite directed weighted graph, with $\Lambda = \{i, j \in m \times m : \alpha_{ij} \geq 0\}$. Let $\mathbf{A}$ denote the Input-Output matrix of G . Decompose $\mathbf{A}_m = \mathbf{V}_m \cup \mathbf{D}_m$ which, by Theorem 3, isolates vertical and horizontal components, and let $\mathcal{W}_{\mathbf{A}_m}$, $\mathcal{W}_{\mathbf{V}_m}$, and $\mathcal{W}_{\mathbf{D}_m}$ being *graphons* – that is, step functions on $U : [0, 1]^2$. Graphon convergence is measured in the “cut norm” $\|U\|_{\square}$, controlling propagation from the spectral norm up to a constant σ : $\|U\|_2 \leq \sigma \|\mathcal{W}_U\|_{\square}$. Given a $1 \times m$ vector of i.i.d. micro-level shocks $\boldsymbol{\epsilon}_m$, under (i) vertical sparsity, $\|\mathcal{W}_{\mathbf{V}_m}\|_{\square} \xrightarrow{m \rightarrow \infty} 0$, and (ii) horizontal convergence to a non-degenerate limit graphon $\mathcal{W} : [0, 1]^2 \rightarrow [0, 1]$ such that $\|\mathcal{W}_{\mathbf{D}_m} - \mathcal{W}\|_{\square} \xrightarrow{m \rightarrow \infty} 0$: continuity transmits these limits to the propagation operator $\|\mathcal{P}_m^{(\mathcal{V})} \boldsymbol{\epsilon}_m\|_2 \xrightarrow{m \rightarrow \infty} 0$ while $\|\mathcal{P}_m^{(\mathcal{D})} \boldsymbol{\epsilon}_m\|_2$ stays bounded away from zero, so $\|\mathcal{P}_m^{(\mathcal{V})} \boldsymbol{\epsilon}_m\|_2 / \|\mathcal{P}_m^{(\mathcal{D})} \boldsymbol{\epsilon}_m\|_2 \xrightarrow{m \rightarrow \infty} 0$. ■

Vertical spillovers disappear relative to the horizontal. For larger production networks (e.g., firm-to-firm), vertical buyer-supplier chains become too sparse and elongated to transmit shocks

effectively, while horizontal complementarities grow denser and increasingly bind nodes together. As more network participants buy and sell, shared buyers/suppliers become more likely, and the more important horizontal complementarities become. In the limit where network size- $m \rightarrow \infty$, horizontal effects dominate: bigger networks transmit shocks not as cascading supply chains, but as clusters of horizontally-related sectors. This dominance persists even when vertical supply chains themselves expand endogenously as the network grows in size.

COROLLARY 2 (Endogenous expansion) *In growing networks, the horizontal geometry asymptotically dominates shock propagation even as vertical chains expand.*

Proof: graphon $\|\mathcal{W}_{\mathcal{V}_m}\|_{\square}$ is non-decreasing in network size- m , and horizontal complementarities must satisfy $\|\mathcal{W}_{\mathcal{D}_m}\|_{\square} \geq a \|\mathcal{W}_{\mathcal{V}_m}\|_{\square}$ for a given $a > 0$. Even if vertical linkages increase with network size- m , horizontal complementarities proliferate, because each new vertical relation creates new horizontal connections via common partners. Given universal constants $0 < b \leq c < \infty$, then $\|\mathcal{P}_m^{(\mathcal{V})}\|_2 \leq c \|\mathcal{W}_{\mathcal{V}_m}\|_{\square}$ and $\|\mathcal{P}_m^{(\mathcal{D})}\|_2 \geq b \|\mathcal{W}_{\mathcal{D}_m}\|_{\square}$. Theorem 4 still holds: as a network of size- m grows, the horizontal component becomes strictly denser than the vertical one, so the denominator grows faster than the numerator. ■

3.3. SECTORAL COMOVEMENT AND AGGREGATE BEHAVIOUR

Equilibrium adjustments in production networks are central to understand the origins of aggregate business cycles. Within this perspective, an important question emerges: to what extent does employment comovement shape the dynamics of aggregate fluctuations in output? From the model's equilibrium conditions, the impact of a certain sector-specific shock on real Gross Domestic Product (GDP) can be decomposed into its propagation on aggregate Total Factor Productivity (TFP) and its effect on aggregate employment – exploiting the Divisia Index (e.g., vom Lehn and Winberry 2022). Fluctuations in aggregate real GDP growth are the result of

$$d \log Y = \sum_{i \in \Phi} \left(\lambda_i d \log z_i + \nu_i d \log n_i \right) \quad (5)$$

where $\lambda_i = \frac{p_i y_i}{P Y}$ is the *Domar weight* of sector- i , and $\nu_i = \frac{p_i^Y y_i^Y}{P^Y Y^Y}$ its *value-added Domar weight*, with $p_i^Y y_i^Y$ representing the nominal value-added. These terms measure, respectively, the sectoral ratios of the gross output and value-added to GDP. As such, both capture each sector's relative importance in the economy and its weight in aggregate fluctuations and business cycle dynamics.

Employment comovement thus constitutes a central phenomenon to investigate under inter-sectoral trade network: the cyclical behaviour of the economy origins from both changes in sectoral TFP, $d \log z_i$, and in sectoral employment, $d \log n_i$. Focusing on employment comovement offers distinct advantages. It provides a directly observable measure of sectoral activity, avoiding the interpretability (whether there is shock transmission or shocks correlation; Huo et al. 2025) and the identification challenges associated with productivity. As such, through the lens of eq. (5), different values of network distances govern the sign of the shock's transmission:

$$d \log Y = \sum_{i \in \Phi} \lambda_i d \log z_i + \sum_{i \in \Phi} \begin{cases} \nu_i \Delta \text{comovement} & \text{under major distance} \\ \nu_i \Delta \text{reallocation} & \text{under minor distance} \end{cases}$$

In fact, the model outlined in Section 2 delivers the following insight into employment comovement.

PROPOSITION 3 (Propagation of sectoral variations) *The overall production network effects on sectoral employment are summarized by:*

$$d \log \mathbf{n} = \Theta \left\{ d \log C + d \log \mathcal{S} + d \log \mathcal{V} + d \log \mathcal{D}(q) \right\} \quad \text{for } q = \{fd, fs\}$$

where $\mathcal{V}$ and $\mathcal{D}(q)$ represent the vertical and horizontal effects, respectively, with $\mathcal{D}(fd) = \mathcal{D}^{fd} \cdot \left[d \log \mathbf{n} - (\mathcal{I}'_{\Phi} d \log \mathbf{n}) \mathbf{1} \right]$ and $\mathcal{D}(fs) = \mathcal{D}^{fs} \cdot \left[d \log \mathbf{p} - (\mathcal{I}'_{\Phi} d \log \mathbf{p}) \mathbf{1} \right]$.

Proof: provided in [Supplemental Appendix A](#). ■

Mitigated by a bundle of structural parameters (Θ), changes in sectoral employment are a function of aggregate consumption (C), both sector-specific productivity and consumption of its good ($\mathcal{S}$), other sectors' productivities, employment levels, and intermediate inputs usage ($\mathcal{V}$) whose magnitudes depend on the Input-Output matrix, and network “economic” distance effects ($\mathcal{D}$) for both demand (through employment differentials) or supply (through relative price differentials) horizontal linkages. Square parentheses denote the difference between the change in all other sectors and the sector's own change. For example, considering sector- i 's employment, $d \log \mathbf{n} - (\mathcal{I}'_i d \log \mathbf{n}) \mathbf{1} = [d \log n_i - d \log n_i, \dots, d \log n_m - d \log n_i]$ where $\mathcal{I}_i$ is an $m \times 1$ indexing vector that selects sector- i and $\mathbf{1}$ an $m \times 1$ vector of ones, both replicating the reference sector's change across all sectors, ensuring that the horizontal propagation term captures relative deviations. For all sectors:

$$\mathcal{D}^{fd} \cdot \left[d \log \mathbf{n} - (\mathcal{I}'_{\Phi} d \log \mathbf{n}) \mathbf{1} \right] = \begin{bmatrix} \sum_{j \in \Phi} d^{fd}[i, j] \cdot (d \log n_j - d \log n_i) \\ \vdots \\ \sum_{i \in \Phi} d^{fd}[m, i] \cdot (d \log n_i - d \log n_m) \end{bmatrix}$$

The interpretation of Proposition 3 follows Subsection 3.1: sectors sharing upstream suppliers exhibit ambiguous comovement, as employment changes ($d \log \mathbf{n}$) enter in both vertical and horizontal components; sectors sharing downstream buyers undergo relative-price reallocation of supply, producing opposite comovement. Moreover, vertical propagation prevails under low proximity (high distance, low theoretical $\mathcal{D}^q$), leading to positive sectoral comovement. For low distances, horizontal forces dominate, thereby inducing opposite comovement.

Finally, it is worth assessing the extent to which the mere horizontal geometry of networked economies contributes to aggregate fluctuations.

THEOREM 5 (On aggregate cycles) *Consider an Input-Output economy made of $i = \{1, \dots, m\} \in \Phi$ sectors. Let λ_i denote sector- i 's Domar weight, and assume no sector is dominant, in the sense that $\lambda_i \asymp \frac{1}{m}$, $\forall i \in \Phi$. Beside vertical propagation, if sectors share upstream suppliers or downstream buyers, then idiosyncratic shocks generate non-vanishing aggregate GDP fluctuations: $\liminf_{m \rightarrow \infty} \text{Var}[Y_m] > 0$.*

Proof: let $G = (m, \Lambda)$ be a finite directed weighted graph, with $\Lambda = \{i, j \in m \times m : \alpha_{ij} \geq 0\}$. Let $\mathcal{A}$ denote the Input-Output matrix of G , associated to its Leontief inverse $\mathcal{L}$. Let $\boldsymbol{\lambda}$ be the $1 \times m$ vector of sectoral Domar weights, and $\boldsymbol{\varepsilon}$ the $1 \times m$ vector of i.i.d. micro-level shocks with variance $\sigma^2 < \infty$. Define aggregate log-output as $Y_m := \boldsymbol{\lambda}' \mathbf{v}$, with $\mathbf{v} = \mathcal{L} \boldsymbol{\varepsilon}$. Then, $\text{Var}[Y_m] = \sum_{i=1}^m v_i^2 \mathbb{E}[\varepsilon_i^2] + \sum_{i \neq j} v_i v_j \mathbb{E}[\varepsilon_i \varepsilon_j]$, so that $\text{Var}[Y_m] = \sigma^2 \boldsymbol{\lambda}' \mathcal{L} \mathcal{L}' \boldsymbol{\lambda}$, which is the pure vertical channel. Adding the symmetric horizontal similarity matrix $\mathcal{D}$ gives the combined operator $\mathcal{P} := \mathcal{L} \mathcal{L}' \cup \mathcal{D}$, with Lowner order $\mathcal{P} \succeq \mathcal{D}$ and

$\mathcal{P} \succeq \mathcal{LL}'$, so that $\text{Var}[Y_m] = \sigma^2 \lambda' \mathcal{P} \lambda$. Restrict the attention to a macroscopic horizontal cluster- Ψ (*i.e.*, a positive fraction of sectors are strongly connected horizontally defined as $|\Psi|/m \rightarrow \text{fraction-}\eta > 0$ with $\{i, j\}$ horizontal distance $d_{ij} \geq \underline{d} > 0$ on Ψ), that is dropping the component $\mathcal{LL}'$ in $\mathcal{P}$: this gives a lower bound to fluctuations in aggregate output (*i.e.*, considering only the sources of aggregate cycles from the horizontal geometry). Then, $x_i = \lambda_i \mathbf{1}_{\{i \in \Psi\}}$ gives $\lambda' \mathcal{P} \lambda \geq \mathbf{x}' \mathcal{P} \mathbf{x}$, the latter given by $\sum_{i,j \in \Psi} \lambda_i \lambda_j d_{ij} \geq d \cdot (\sum_{i \in \Psi} \lambda_i)^2$ using $d_{ij} \geq \underline{d}$ on Ψ . Under uniform Domar weights $\lambda_i = 1/m, \forall i \in \Psi$, then $\sum_{i \in \Psi} \lambda_i = |\Psi| \cdot \frac{1}{m} \equiv \eta$. Hence $\text{Var}[Y_m] \geq \sigma^2 \cdot \underline{d} \cdot \eta^2$ so that $\liminf \text{Var}[Y_m] > 0$: aggregate volatility does not vanish under horizontal propagation only. Note that with no horizontal clusters, $\underline{d} \rightarrow 0$, so that $\text{Var}[Y_m] \rightarrow 0$. This result is uniform in m : since horizontal clusters scale proportionally with the size of the economy (as resulting from Corollary 2, Theorem 4), that is $\Psi \propto m$, then $\liminf_{m \rightarrow \infty} \text{Var}[Y_m] > 0$. ■

Aggregate cycles need not originate from economically dominant sectors nor from vertical production chains: as many horizontally-related sectors move in response, shocks do not average out, and aggregate volatility survives even as the economy grows. Even when sectors are small, idiosyncratic shocks generate persistent aggregate volatility under horizontal linkages: fluctuations from sectoral responses are induced by shared Input-Output structures rather than through usual economic centrality or vertical propagation (more in Subsection 4.1). Horizontal interdependences thus prevent shocks from averaging out in aggregate.

COROLLARY 3 (Granular sectoral weights) *Economically large sectors amplify aggregate fluctuations induced by horizontal network propagation.*

Proof: the proof of Theorem 5 has shown the limiting case where Domar weights are uniform, so that large nodes do not exist: $\lambda_i = 1/m, \forall i \in \Psi$, so $\sum_{i \in \Psi} \lambda_i = |\Psi|/m$. With granular Domar weights, some λ_i are larger than others. If a shock hits one of these sectors, then $\sum_{i \in \Psi} \lambda_i \geq \bar{\lambda} \gg |\Psi|/m$, and $\text{Var}[Y_m] \geq \sigma^2 \cdot \underline{d} \cdot \bar{\lambda}^2$. ■

Allowing for heterogeneous Domar weights – so that some sectors represent a larger fraction of aggregate GDP – does not alter the horizontal transmission mechanism identified in Theorem 5. Instead, they strengthen its aggregate impact: when a highly horizontally connected sector is also economically large, its shock propagates across the horizontal cluster and carries greater weight in aggregate output.

Horizontality also shapes the effectiveness of the aggregate diversification rate.

COROLLARY 4 (Rate of decay) *Let Ψ be the set of horizontally connected sectors whose cardinality is $|\Psi|$. Horizontal geometry alone breaks the standard diversification argument: $\text{Var}[Y_m] = \Omega(|\Psi|/m)$. Consequently, aggregate volatility decays at slower rate than $\mathcal{O}(1/m)$, the standard diversification rate.*

Proof: variance decomposition of aggregate output is also $\text{Var}[Y_m] = \sum_{i=1}^m \lambda_i^2 \text{Var}[y_i] + \sum_{i \neq j} \lambda_i \lambda_j \text{cov}(y_i, y_j)$. When $\lambda_i \asymp \frac{1}{m}$, by the Lucas (1977)'s argument the second term is zero, and $\sum_{i=1}^m \lambda_i^2 \text{Var}[y_i] \asymp \frac{1}{m}$ where, if $m \rightarrow \infty$, $\text{Var}[Y_m] \rightarrow 0$. Let $|\Psi|$ denote the size of the set- Ψ of horizontal clusters. The number of correlated pairs is $m |\Psi|$. Since covariance is weighted by $\lambda_i \lambda_j$, which satisfies $\lambda_i \lambda_j \asymp \frac{1}{m^2}$ then, given the number of correlated pairs, $\sum_{i \neq j} \lambda_i \lambda_j \text{cov}(y_i, y_j) \asymp \frac{|\Psi|}{m}$. Plugging the variance and covariance contributions into the variance decomposition: $\text{Var}[Y_m] = \mathcal{O}\left(\frac{1+|\Psi|}{m}\right)$, where $\mathcal{O}(\cdot)$ describes the asymptotic rate at which aggregate volatility decays as the number of network participants increases. Hence, aggregate volatility is positive function of horizontal propagation, and variance cannot decay faster than $\text{Var}[Y_m] = \Omega(|\Psi|/m)$. ■

The declining speed of aggregate GDP volatility reflects the strength of horizontal complementarities. When small, sectors behave almost independently and horizontal propagation diversifies, with negligible aggregate outcomes (*e.g.*, Lucas 1977). As horizontal cluster expands, shocks affect more sectors simultaneously: diversification weakens as aggregate volatility is

bounded below $\Omega(\cdot)$, but exceeding the standard diversification rate $\mathcal{O}(\cdot)$. When the network size scales horizontal similarities, sectoral responses are broadly synchronized and aggregate volatility persists.

Vertical and horizontal transmissions thus reflect distinct but complementary mechanisms. Vertical propagation operates through the Input-Output matrix: shocks cascade along supply chains via observed linkages, generating amplifications that scale up mechanically with sectoral importance. Granularity therefore magnifies aggregate responses by strengthening vertical cascades. Horizontal propagation, instead, acts on comovement: sectors sharing many suppliers or buyers respond jointly, without inducing cascades but generating systematic correlation that survives aggregation. Granularity amplifies these spillovers if large sectors belong to dense horizontal clusters. As in the vertical framework (*e.g.*, Acemoglu, Carvalho, et al. 2012), aggregate fluctuations decay more slowly under network interconnections: the law of large numbers implied by the standard diversification argument is weakened due to novel horizontal effects, and aggregate volatility is more persistent.

4. DATA AND HORIZONTAL ANATOMY

From the United States' Bureau of Economic Analysis (BEA), available data display information, for 3-digit U.S. 2017 North American Industrial Classification System (NAICS) sectors, on value-added, employment, net exports, and inter-sectoral Input-Output (I-O) tables. The (balanced) panel data built is made of 65 private sectors over the period 1998-2022. For additional details refer to [Supplemental Appendix B](#).

Distance measures are derived from the *Total Requirements Table*, which captures both direct and indirect Input-Output trade along each supply chain. This representation allows a comparison beyond immediate network relationships, thereby accounting for the complete production network structures. Accordingly, the distance between two sectors reflects the extent to which they share similar buyers/suppliers locally and throughout the depth of production linkages. Each cell $d^q[i, j] \in [0, 1]$ of matrix $\mathcal{D}^q$, for $q = \{fd, fs\}$ and $\forall i, j \in \Phi$, identifies the Euclidean distance (square root of the sum for squared differences of network intensities; see Section 2) between dyads of sectors in demand or supply relative to common sectors.

The empirical estimation of Section 5 will consider network distances at their *extensive margin* for sectoral classification purposes. Such matrices are labelled as $\mathcal{D}_{ext}^q$, whose generic entry is $d_{ext}^q[i, j] = \{1, 2, \dots, d_{max}\}$. Differently from the theory, values closer to one correspond to shorter horizontal distances. As for their characterization, network distance- d among any two sectors $\{i, j\}$ is computed as a *shortest path problem* from *intensive margin* distance matrices, $\mathcal{D}^q$, implementing an algebraic version (*i.e.*, using matrix multiplication) of the traditional Dijkstra (1959)'s or Roy (1959)-Floyd (1962)-Warshall (1962)'s types of algorithms, since it is (i) suitable for discrete-value adjacency matrices – as the ones for intensive margin distances –, and (ii) fairly efficient for small to medium-sized graphs – as a sector-level production network. [Supplemental Appendix B](#) reports the technical version of the implemented algorithm which, in plain words, reads as follows.

(Shortest path algorithm) *Central idea is to find, for each single node, all the other nodes*

that can be reached in one step, then in two steps, and so on. Each time a node is reached for the first time, it records the number of steps it took to get there – this is the shortest path approach. In other words, find the shortest number of steps it takes to get from every node to every other node in a network. In particular:

- (a) identify the direct (one-step) connections between any pair of nodes;
- (b) for not-connected pairs, increase the path length by 1 in each round (two steps, then three steps, etc.). Multiply the graph by itself to discover which new pairs of nodes are now connected through longer paths;
- (c) if two nodes becomes connected, record the length as the shortest distance;
- (d) repeat this process until any new reachable pairs of nodes is found. For any pair of nodes that are still uncategorised, set their distance to infinity.

4.1. AGGREGATE DYNAMICS

To illustrate the implications of horizontal geometry for business cycles, I simulate aggregate GDP responses to sectoral shocks exploiting the network’s vertical (Input-Output chains) and horizontal (shared buyers/suppliers) structures, relying on observed Input-Output and intensive margin distance matrices from the 2007 BEA Input-Output tables. Both network dimensions remain unweighted, as dictated by Theorem 3.

Sectoral outputs evolve as $\mathbf{y}_t = \mathcal{P}^{(\mathcal{V}, \mathcal{D})} \boldsymbol{\varepsilon}_t$, where matrix $\mathcal{P}^{(\cdot)}$ encodes how sector-specific shocks in $\boldsymbol{\varepsilon}$ propagate, following a mean-reverting AR(1) process with persistence ρ . Aggregate Impulse Response Functions (IRFs) trace the path $\{Y_t\}_{t \geq 0}$ following an initial shock in $t = 0$, and evolve over time according to

$$Y_t = \boldsymbol{\lambda}' \mathbf{y}_t \iff Y_t = \boldsymbol{\lambda}' \mathcal{P}^{(\mathcal{V}, \mathcal{D})} (\rho_t \boldsymbol{\varepsilon}_0)$$

where $\boldsymbol{\lambda}$ bundles Domar weights (*i.e.*, relative sectoral contribution to GDP). The resulting series for Y_t provide dynamic responses that summarize either cascading or distance-lead effects across the observed network structure, shown in Figure 4.

Panel 4a directly mirrors the logic of Theorem 5 and its Corollary 3. Vertical propagation, governed by the Leontief inverse, amplifies shocks primarily through parallel supplier-customer chains.⁸ Horizontal propagation, instead, does not rely on Input-Output flows but depends on the geometry of production similarities: idiosyncratic shocks diffuse strongly across sectors with common network linkages, even in the absence of direct trade, and aggregate effects depend on the distribution of economic proximity. In both cases, granularity mechanically increases aggregate volatility because economically large sectors transmit shocks proportionally more strongly. Horizontal spillovers are mitigated when large Domar weights are associated with sectors embedded in large horizontal clusters: a sector that is not central in the standard Input-Output network but is strongly connected to many others through shared linkages (the “*h*-cluster”) generates milder aggregate effects than a scenario in which equally large Domar

⁸ Sectoral shocks average out in aggregate for “balanced” – *i.e.*, almost symmetric – Input-Output networks (*e.g.*, Acemoglu, Carvalho, et al. 2012), and not necessarily for equal Domar weights.

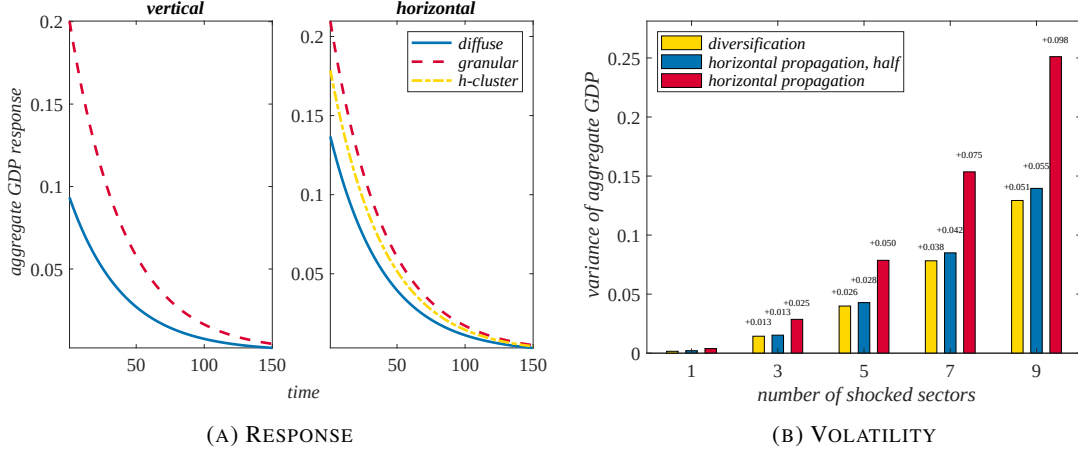


FIGURE 4: AGGREGATE FLUCTUATIONS

Note: under “diffuse”, sectoral Domar weights are evenly distributed; if concentrated, the economy is vertically “granular”; a not vertically dominant yet strong horizontally connected sector lies under “h-cluster”. Numbers above bars indicate the incremental change in aggregate output variance relative to the previous one. Source: BEA and own calculations.

weights are concentrated in vertically central sectors (the “granular” setup). Indeed, this transmission mechanism is not driven by upstream/downstream similarity alone, but by the alignment between sectoral importance and the network’s horizontal geometry.

Additionally, from the y -axis, horizontal propagation generates slightly larger aggregate responses than vertical connections. This outcome is not a numerical artifact, but a direct consequence of the economic mechanisms embedded in: reminiscent from Figure 3, vertical transmission amplifies shocks through recursive cascades but produces “localized” effects along specific supply chains, whereas horizontal propagation increases systemic covariance, creating widespread comovement throughout the production network without confined feedback loops or cascading amplification.

Panel 4b illustrates how horizontal propagation modifies the conventional diversification argument. When shocks are transmitted through horizontal spillovers, aggregate volatility increases more sharply than in the diversification scenario. Even a partial (50%) horizontal spread amplifies fluctuations beyond such benchmark with no network effects, while full horizontal propagation generates the largest variance, resulting in the slowest rate of decay. Corollary 4 finds support: horizontal complementarities prevent idiosyncratic shocks from averaging out, so aggregate fluctuations persist and intensify with the strength of network interconnectedness.

An alternative test is to decompose the contribution of the network distance structure to aggregate GDP variance. I compute the spectral norm of the horizontal propagation matrix, which measures the maximal amplification of idiosyncratic shocks through the network; larger values imply a slower rate of decay. The results indicate that horizontal complementarities generate aggregate dynamics beyond the benchmark predicted by the standard diversification argument. Specifically, the fact that $\|\mathcal{P}^{(D)}\|_2 = 0.156 > 0.124 = \|\mathcal{P}^{(Lucas, 1977)}\|_2$ illustrates how structural interdependence among sectors sharing suppliers or buyers prevents idiosyncratic shocks from averaging out, in contrast to the classical $1/\sqrt{m}$ convergence predicted by Lucas (1977). This comparatively slow decay implies that aggregate volatility is less likely to be smoothed out in economies with strong horizontal complementarities, highlighting systemic vulnerabilities,

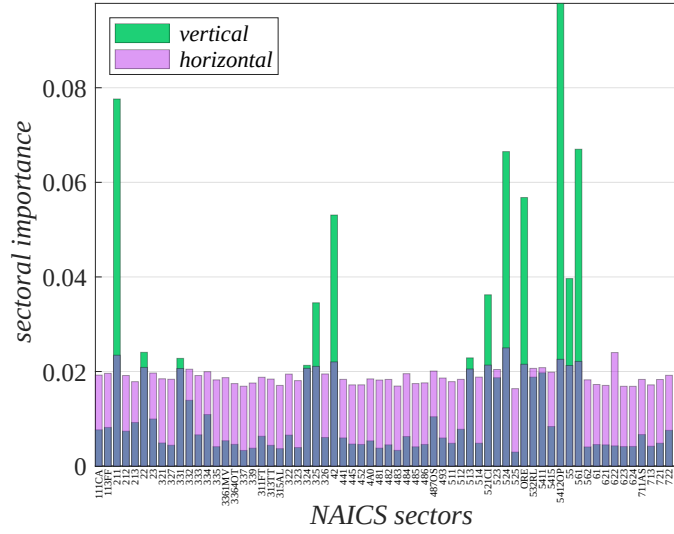


FIGURE 5: SECTORAL INFLUENCE IN SHOCK TRANSMISSION

Note: importance of each sector in reverberating idiosyncratic shocks through their vertical and horizontal network structures as dictated by $\mathcal{P}^{(V,D)}$ U.S. Input-Output propagation matrices in year 2007. Source: BEA and own calculations.

and supporting the claim of Corollary 4.

4.2. PROPAGATING PATTERNS

I turn to analyse the mechanics of the horizontal transmission. Exploiting the observed propagation matrices, Figure 5 visualizes the sectoral propagation patterns using centrality measures from spectral norms, to be read in terms of how important each sector is for the transmission of shocks considering the dual network structures. The vertical transmission highlights few key sectors that act as propagation hubs:⁹ asymmetric shocks originating in these sectors are recursively amplified along parallel supply chains, generating strong cascading effects, while most other sectors transmit shocks only weakly. In contrast, the horizontal propagation system spreads shocks uniformly across sectors: each sector contributes to comovement with others, but without recursive amplification, producing widespread responses. Overall, the image demonstrates the distinction between cascade-driven amplification along key nodes and diffuse propagation across economically similar sectors: vertical chains concentrate and amplify shocks along central nodes, whereas horizontal complementarities induce diffuse comovement without feedback loops.

Table 1 investigates how vertical and horizontal geometries respond to varying levels of network sparsity. By progressively removing weaker transactions (increasing the threshold- τ for the volume of total sectoral shares), the network becomes sparser and vertical propagation declines, reflecting the diminishing role of cascade effects. In contrast, horizontal propagation remains remarkably stable across thresholds, and it gradually becomes the predominant channel of shock transmission as τ increases, so that horizontal complementarities dominate in

⁹ Systemic role in providing inputs – e.g., energy (211), risk intermediation (524), knowledge-intensive services (5412OP), and organizational support (561) – that underpin a wide range of production activities.

TABLE 1: DIMENSIONAL DOMINANCE

	<i>network sparsity</i>				
	15% $\tau = 0.5\%$	30% $\tau = 1\%$	40% $\tau = 1.5\%$	50% $\tau = 2\%$	95% $\tau = 5\%$
<i>vertical propagation</i>	.2133	.1995	.1803	.1497	.0762
<i>horizontal propagation</i>	.1371	.1559	.1718	.1731	.1746

Spectral norms for a 3-digit U.S. Input-Output network in year 2007. Due to thresholding- τ , transactions are removed and sparsity is approximated. Source: BEA and own calculations.

sparser networks. These results illustrate that, even when vertical linkages (*i.e.*, observed trade intensities) are trimmed, horizontal interdependencies sustain substantial propagation, empirically clarifying Theorem 4: as Input-Output economies become sparser, shocks increasingly travel along clusters of horizontally related sectors rather than along elongated vertical chains, confirming the principle of horizontal dominance.

4.3. NETWORK EXPOSURE AND AGGREGATE SHOCKS

The discussion so far has shown that horizontal geometries capture dimensions of economic exposures that are unconditionally distinct from vertical Input-Output linkages. If these capture economically meaningful dimensions of interdependence, they should also explain differences in sectoral responses to aggregate shocks “above and beyond” the information already embedded in Leontief inverse measures, retaining a predictive power that conditionally complements them.

A Bartik (1991) shift-share design comes to the rescue. Consider an aggregate shock that affects all sectors simultaneously. Since the shock is widespread, identification cannot come from time-series variation alone. Instead, the relevant source of variation is cross-sectional: sectors may be differently exposed to the same aggregate disturbance because of their Input-Output exposures. Formally, I first construct a measure of exposure by combining the weight- ω of each network structure with a measure of predetermined sectoral sensitivity to the aggregate disturbance

$$exposure_i^{(\mathcal{V}, \mathcal{D}^j)} = \sum_{j \in \Phi_i} \omega_{i,j}^{(\mathcal{V}, \mathcal{D}^j)} sensitivity_j$$

for $q = \{fd, fs\}$. This formulation encompasses all network structures and, depending on the application, the weight may correspond to the vertical production network, or the horizontal demand- and supply-based network. Intuitively, a sector is highly exposed when it is strongly connected to sectors that are themselves highly sensitive to the aggregate disturbance. Exposure therefore measures how embedded a sector is to the other sectors’ sensitivities relative to the aggregate disturbance.

A natural concern is that horizontal and vertical exposures may be highly collinear and partly reflect similar features, since both belong to the same primitive Input-Output matrix (Theorem 1). Hence, the component of horizontal exposure that is orthogonal to vertical one is residualised from the auxiliary projection

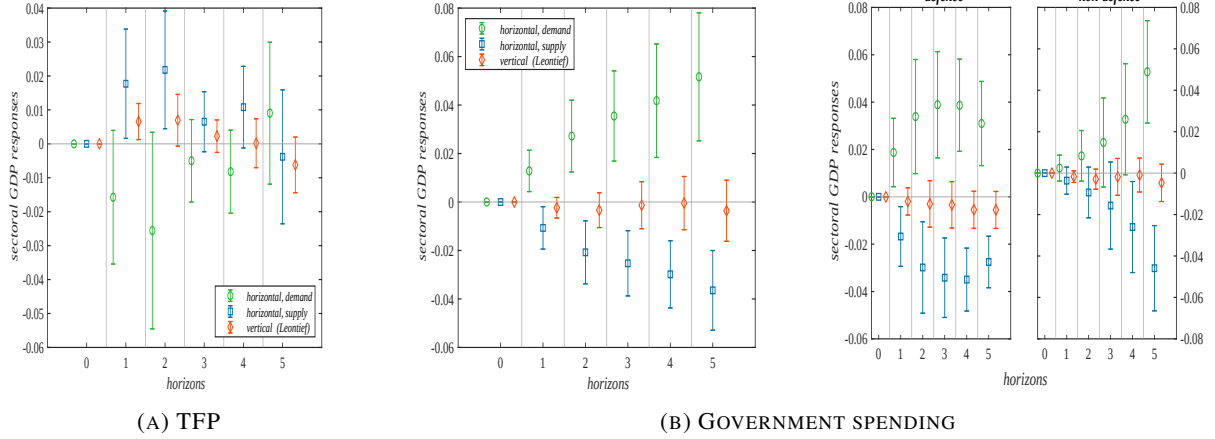


FIGURE 6: “NETWORKING UP” AGGREGATE SHOCKS

Note: cross-sectional coefficients measure how differently sectors respond positively (amplification) or negatively (attenuation), due to vertical and horizontal network exposures, to the same aggregate shock, as in one standard deviation. Bars correspond to intervals at 90% of bootstrapped confidence level, clustered by sectors. *Source:* various and own calculations.

$$exposure_i^{\mathcal{D}^j} = \beta_{exposure}^{cons} + \beta_{\mathcal{V}} exposure_i^{\mathcal{V}} + \gamma_i' \mathcal{Z}_i + r_i^{\mathcal{D}^j}$$

where $\mathcal{Z}_i$ are sectoral controls (*e.g.*, size, Domar weights, network importance). Then, let G_t denote an aggregate shock. The Bartik exercise is the Local Projection (LP)

$$\begin{aligned} \Delta y_{i,t+h} = & \beta_h^{\mathcal{D}^{fd}} \left(G_t \times \hat{r}_i^{\mathcal{D}^{fd}} \right) + \beta_h^{\mathcal{D}^{fs}} \left(G_t \times \hat{r}_i^{\mathcal{D}^{fs}} \right) \\ & + \beta_h^{\mathcal{V}} \left(G_t \times exposure_i^{\mathcal{V}} \right) + \phi_{t+h} + \phi_i + v_{t+h}^{\mathcal{V}_i} \end{aligned}$$

given $h = \{1, \dots, \bar{h}\}$ time-horizons, where ϕ_i and ϕ_{t+h} denote sector and time fixed effects, respectively. The interaction term captures the identified intensity with which the aggregate disturbance affects each sector through the network. Importantly, the coefficients β do not measure the dynamic response of the economy to the aggregate shock. Instead, each measures whether sectors with greater vertical/horizontal network exposure systematically react more or less strongly to the same shock than sectors with lower exposure. Consequently, the identifying variation is cross-sectional: a positive coefficient implies that highly exposed sectors react more strongly to the aggregate disturbance, whereas a negative coefficient indicates that network exposure dampens the transmission of the shock; the sequence of such coefficients across horizons traces the evolution of these cross-sectional differences over time. Therefore, the key empirical content of Figure 6 is not the dynamic unconditional effect of horizontal distance, but rather the per-period conditional effect of horizontal distance after controlling for vertical Input-Output exposure. Note that the aggregate shock itself is absorbed by time fixed effects, implying that the identification comes entirely from differences in exposure across sectors.

Aggregate TFP shock.— I first consider an aggregate TFP disturbance, obtained as the unexpected component of aggregate total factor productivity after removing persistence and low-frequency trends, where sectoral sensitivity measures capture how close each sector is to sectors

that are historically sensitive to aggregate TFP innovations.¹⁰ The estimated cross-sectional LP coefficients in Panel 6a reveal a sharp contrast across transmission channels. While the Leontief-based exposure displays some predictive power at short horizons, its magnitude is smaller and its effects dissipate rapidly. By contrast, the supply-based measures generate larger and more persistent differences in sectoral responses: sectors that are more closely connected, through common downstream buyers, to TFP-sensitive sectors react more strongly to aggregate productivity shocks over several horizons. Demand-based horizontal exposure does not generate statistically significant effects. A positive aggregate TFP shock hits asymmetrically due to pre-determined exposures, increasing productivity more in some sectors than others. Through the horizontal supply channel, more productive sectors expand production and capture a larger share of downstream demand, leading to a reallocation of expenditure across competing suppliers. This competitive displacement implies that aggregate demand is satisfied by increasingly efficient producers, so that total economic activity rises despite unchanged downstream demand, as production is concentrated in higher-productivity sectors. By contrast, the horizontal demand channel does not play a significant role, since TFP shocks operate through productivity differences rather than own demand expansion: they do not systematically increase intermediate input demand, making upstream propagation through shared suppliers weak, non-systematic, and of second-order importance. These findings suggest that horizontal network structures constitute an independent and economically relevant dimension of production networks.

Government spending shock.— Panel 6b, instead, plots the coefficients for fiscal disturbances, constructed similarly to the TFP case.¹¹ The results reveal a markedly different pattern across propagation channels. Government spending shocks diffuse not only through directly affected industries but also through common Input-Output relationships, and appear to diffuse across sectors sharing common suppliers, generating positive horizontal spillovers in consistency with the intrinsic demand-based nature of these shocks, while sectors sharing common buyers exhibit weaker responses, consistent with competitive reallocations of downstream demand. Reassuringly, the results are also robust to disentangling aggregate government expenditure into defence and non-defence components, whose pre-determined spending categories are concentrated in targeted industries: despite the different cross-sectional allocation of network exposures, the pattern of sectoral responses is preserved. These type of shocks operate as positive demand shocks that propagate through the Input-Output network via two distinct horizontal channels. On the demand side, higher input requirements from directly affected sectors increases demand faced by shared upstream suppliers, which can adjust either through quantities or prices. If they respond by expanding production at given prices, the increase in input supply can benefit other sectors connected through the same supplier, or leave their production unaf-

¹⁰ Using the annual series from Fernald (2014), I extract the TFP innovation as the fitted residual from the AR(1) regression with time trend: $\log TFP_t = \beta_{tfp}^{cons} + \beta_{tfp} \log TFP_{t-1} + \beta_{time} \cdot t + \varepsilon_t^{tfp}$. This series is also exploited to construct sectoral exposures. First, I estimate sector-specific historical sensitivities to aggregate TFP innovations, capturing how strongly the sector tends to react when aggregate productivity unexpectedly changes. These sensitivities are then propagated through the alternative network structures considered.

¹¹ Pre-determined exposures are constructed from the 1997 U.S. Input-Output table as the government purchases in the sectoral supply of intermediates, capturing the extent to which a sector is connected to government demand and therefore likely to benefit from expansions in aggregate public expenditure.

affected. If instead they raise prices, downstream sectors may either continue purchasing at higher cost or substitute toward alternative suppliers, reallocating production across the network. In both cases, the adjustment of common upstream suppliers to higher downstream demand generates positive spillovers as it propagates through Input-Output linkages or substitution effects, and leads to an overall expansion in economic activity. On the supply side, targeted spending induces sectors to expand their production to meet higher government demand. This allows these sectors to meet their supply more effectively, affecting how downstream expenditure is allocated across upstream suppliers. Given the unaltered demand of common buyers, higher absorption by directly exposed sectors crowds out others serving the same customers: horizontally exposed sectors experience negative spillovers through competitive pressures for downstream markets. In this case, there is no compensating expansion from shared downstream demand, so that effects are driven by within-buyer reallocation, resulting in overall dampened responses in horizontally exposed sectors.

Taken together, these results reinforce the broader message of the paper. Aggregate productivity shocks primarily propagate through the supply-based horizontal structure, while aggregate fiscal shocks operate through a distinct network dimension, namely the demand-based horizontal geometry. The fact that the dominant propagation channel varies across shocks suggests that horizontal geometries do not merely replicate classical Input-Output effects, but instead capture economically meaningful dimensions of sectoral interdependence that become relevant under different types of aggregate disturbances, and contain additional information on shock propagation that is not fully captured by standard vertical production linkages.

5. EMPIRICS

In this empirical section sector-level data for the United States economy are exploited to test the predictions about sectoral comovement in Section 3. Estimation relies on the two-stage procedure outlined by Barattieri and Cacciatore (2023): first, I perform a series of sectoral panel Fixed-Effect (FE) regressions to isolate the exogenous dynamics in a given input of production not due to other (sector-specific, other sectors-specific and aggregate) factors; then, each estimated residual is exploited as an identified structural shock into a Local Projection (LP) analysis, in order to assess the network-based short-run cumulative variations in sectoral employment.

5.1. INTERMEDIATE INPUTS AND SECTORAL EMPLOYMENT

Sections 2 and 3 rest on the premise that adjustments in the stock of circulating intermediate inputs lead to synchronized sectoral employment responses. Considering variations within a sector and across its economic neighbours, the first step isolates changes in input sourcing plausibly orthogonal to employment dynamics. The following panel fixed-effect regression over $i \in \Phi$ sectors is estimated:

$$\begin{aligned}
x_{i,t} = & \beta_x^{cons} + \beta_n \dot{n}_{i,t} + \sum_d \beta_{n(d)} \dot{n}_t(\Phi_i, d) + \sum_k \beta_{n(k)} \dot{n}_t(\Phi_i, k) \\
& + \gamma'_i \mathcal{Z}_{i,t} + \Gamma'_i \dot{\mathcal{Z}}_{i,t} + \gamma' \mathcal{Z}_t + \Gamma' \dot{\mathcal{Z}}_t + \phi_i + u_{i,t}^x
\end{aligned} \tag{6}$$

where $x_{i,t}$ identifies the set of intermediate inputs bought by sector- i (as a share of its total production), while $\dot{n}_{i,t}$, $\dot{n}_t(\Phi_i, d)$, and $\dot{n}_t(\Phi_i, k)$ represent its employment growth, that of its economically closer and further sectors, and that of its upstream and downstream sectors, respectively. The latter two measures are defined as

$$\begin{aligned}
\dot{n}_t(\Phi_i, d) &= \sum_{j \in \Phi_i} \Delta n_j(d) \quad , \quad \forall d \in \mathcal{D}_{ext}^q \quad \text{and} \quad q = \{fd, fs\} \\
\dot{n}_t(\Phi_i, k) &= \sum_{j \in \Phi_i} \ell_{ij} \Delta n_j(k) \quad , \quad \forall k = \{up, dw\}
\end{aligned}$$

where $\Delta n(\cdot) = n_t(\cdot) - n_{t-1}(\cdot)$. As for vertical propagation, each change is weighted by the sales-based Leontief inverse relation, $\ell_{ij} = [1 - \alpha_{ij}]^{-1}$, between sector- i and each of the other sectors whenever these are located upstream or downstream to sector- i , thereby capturing direct and indirect network exposure. Differently, changes in employment under extensive margin network demand or supply distances seize all the horizontal relationships across sectors which, in line with Theorem 3, are not leveraged by the Leontief inverse weights.

A set of control variables characterizes the second line of eq. (6). $\mathcal{Z}_{i,t}$ and $\dot{\mathcal{Z}}_{i,t}$ comprise the labour force size of the sector, both its level and growth rate of value-added, and its net exports that control for changes in demand/supply not due to internal factors. The set of aggregate controls, $\mathcal{Z}_t$ and $\dot{\mathcal{Z}}_t$, is made of aggregate value-added and aggregate employment in size as well as in growth terms. Finally, while β 's are coefficients to be estimated, ϕ_i imposes fixed-effects to control for unobserved heterogeneity across sectors. Standard errors are clustered at sector-level. For each sector, the estimated residual $\hat{u}_{i,t}^x$ identifies the exogenous variation in the share of production of sector- i that is bought from Input-Output connections.

Impulse Response Functions (IRFs) of sectoral employment to changes in the sector-specific intermediate inputs rely on Local Projections. Given $h = \{1, \dots, \bar{h}\}$ time-horizons, and for all sectors $i, j = \{1, \dots, m\} \in \Phi$, a battery of panel FE predictive regressions, consisting in estimating the identified sector-specific structural shock, $\hat{u}_{i,t}^x$, on the cumulative difference in sectoral levels, is performed:¹²

$$\dot{n}_{i,t+h}^{cum} = \beta_{\hat{u}^x, h} \hat{u}_{i,t}^x + \phi_{t+h} + \phi_i + v_{i,t+h}^x \tag{7}$$

with $\dot{n}_{i,t+h}^{cum} = n_{i,t+h} - n_{i,t-1}$ denoting the cumulative change in sector- i employment at each horizon. Estimated coefficient $\beta_{\hat{u}^x, h}$ is the response of the variation in the cumulative difference of sectoral employment at time $t+h$ given the identified shock hitting in period- t . Prediction error term $v_{i,t+h}^x$ is horizon-specific, while standard errors and bootstrapped Con-

¹² Cumulative variables are likely to be highly correlated with the error term because they embed past values – bequeathing past errors – of the variable of interest, leading to biased and inconsistent estimates. Differencing cumulative measures soothes this endogeneity concern.

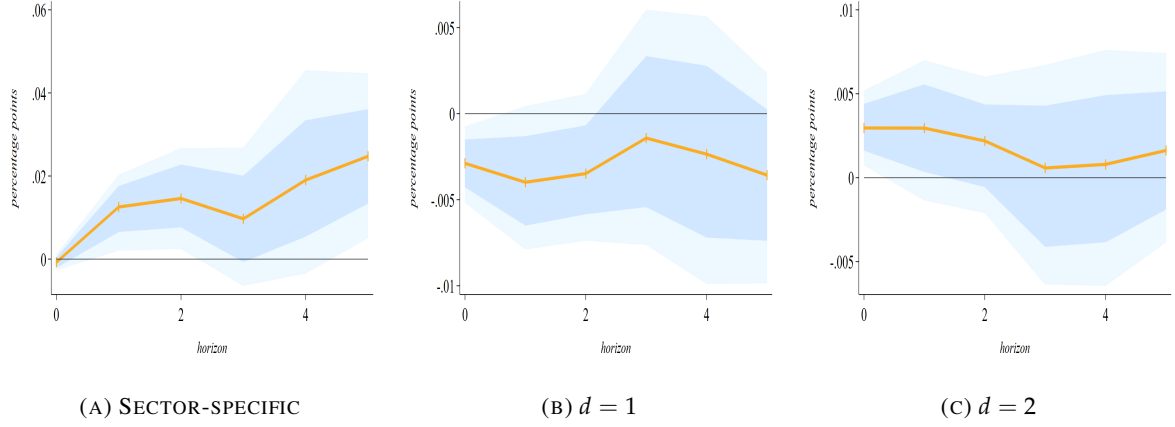


FIGURE 7: SECTORAL EMPLOYMENT RESPONSE TO INTERMEDIATE INPUTS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the sector-specific set of intermediate inputs as a share of its value-added for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. In particular, Panel 7a represents the sector-level employment response to changes in its intermediate inputs, while Panel 7b-7c to changes in the set of intermediates in closer (distance equal to 1) and further (distance equal to 2) sectors. The solid-orange line corresponds to the average response of employment across sectors, while shadow-blue and shadow-light blue areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eqs. (7)-(8). *Source:* BEA and own calculations.

fidence Interval (CI) of $\beta_{\hat{u}^x, h}$ are clustered by sectors. To avoid measurement errors due to unobserved heterogeneity in response, sectoral fixed-effects ϕ_i , and horizon fixed-effects ϕ_{t+h} – that remove common trends in the comovement of sectoral employment – are imposed.¹³

Outcomes from eq. (7) estimate the cumulative responses of sectoral employment to changes in their set of intermediate inputs. Analogous estimation is performed to variations in nearby sectors, considering the d^{th} -distance between sectors:

$$\dot{n}_{i,t+h}^{cum} = \beta_{\hat{u}^x(d), h} \sum_{j \in \Phi_i} \hat{u}_{j,t}^x(d) + \phi_{t+h} + \phi_i + v_{i,t+h}^x(d) \quad (8)$$

Performed estimation of eqs. (7) and (8), under *Leontief inverse* network structure and *factor input supply* distances, delivers Figure 7. The cumulative (short-term) response of sector-specific employment levels to a 1% change in its output share of intermediate input is in the first panel: averaging across sectors, the increase induces a positive statistically-significant change in sectoral employment, the central rebalancing effects from circulating inputs in Sections 2-3. Each additional unit of intermediate inputs requires less than one worker so that factors of production are at least substitute, as the Cobb-Douglas specification instructs.

Changes in intermediate inputs from other sectors do not yield consistent outcomes. Results are almost always not statistically significant at 90% except when the shock hits, and the magnitude of the variation is negligible. Interesting is to note the different response between Panels 7b and 7c: for intermediate inputs changes in closer sectors there is an average negative effect, while changes in further sectors are positively related with variations in sectoral employ-

¹³ Differently from the first-stage, where I control common trends over time using aggregate variables, ϕ_{t+h} removes the possibility of common trend in the *response* of sector-specific employment to idiosyncratic shocks; the possibility that employment responses are influenced by a common trend is ruled out.

ment.¹⁴ Such observations might suggest that certain types of intermediate inputs are more essential than others in production (*e.g.*, Carvalho and Voigtländer 2015), but it is actually the type of economic distance that determines their importance both in the production of sectoral output and in their network centrality of being larger buyers or suppliers.

5.2. SECTORAL COMOVEMENT OF EMPLOYMENT

I now analyse how sectoral employment adjusts in response to employment changes in other sectors, depending on their network economic distance. The previous identification and estimation scheme is developed, but now purging the dynamics of sector-specific employment with changes in its and other sectors' value-added, since employment is not exogenous to growth rates of sectoral output (*e.g.*, Hornstein and Praschnik 1997; Acemoglu, Akcigit, et al. 2015; Sandqvist 2017).

Identification.— To identify exogenous variations in sectoral employment, $n_{i,t}$, a panel fixed-effects regression over each sector $i \in \Phi$ is estimated:

$$\begin{aligned} n_{i,t} = & \beta_n^{cons} + \beta_y \dot{y}_{i,t} + \sum_d \beta_{y(d)} \dot{y}_t(\Phi_i, d) + \sum_k \beta_{y(k)} \dot{y}_t(\Phi_i, k) \\ & + \gamma'_s \mathcal{Z}_{i,t} + \Gamma'_i \dot{\mathcal{Z}}_{i,t} + \gamma' \mathcal{Z}_t + \Gamma' \dot{\mathcal{Z}}_t + \phi_i + u_{i,t}^n \end{aligned} \quad (9)$$

where $\dot{y}_{i,t}$ is the value-added growth for sector- i , $\dot{y}_t(\Phi_i, d)$ identifies the value-added growth of its related sectors according to network distances, and $\dot{y}_t(\Phi_i, k)$ represents the value-added growth of its upstream and downstream sectors. Each measure is defined as that of eq. (6) but using y_i instead of employment. Moreover, for this specification, sector-level controls, $\mathcal{Z}_{i,t}$ and $\dot{\mathcal{Z}}_{i,t}$, comprise the labour force size of the sector, its value-added level and net exports, while aggregate controls, $\mathcal{Z}_t$ and $\dot{\mathcal{Z}}_t$, are analogous to eq. (6). Estimated coefficients are the β 's, given sectoral fixed-effects, ϕ_i , and sectoral standard errors.¹⁵ Sector-specific estimated residual $\hat{u}_{i,t}^n$ from eq. (9) identifies the exogenous variation in sectoral employment.

Local Projection analysis.— Sectoral comovement is defined as the response of employment in a given sector to changes in employment of other sectors:

$$\dot{n}_{i,t+h}^{cum} = \beta_{\hat{n}(d),h} \sum_{j \in \Phi_i} \hat{u}_{j,t}^n(d) + \phi_{t+h} + \phi_i + v_{i,t+h}^n(d) \quad (10)$$

with $\dot{n}_{i,t+h}^{cum} = n_{i,t+h} - n_{i,t-1}$ denoting the cumulative change in sector- i employment at each horizon given the identified employment shock in closer or further sectors at the d^{th} -distance. Coefficient $\beta_{\hat{n}(d),h}$ is the h -step-ahead response of sector-specific employment in cumulative difference due to the identified shock $\hat{u}_{j,t}^n(d)$. Still, sector and horizon fixed effects, ϕ_i and ϕ_{t+h} respectively, are imposed to remove unobserved heterogeneity and common trends across sectors in the comovement of sectoral employment. Prediction error term $v_{i,t+h}^n(d)$ is horizon-

¹⁴ In Appendix C, under *Leontief inverse* and *factor input demand* network distances, I show how that the same impact of Panel 7a still holds (Panel C.1a of Figure C.1), while results for distances' effects (Panels C.1b and C.1c) identify stronger but opposite results: increasing for closer sectors, declining for further ones.

¹⁵ In both identifications I control for time fixed-effect through variations in aggregate variables, as comovement between aggregate and sectoral variables is well established (*e.g.*, Stock and Watson 1999; Huffman and Wynne 1999; Rebelo 2005; DiCecio 2009).

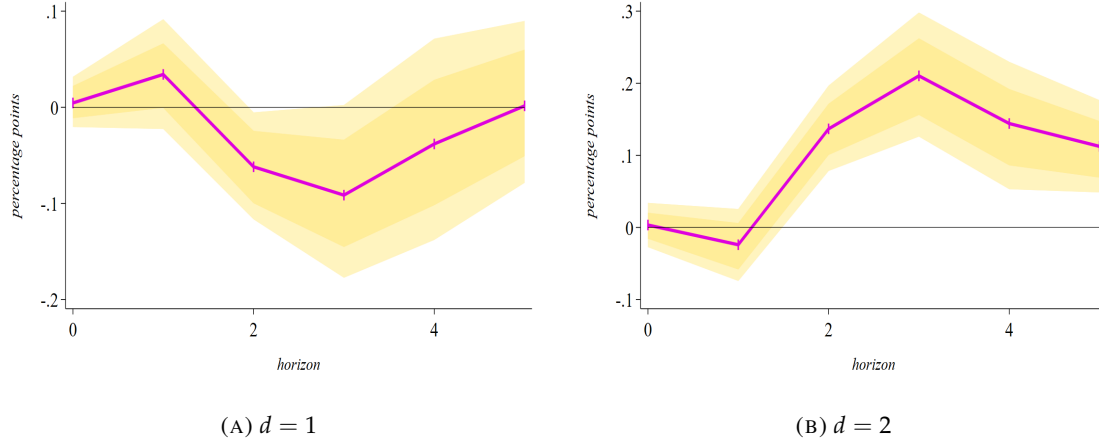


FIGURE 8: EMPLOYMENT COMOVEMENT UNDER DEMAND LINKAGES

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998–2022. Panel 8a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 8b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (10). *Source:* BEA and own calculations.

and distance-specific in each predictive panel regression, with standard errors and bootstrapped Confidence Interval (CI) of $\beta_{\hat{n}(d),h}$ clustered by sectors.

5.2.1. Results for factor input demand distances

Considering sectors whose distance is defined in terms of buying from the same set of upstream suppliers, estimating eq. (10) yields Figure 8. As already emphasized in discussing Proposition 1, demand-based complementarities are inherently ambiguous to interpret since they depend on the joint action of vertical propagation and horizontal spillovers: when buyers can easily substitute inputs, a positive shock in one sector strengthens the demand for the shared supplier and raises the upstream price, thereby crowding out the other sector; by contrast, when inputs are complements, the increase in the supplier’s price erode the demand competitiveness of the other sector, dampening comovement. Where these forces conflict or remain weak, observed correlations are muted or ambiguous.

These mechanisms echo in observed responses. Panels 8a and 8b show that a 1% increase in employment in closely linked sectors ($d = 1$) produces an ambiguous response, whereas more distant sectors ($d = 2$) consistently display positive comovement. However, sectoral shocks in Panel 8a do not produce any statistically significant effects, as the estimated responses fail to reach significance at the 90% confidence levels. This lack may be driven by including all sectors indiscriminately, as sectors with fewer inter-sectoral linkages might introduce noise or bias to the estimation, masking and diluting the true magnitude of measured comovement in sectoral employment: restricting the sample to sectors with most Input-Output linkages could reveal more meaningful propagation effects, in line with Theorem 4, where network effects are more pronounced.

Nevertheless, ambiguous patterns emerge when partitioning sectors into finer subsets by

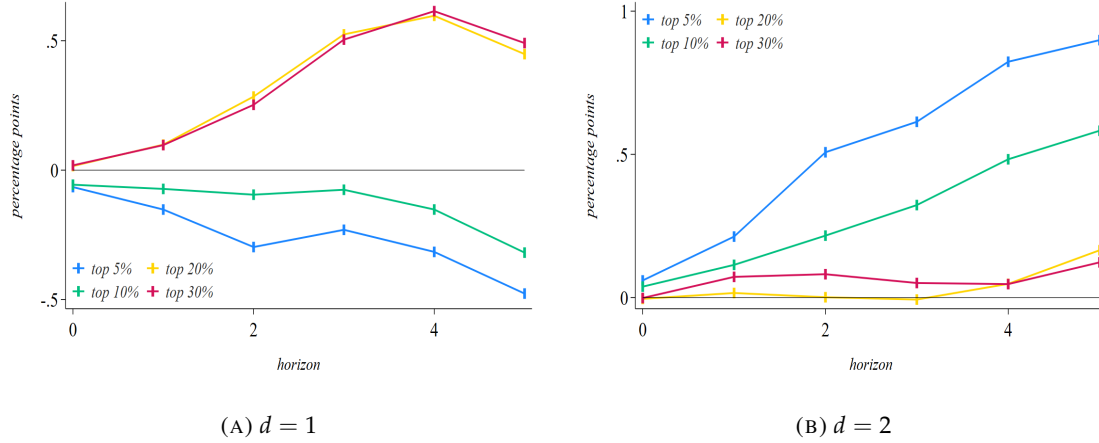


FIGURE 9: DEMAND LINKAGES AND COMOVEMENT IN HIGHLY INTERLINKED SECTORS

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel 9a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 9b plots the response to employment changes in further (distance equal to 2) ones. Each line corresponds to the average response of employment across sectors with different numbers of linkages, robust to 68% and 90% significance levels of bootstrapped Confidence Interval (CI) computed from eq. (10). If a plot is not appearing, it means there is no response for the specified distance value. *Source:* BEA and own calculations.

the number of Input-Output linkages (Figure 9): comovement depends on the specific cluster considered. This confirms that demand-driven horizontal complementarities and vertical propagation operate jointly, often obscuring one another. Such mixed evidence ultimately reflects the inherent ambiguity of demand-based distances discussed in Subsection 3.1: because they capture substitution patterns across upstream intermediate goods rather than genuine horizontal complementarities, horizontal propagation remains weak and easily masked by vertical effects.

Observed demand-driven comovement broadly aligns with the theory developed in Sections 2 and 3. Positive comovement is most likely to emerge when sectors are “economically” distant, since vertical effects dominate and the conflicting role of horizontal spillovers is attenuated. For mostly connected sectors, however, the balance between vertical and horizontal channels is more delicate, producing the mixed responses observed in the data. This suggests that observed employment comovement mirrors an interplay of vertical and horizontal propagation, thereby highlighting the inherent complexity in interpreting demand-based complementarities.

5.2.2. Results for factor input supply distances

Considering now sectors whose distance is defined in terms of selling to the same set of downstream buyers, Figure 10 plots the responses from the estimation of eq. (10). Interpreting the effect of further ($d = 2$) sectors’ variations in employment yields the same results of the factor input demand case: from Panel 10b, a 1% increase in their employment increases sectoral employment as well, thus determining positive comovement and a positive transmission of the sector-idiosyncratic shock. Different is the case of closer ($d = 1$) sectors, where it appears a positive and then a negative cumulative effect on sector-specific employment with a small magnitude.

The lack of significance in Panel 10a might have two complementary interpretations: (i)

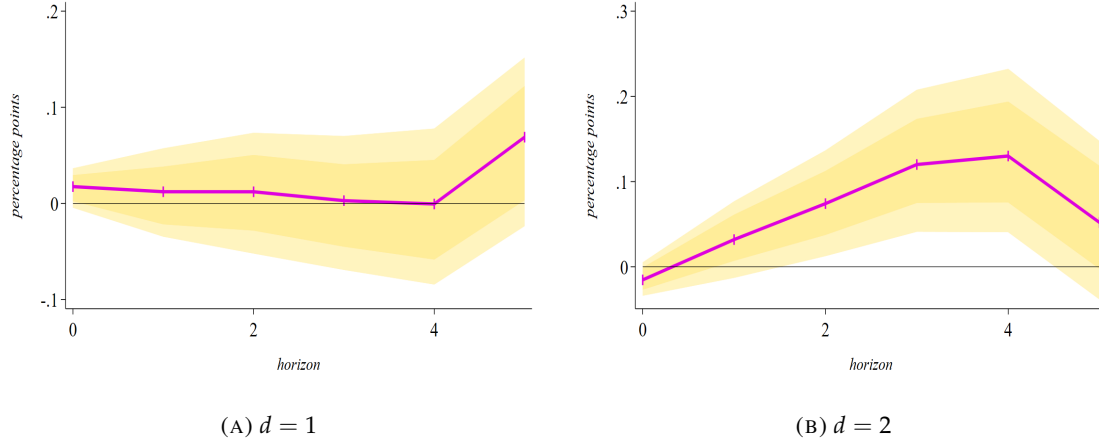


FIGURE 10: EMPLOYMENT COMOVEMENT UNDER SUPPLY LINKAGES

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998–2022. Panel 10a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 10b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (10). *Source:* BEA and own calculations.

changes in employment in nearby sectors have no measurable impact on others when the sectors are linked through a common set of downstream buyers; and (ii) the factor input supply network distance seems to dampen the vertical propagation of shocks across the production network, thereby weakening the expected comovement in production inputs across sectors. Specifically, when sectors share the same downstream buyers, a change in their relative price faced by the common buyer leads to opposite changes in their relative supply of intermediate inputs. This generates an inverse comovement in sectoral production levels and, consequently, an opposite comovement in sectoral employment levels. In the context of Panel 10a, these opposing forces roughly cancel each other out when averaged across all sectors, resulting in the non-significant response. Accordingly, clearer effects might be correctly identified when considering highly interlinked sectors.

Impulse responses for the clustered analysis are presented in Figure 11. The results for major network distances, shown in Panel 11b, largely mirror those found using the full sample, but with a notably higher magnitude in the point estimates responding to the structural employment shock. More strikingly, minor supply distances in Panel 11a fully confirm Proposition 3: any increase in employment levels within closely linked sectors induces a negative transmission to other nearby sectors, resulting in an opposite comovement in sectoral employment. This inverse relationship grows stronger as the analysis narrows to an even smaller group of sectors with increasingly dense linkages: the closer the sectors are in the production network when supplying downstream, the more pronounced these opposing employment responses become. In sum, restricting the analysis to highly interlinked sectors (Theorem 4) reveals that the Input-Output structure and its horizontal geometry fundamentally condition either the direction and the strength of sectoral comovement.

All these theoretical and empirical results are in line with the non-linear theories in Atalay (2017) and Baqaee and Farhi (2019): downstream complementarities in intermediate inputs

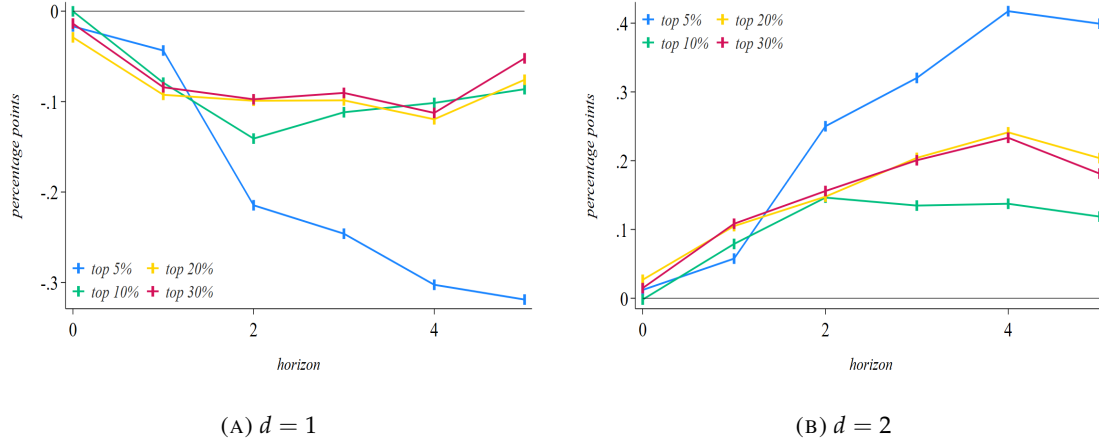


FIGURE 11: SUPPLY LINKAGES AND COMOVEMENT IN HIGHLY INTERLINKED SECTORS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel 11a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 11b plots the response to employment changes in further (distance equal to 2) ones. Each line corresponds to the average response of employment across sectors with different numbers of linkages, robust to 68% and 90% significance levels of bootstrapped Confidence Interval (CI) computed from eq. (10). *Source:* BEA and own calculations.

are crucial to generate opposite responses from an idiosyncratic shock – *i.e.*, the downstream demand pass-through effect from the supply-driven network economic distance is accordingly generating opposite comovement (Proposition 2). Yet, such pass-through is found to be weaker when sectors have dissimilar Input-Output structure in supply.

5.3. ROBUSTNESS

Results on comovement hold along different “economically-based” robustness checks, sequentially presented in Appendix C. Initially, I shift the focus towards peripheral sectors within the production network. Much of the existing literature holds on the role of central sectors, typically measured using Bonacich-Katz centrality (*e.g.*, Carvalho 2014). The underlying premise is that a positive shock in a central sector transmits outward, potentially inducing broader economic expansions – *i.e.*, microeconomic shocks can scale up to macroeconomic consequences through network centrality. From a comovement perspective, periods of economic expansion (contraction) are expected only when a positive (negative) shock in a central sector effectively propagates to more peripheral sectors: fluctuating aggregate activity is not merely a direct realization of shocks to central nodes, but rather it stems from the manner in which such asymmetric shocks ripple across the network to other, more peripheral sectors.¹⁶ If network distances were irrelevant, comovement would be expected universally. However, should demand- and supply-side linkages prove significant, the comovement patterns ought to remain consistent with those documented thus far.

I further account for changes in sectoral inventories, since they introduce a short-run margin that may buffer the network-based transmission: when inventories are available, sectors smooth

¹⁶ Stated differently, it is not the initial shock itself that generates business cycle fluctuations, but the subsequent consequences it produces as it transmits across the network to less central nodes.

shocks through previously accumulated stocks, temporarily decoupling production adjustments from contemporaneous changes in the purchased intermediate input bundle. As a result, the transmission of shocks through shared Input-Output structures should be attenuated, breaking the simultaneity of horizontal complementarities and weakening the role of both demand- and supply-based network economic distances in shaping short-run comovement. By contrast, in the absence of inventories, changes in economically nearby sectors are expected to translate directly into the usual horizontal propagation effects.

Finally, I test the robustness of the presented findings by employing alternative base years for the U.S. production network. My main analysis is anchored at the 2007 Input-Output structure, which conveniently divides the sample period into two equal sub-periods. Theoretically, the results should remain valid provided that the production network structure exhibits limited evolution over time.¹⁷ Empirically, I replicate the full set of analyses using network data from other benchmark years.

Centrality scores.— Figures C.6 and C.7 present the impulse responses of employment to factor input demand and supply distances, respectively, considering only the peripheral sectors of the U.S. production network. Results are broadly consistent with those discussed in earlier sections: opposite comovement under supply distances, and positive comovement for shocks originated in further sectors. The only noteworthy exception concerns the response of employment to changes in closer sectors under demand-based distances: rather than exhibiting the ambiguity observed for all sectors (overlapping of vertical and horizontal dimensions), inspecting horizontal complementarities in demand for more peripheral sectors generate effects that resemble those observed under supply-based distances, with opposite comovement occurring across sectors sharing the same network structure in terms of buying from common suppliers. This refines the observed pattern under direct demand-based measures of Figure 8, suggesting that opposite employment comovement among sectors purchasing from similar upstream suppliers largely occurs in sectors not playing a key role in the propagation of shocks.¹⁸ These outcomes create a bridge between eq. (5) and the insights of Theorem 5: positive shocks in central sectors transmit unevenly to peripheral ones according to different network economic distance values, thereby inducing different types of aggregate behaviours.

Inventories.— The outlined hypothesis is strongly supported by the supply-based patterns in Figure C.8. Horizontal complementarities vanish for sectors holding inventories, and the transmission of a positive shock reverts to its vertical logic.¹⁹ By contrast, in contexts without inventories, the baseline results are preserved, with positive shocks generating opposite comovement across sectors with nearly similar Input-Output structures. This sharp contrast indicates that horizontal effects are operative when production adjustments are directly tied to

¹⁷ A condition broadly supported in the empirical literature. For instance, Carvalho (2014) and Acemoglu, Ozdaglar, et al. (2016) document relative stability of U.S. sectoral production network and its role on shock propagation. Classical Input-Output analysis (e.g., Leontief 1986; Miller and Blair 2022) also emphasises the persistence of production linkages in the short to medium run.

¹⁸ In line with the discussion in Baqaee and Farhi (2019): within a sector, complementarities (“horizontal” in my framework) across intermediate inputs attenuate the aggregate effects of a positive idiosyncratic shock.

¹⁹ Over the sample period, inventories are observed in NAICS agriculture (11), extraction (21), and manufacturing (31-33) sectors, as well as in transportation (481-484), publishing (511), and recording (512) industries.

contemporaneous changes in circulating intermediate inputs, rather than when accumulated in storage. Inventories act as a short-run buffer that temporarily deactivates the horizontal propagation channel without altering the underlying network structure. These findings provide direct evidence that the documented horizontal complementarities reflect structural economic mechanisms, arising from contemporaneous adjustments through shared linkages, and not mechanical features of the production network.

Different base years.— All the documented results, derived using the 2007 U.S. tables, remain robust if replicated for alternative benchmark years (2002 and 2012), as depicted in Figures C.9, C.10, and C.11, supporting the inherent ambiguity of demand-based distances and the clearer, more systematic comovement patterns under supply-based distances. This temporal consistency suggests that the structural features of the U.S. production network have remained remarkably stable over time, documenting the enduring nature of Input-Output linkages and the slow-moving evolution of production networks, thus lending further credibility to the analysis.

6. FUTURE POLICY PERSPECTIVES

The Input-Output structure of the economy shapes how shocks propagate and contributes to business cycle dynamics. This paper has explored how the structure of production networks allows to push further the traditional focus on vertical transmission to emphasize the role of horizontal economic relationships. By analysing shared network connections, the analysis reveals that it is not merely the existence of Input-Output linkages, but rather the demand and supply geometry of these linkages that fundamentally governs the transmission of micro-originated shocks and the resulting macroeconomic comovement patterns. Several are the contributions that horizontal complementarities can provide to the ongoing literature exploiting a production network perspective to form and deliver policy prescriptions.²⁰

Fiscal policy.— Horizontal complementarities play a critical role when considering the specific composition of public expenditure since certain sectors may be disproportionately exposed to changes in government demand. The evidence presented in Subsection 4.3 suggests that fiscal shocks propagate differently across horizontal geometries, implying that two spending programs of identical magnitude may therefore generate substantially different outcomes depending on the targeted industries. Incorporating demand- and supply-driven network distances into fiscal policy analysis could thus provide a more accurate characterization of how sector-specific government interventions diffuse throughout the economy than traditional Input-Output measures alone (*e.g.*, Barattieri, Cacciatore, and Traum 2023; Bouakez et al. 2025). Tariff policies operate through similar network mechanisms, not by raising demand directly, but rather altering relative prices (*e.g.*, Barattieri and Cacciatore 2023; Antonova et al. 2025; Clausing and Obstfeld 2025), effectively acting as revenue reallocation shocks (*i.e.*, my supply-based

²⁰ On the theory side, horizontal complementarities can provide valuable insights into: (i) the functioning of global production networks (*e.g.*, Caliendo et al. 2022; Huo et al. 2025), extending beyond merely vertical supply chains; (ii) the complexity of firm-to-firm production networks (*e.g.*, di Giovanni et al. 2018; Boehm, Flaaen, et al. 2019), as demand- and supply-based distances help to reveal relationships between nodes that would otherwise appear unconnected; and (iii) the endogenous formation of production networks, since “economically” closer sectors tend to share similar Input-Output structures and might rely on common intermediate inputs that are more essential than others (*e.g.*, Carvalho and Voigtländer 2015).

distances), lending further perspectives from network’s horizontal geometries.

Industrial policy.— Two contemporaneous papers analyse how production networks amplify the impact of industrial policies supporting strategic sectors to promote sustained economic growth:²¹ central are the notions of “distortionary effects” (market imperfections accumulate through backward demand linkages; Liu 2019) and “downstream spillovers” (positive effects on buyers of targeted sectors; Lane 2025), and both perspectives exploit the verticality of sectoral connections. Yet, how do industrial policies affect untargeted sectors exhibiting similar Input-Output structures to the targeted ones? This is where the horizontal geometry matters.

Keynesian transmission mechanism.— Complementarities (either in consumption or production) might turn an asymmetric supply shock, affecting a subset of sectors and reducing their demand for other sectors, into a demand-like shock at the aggregate level (the “Keynesian supply shock”; Guerrieri et al. 2022). Production network is a natural field to study this mechanism (e.g., Cesa-Bianchi and Ferrero 2021) and, reminiscent from Subsection 2.1 (Figure 2b), horizontal complementarities among sectors can help to clearly, easily and linearly identify demand and supply forces within the network that guide the Keynesian transmission. This extended mechanism is important for interventions aimed at stabilizing business cycles, acting as a suitable extension of the production network framework for monetary policy (e.g., Pasten et al. 2020; Kalemli-Özcan et al. 2025).

Finally, by embedding similarities in common structures of Input-Output linkages, the horizontal dimension transcends the confines of production networks, rendering it applicable to a broader class of economic interdependencies including studies analysing investment (e.g., vom Lehn and Winberry 2022), credit (e.g., Delli Gatti et al. 2010), or financial (e.g., Elliott et al. 2014; Huremovic et al. 2025) networks.

7. CONCLUDING REMARKS

Initial inquiries into production networks chiefly centred on how disaggregation induced by inter-sectoral trade generates comovement of macroeconomic variables. Reviving this foundational perspective, this paper moves beyond traditional vertical linkages by theoretically introducing and empirically validating the horizontal geometry of Input-Output economies, measured through network “economic” distances reflecting similarities between sectors in terms of common upstream suppliers or downstream buyers. Constitutive premise is that sectors are densely interconnected not only by their observed trade flows but also through shared inter-sectoral relationships, providing a richer understanding of the complex architecture underlying any capitalist economy. Demand-based distances promote an overlap of horizontal and vertical propagation, as shocks transmitted through shared upstream suppliers interact with parallel supply chain effects. By contrast, supply-based distances directly threaten the design of vertical transmission, as revenue reallocation among sectors selling to the same downstream buyers counteracts the standard propagation of shocks. These mechanisms operate independently of

²¹ Explanations related to cross-country income differences, Input-Output economies and industrialization are presented in Ciccone (2002) and Fadinger et al. (2022). Conley and Ligon (2002) study how cross-country dependence and GDP growth rates are shaped by “economic distance”.

trade intensities, offering a powerful refinement to prevailing network analysis. As for aggregate fluctuations, horizontal propagation induces correlated adjustments across sectors sharing suppliers or buyers, generating persistent aggregate comovement absent any cascading effect. By conceptualising these horizontal complementarities, the paper enriches the understanding of how independent and idiosyncratic shocks permeate the production system. Ultimately, it reveals that the demand/supply architecture of an Input-Output structure and its horizontal geometry critically shape the transmission of shocks, offering novel insights into the networked nature of economic activity.

APPENDIX

A. Proof of Theorem 3: on the weighting of network “economic” distances

Let $G = (m, \Lambda)$ be a finite directed weighted graph, with indicator- m representing the size of the network, and $\Lambda = \{i, j \in m \times m : \alpha_{ij} \geq 0\}$ the set of directed edges representing non-negative linkages $i \leftarrow j$. Let $\mathcal{A} = [\alpha_{ij}] \in \mathbb{R}^{m \times m}$ denote the adjacency (Input-Output) matrix of G , associated to the Leontief inverse $\mathcal{L} := (\mathbf{I} - \mathcal{A})^{-1}$. Production is sequential at the k -th round of input requirements, $\mathcal{L} \equiv \sum_{k=0}^{\infty} \mathcal{A}^k$.

Matrix $\mathcal{D} = [d_{ij}] \in \mathbb{R}^{m \times m}$ is called an “horizontal (distance) operator” if $d_{ij} = f^{\mathcal{D}}[\alpha_{.i}, \alpha_{.j}]$, where $\alpha_{.i}$ denotes the immediate neighbourhood of node- i (common upstream suppliers or downstream buyers), and $f^{\mathcal{D}}[\cdot]$ depends only on the overlap or similarity of these neighbourhoods. Differently, vertical operators are fundamentally different from profile comparison and involve repeated application of $\mathbf{V} = \{\mathcal{A}, \mathcal{L}\}$. Indeed, an operator is called “vertical” if it aggregates information along paths of length greater than or equal to one $\sum_{k \geq 1} \mathbf{V}^k$.

The following provides vertical and horizontal anatomies.

LEMMA 3 (Locality of horizontal operators) *If a certain horizontal entry is identified by $d_{ij} = f^{\mathcal{D}}[\alpha_{.i}, \alpha_{.j}]$, then horizontal matrix $\mathcal{D}$ depends only on a finite-dimensional representation of $\mathbf{V}$ and does not involve recursive application of $\mathbf{V}$.*

Proof: each entry d_{ij} is constructed from the vectors $\alpha_{.i}$ and $\alpha_{.j}$ through a fixed functional $f^{\mathcal{D}}[\cdot]$. No term involves powers $\mathbf{V}^k$ with $k \geq 2$, nor any infinite series. ■

LEMMA 4 (Non-locality of the Input-Output weights) *Any vertical operator involving $\mathbf{V}$ depends on all powers $\mathbf{V}^k$, for $k \geq 0$.*

Proof: by definition, $\mathbf{V} \equiv \sum_{k=0}^{\infty} \mathbf{V}^k$. Each power $\mathbf{V}^k$ represents k -fold composition of technologies. Hence, any object involving $\mathbf{V}$ depends on recursively propagated Input-Output relations. ■

LEMMA 5 (Incompatibility) *No local operator contains a global path operator.*

Proof: assume by contradiction that a local operator $\mathcal{D}$ could be written as an equilibrium object $\mathbf{B} = \mathbf{V}'\tilde{\mathcal{D}}$ for some $\tilde{\mathcal{D}}$, so that horizontal effects enter the equilibrium multiplied by the Input-Output weights. By contrast, given the non-locality of $\mathbf{V}$, the object $\mathbf{B}$ depends on arbitrarily long production paths on all recursive compositions $\mathbf{V}^k$, besides the local neighbourhood similarity of $\mathcal{D}$. Then, $\mathbf{B}$ inherits the non-local dependence of $\mathbf{V}$, contradicting the established locality: by the incompatibility lemma, such a representation is impossible. ■

A re-written version of the incompatibility lemma delivers Theorem 3: horizontal distance operators cannot be multiplicatively weighted by the Input-Output weights.

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SUPPLEMENTAL APPENDIX

(Outline) In this appendix I report all the material complementary to the main text; it is made of additional tables and figures, and of discussions on further analytical results. Section **A** deepens further all the theoretical findings; in particular, Subsections **A.1-A.3** derive and prove all the characterizing features of the outlined Input-Output general equilibrium model. Section **B** consolidates and further discusses the data presented in Section 4. Finally, Section **C** complements and enriches the exposition of the empirical findings in Section 5. All the proofs are presented to be intuitive and informative, expanding that in the main text.

A. MODEL DERIVATION AND FURTHER THEORETICAL RESULTS

A.1. Model derivation

(Household problem) The household's utility problem is to maximize utility function subject to its budget constraint:

$$\max_{c_i, n_i} \mathcal{U}[c_i; n_i]_{\forall i \in \Phi} := \prod_{i \in \Phi} \left(c_i^{\omega_i} - \frac{n_i^{1+\phi}}{1+\phi} \right) \quad \text{s.t.} \quad \sum_{i \in \Phi} p_i c_i = w_i n_i + \sum_{i \in \Phi} \pi_i$$

Sectoral consumption and labour supplied are c_i and n_i , respectively, and each is determined by a price, p_i and w_i . Parameter $\omega_i \in (0, 1)$ identifies the weight of each sectoral good in household's consumption basket and $\sum_{i \in \Phi} \omega_i = 1$, ϕ is the inverse Frisch labour supply elasticity, measuring the elasticity of hours worked to the wage rate under a constant marginal utility of income. Finally, π_i is the constant share of sector-specific profits flowing from sector- i to household.

Utility maximization implies the Lagrangian function to be

$$\mathcal{L}_{c_i, n_i} := \prod_{i \in \Phi} \left(c_i^{\omega_i} - \frac{n_i^{1+\phi}}{1+\phi} \right) - \psi^L \left[\sum_{i \in \Phi} p_i c_i - \left(w_i n_i + \sum_{i \in \Phi} \pi_i \right) \right]$$

with ψ^L being the penalty multiplier. Optimality conditions are in order

$$\frac{p_i c_i}{\omega_i} = \frac{p_j c_j}{\omega_j} \quad \text{and} \quad w_i = n_i^\phi p_i c_i [\omega_i c_i]^{-1}$$

which report the sector-level optimality conditions for consumption and labour supply. The first condition for consumption defines how the total expenditure from households for sectoral goods gives rise to the relative importance of each good in total consumption, C . This is an aggregator over sectoral consumption levels:

$$C = f^C \left[c_i, c_j, \dots, c_m ; \{ \omega_i \}_{i=1}^m \right]$$

ASSUMPTION 2 (Final consumption requirements) As main characteristics of a final consumption aggregator: (i) it is strictly quasi-concave, non-decreasing and homogeneous of degree one in each of the sectoral consumption levels; (ii) consumption goods are normal, so that their demand increases with households' income; and (iii) the weight of sectoral goods

in total consumption is driven by ω_i , which is the relative weight that households give in the consumption of the good produced by sector- s .

Moreover, denote by $\mathbf{c} = [c_i]$ the $m \times 1$ vector of consumption levels over sectoral goods, with generic element defined as $c_i > 0, \forall i \in \Phi$. Under this specification and the regularities in Assumption 2, it holds that $C = \prod_{i \in \Phi} c_i^{\omega_i} \propto \omega' \mathbf{c}$.

□

(Sectoral optimization) The production side of the economy is made of a finite set of sectors, $i, j, s = \{1, \dots, m\} \in \Phi$, populated by a representative firm whose output is the result of a Cobb-Douglas technology satisfying Assumption 1:

$$y_i = z_i n_i^{\alpha_i} \prod_{j \in \Phi} x_{ij}^{\alpha_{ij}} \mathcal{K}_i^q \quad (\text{A.1})$$

where $\mathcal{K}_i^q$, for $q = \{fd, fs\}$, is a normalization constant ruling out double counting of intermediate inputs when considering both factor input demand and supply network distances between sectors; absent any of these two cases, then $\mathcal{K}_i^q = 1$.

A perfectly competitive representative firm in sector- i maximizes total revenues from production net of costs of inputs of production:

$$\max_{n_i, x_{ij}} y_i = z_i n_i^{\alpha_i} \prod_{j \in \Phi} x_{ij}^{\alpha_{ij}} \mathcal{K}_i^q \quad \text{s.t.} \quad C_i := w_i n_i + \sum_{j \in \Phi} p_j x_{ij}$$

where $\alpha_i + \sum_{j \in \Phi} \alpha_{ij} = 1$, and $\mathcal{K}_i^q = 1$ for $q = \{fd, fs\}$. Profit maximization then implies that optimal competitive factor demands of sector- i relative to sector- j are given by $n_i = \alpha_i \frac{p_i y_i}{w_i}$ and $x_{ij} = \alpha_{ij} \frac{p_i y_i}{p_j}$.

□

(Equilibrium characterization) In equilibrium, the model should specify the clearing conditions of labour, circulating intermediate inputs, and goods markets. Starting from the labour market, equating labour demand and labour supply would deliver the labour market equilibrium condition, $n_i = \left(\alpha_i \frac{C}{c_i} \omega_i y_i \right)^{1/(1+\phi)}$. Moreover, total labour demand is found by aggregating labour optimality condition of sectors, $N^{dem} = \sum_{i \in \Phi} n_i^{dem}$, and total labour supply is found by aggregating labour optimality condition of households, $N^{sup} = \sum_{i \in \Phi} n_i^{sup}$. Labour market clearing then requires that $N^{dem} = N^{sup}$. For the equilibrium in the circulating intermediate inputs market, total demand of intermediates from sectors is $x_i^{dem} = \sum_{j \in \Phi} x_{ij}$. Analogously, its total supply of intermediate inputs is $x_i^{sup} = \sum_{j \in \Phi} x_{ji}$ so that, in equilibrium it must be true that $x_i^{dem} = x_i^{sup}$ which, in aggregate, simply states that $\sum_{i \in \Phi} x_i^{dem} = \sum_{i \in \Phi} x_i^{sup} \equiv X^{sup} = X^{dem}$. The equilibrium in the good market is given by summing over all sectors the sectoral equilibrium in which total production must equal its total final consumption (by households) and its circulating intermediate input bought by other sectors: $\sum_{i \in \Phi} y_i = \sum_{i \in \Phi} c_i + \sum_{i \in \Phi} \left(\sum_{j \in \Phi} x_{ij} \right)$. Finally, by aggregating the households' budget constraints over sectors, and imposing the clearing conditions so far, the aggregate resource constraint for this economy reads as $P^C C = WN + \pi$ which equals total output, defined by $Y \propto \omega' \mathbf{y}$.

DEFINITION 1 (Equilibrium) According to the structure of the model, equilibrium conditions read as follows. A competitive equilibrium for this efficient economy is defined by a set of sectoral prices for labour and intermediate inputs, $\{w_i, p_i\}_{i \in \Phi}$, a set of sectoral production input quantities, $\{n_i, x_{ij}\}_{i,j \in \Phi}$, exogenous sectoral productivities, $\{z_i\}_{i \in \Phi}$, and a set of aggregate quantities, $\Omega = (Y, C, N, X, \pi)$, such that (i) each household satisfies its optimality conditions, (ii) the representative firm of each sector maximizes profits, and (iii) all markets clear, shaping Ω .

□

(Derive aggregate fluctuations result of eq. (5)) Following exactly the same operating steps in vom Lehn and Winberry (2022) to characterize the dynamical changes in aggregate Gross Domestic Product (GDP), it is first necessary to determine the existing relation between sectoral intermediate output, y_i , and the Divisia Index.¹ Constant returns to scale and homogeneity of degree one in Assumption 1 and in the production function in eq. (A.1) implies the following zero-profit condition

$$p_i y_i = w_i n_i + \sum_{j \in \Phi} p_j x_{ij}$$

where $p_i y_i$ is the (gross) nominal output of sector- i , and from where the national accounting definition of nominal value added is just $p_i y_i - P_i X_i = w_i n_i$, where I define $P_i X_i = \sum_{j \in \Phi} p_j x_{ij}$ the total expenditure of sector- i for intermediate inputs from other connected sectors. Define the left-hand side of the previous equation as $p_i^Y y_i^Y \equiv p_i y_i - P_i X_i$, i.e., the nominal value-added.

In order to construct a single, aggregate measure of real output (or value added) across multiple sectors, I use a Divisia index. This index combines the growth rates of individual sectoral outputs into an overall growth rate, but does so in a way that accounts for each sector's economic importance – measured by its share in total nominal value added. Specifically, the Divisia index computes a weighted average of the growth rates of sectoral real outputs, where the weights are each sector's share in total nominal value added. This allows to track how the economy's output is changing over time, adjusting dynamically as the composition of output across sectors evolves.

Henceforth, the Divisia Index requires to differentiate the nominal value added under constant price levels associated to variables:

¹ This indicator provides a theoretically consistent measure of growth. It is a composite index, particularly suited to analyse variables made of multiple and differentiated elements, measuring changes in certain aggregate quantity of a given variable, and weighting all its defining components according to their relevance on that variable, that is it expresses the overall rate of output change as a weighted average of the growth rates of individual components, with time-varying weights corresponding to each component's share in total output. Conceptually, it captures the notion that aggregate GDP growth reflects both the differential growth of its components and their evolving relative importance within the economy. This formulation offers an improvement over fixed-weight indices by accommodating structural changes in the composition of economic activity over time. Refer to Oulton (2022) for further details.

$$\begin{aligned}
p_i^Y d \log y_i^Y &= p_i d \log y_i - P_i d \log X_i \\
\rightarrow p_i^Y y_i^Y d \log y_i^Y &= p_i y_i d \log y_i - P_i X_i d \log X_i \\
\rightarrow \alpha_i d \log y_i^Y &= d \log y_i - \alpha^i d \log X_i \\
\rightarrow \alpha_i d \log y_i^Y &= \left(d \log z_i + \alpha_i d \log n_i + \alpha^i d \log X_i \right) - \alpha^i d \log X_i \\
\rightarrow d \log y_i^Y &= \frac{1}{\alpha_i} d \log z_i + d \log n_i
\end{aligned}$$

where $\alpha^i = \sum_{i \in \Phi} \alpha_{ij}$. Plug the above sector-level nominal value added growth, $d \log y_i^Y$, into the Divisia Index, $d \log Y = \sum_{i \in \Phi} \left(\frac{p_i^Y y_i^Y}{P^Y Y} \right) d \log y_i^Y$, so that it is possible to obtain a non-complete equation for real GDP growth:

$$d \log Y = \sum_{i \in \Phi} \left(\frac{p_i^Y y_i^Y}{P^Y Y} \right) \left(\frac{1}{\alpha_i} d \log z_i + d \log n_i \right) \quad (\text{A.2})$$

Summing over all sectors the sectoral optimality conditions for intermediate inputs, $\sum_{j \in \Phi} x_{ij} = \sum_{j \in \Phi} \alpha_{ij} [p_i y_i] (p_j)^{-1}$, noticing that $\alpha_i = 1 - \alpha^i$, and inserting the zero-profit condition would yield $\alpha_i = \frac{p_i^Y y_i^Y}{p_i y_i}$, which is the ratio of sectoral value-added to sectoral intermediate (gross) output.

By plugging into eq. (A.2) the condition for α_i , and making all the adjustments to simplify the elements multiplying the sectoral productivity, it is then possible to get eq. (5), which determines the drivers of real aggregate GDP fluctuations.

□

(Labour market equilibrium with network “economic” distances) Beyond the equilibrium definition, rearranging optimality conditions for labour market variables from both households and sectors, the following equilibrium condition for sectoral labour force is detected:

$$n_i = \left(\alpha_i \frac{C}{c_i} \omega_i y_i \right)^{\frac{1}{1+\phi}} \quad (\text{A.3})$$

It defines that employment is a positive function of both output and consumption (recall that total consumption C increases with sector-specific consumption).

To account for changes to other sectors’ employment on sectoral employment level, notice that when sector- i buys intermediate inputs from sector- j , it is introjecting also the quantities of production inputs in such sector. Henceforth, each element in the intermediate inputs bundle of sector- i can be taught as $x_{ij} = \vartheta_{ij} y_j$, $\forall j \in \Phi$, which states that each intermediate input is just a given share ϑ_{ij} of other sector’s output. Labour market equilibrium in eq. (A.3) is thus

$$n_i = \left[\alpha_i \frac{C}{c_i} \omega_i z_i n_i^{\alpha_i} \prod_{j \in \Phi} (\vartheta_{ij} y_j)^{\alpha_{ij}} \right]^{\frac{1}{1+\phi}} \quad (\text{A.4})$$

so that, once plugging in the output of sector- j , sectoral employment levels are transparently

related through the production network structure. As a result, the above labour market condition allows to study the general equilibrium comovement of employment levels across sectors.

Once inserting the production function of eq. (A.1) in the labour market equilibrium condition of eq. (A.4), then optimal quantities of a sector are related to that of the other ones. To incorporate factor input demand distance, manipulation in general equilibrium would result in

$$n_i = \left(\alpha_i \frac{C}{c_i} \omega_i z_i n_i^{\alpha_i} \mathbf{N}^{fd} \mathbf{\Lambda}^{fd} \mathbf{\Xi}^{fd} \right)^{\frac{1}{1+\phi}} \quad (\text{A.5})$$

where $\mathbf{N}^{fd} = \prod_{j \in \Phi} \left(\prod_{i \in \Phi} x_{ji}^{\alpha_{ji}} \right)^{\frac{\alpha_{ij}}{1+\phi}}$ identifies the set of intermediate inputs bought by sector- i and all the other sectors connected to it, the element $\mathbf{\Lambda}^{fd} = \prod_{j \in \Phi} \left(\vartheta_{ij} z_j n_j^{\alpha_j} \right)^{\alpha_{ij}}$ captures the relevance of other sectors' productivities and employment levels on sector- i working through the production network, and the component $\mathbf{\Xi}^{fd} = \prod_{j \in \Phi} \left[\prod_{i \in \Phi} \left(\frac{x_{ji}}{x_{ii}} \right)^{\frac{\alpha_{ji}}{\alpha_{ii}}} \right]$ determines the ratio among the intermediate inputs between sector- i and each of the other sectors, when these are buying from the same upstream sector.

Manipulation in general equilibrium to include factor input supply distance for pairs of sectors would result in

$$n_i = \left(\alpha_i \frac{C}{c_i} \omega_i z_i n_i^{\alpha_i} \mathbf{N}^{fs} \mathbf{\Lambda}^{fs} \mathbf{\Xi}^{fs} \right)^{\frac{1}{1+\phi}} \quad (\text{A.6})$$

where $\mathbf{N}^{fs} = \prod_{j \in \Phi} \left(\prod_{j \in \Phi} x_{jj}^{\alpha_{jj}} \right)^{\frac{\alpha_{ij}}{1+\phi}}$ is the compounded set of intermediate inputs, $\mathbf{\Lambda}^{fs} = \prod_{j \in \Phi} \left(\vartheta_{ij} z_j n_j^{\alpha_j} \right)^{\alpha_{ij}}$ considers sectoral productivities and employment levels, and $\mathbf{\Xi}^{fs} = \prod_{j \in \Phi} \left[\prod_{i \in \Phi} \left(\frac{x_{ji}}{x_{jj}} \right)^{\frac{\alpha_{ji}}{\alpha_{jj}}} \right]$ determines the ratio among intermediate inputs when sectors are selling to the same downstream sector.

□

A.2. Network-based propagation to sectoral employment

PROPOSITION 4 (Propagating variations to intermediate inputs) Consider a market economy characterized by Input-Output linkages as in eq. (A.1), whose labour market equilibrium is eq. (A.3). Then, the response of sectoral employment to changes in sectoral intermediate inputs is a first-order (log-linear) approximation

$$d \log \mathbf{n} = \Theta \left\{ d \log C + \underbrace{d \log \mathbf{z} - d \log \mathbf{c}}_{d \log \mathcal{S}} + \mathbf{A} d \log \mathbf{X} \right\} \quad (\text{A.7})$$

where:

- (a) $\mathbf{n}$ identifies sectoral employment levels;
- (b) $\Theta = \frac{1}{1+\phi+\alpha}$ is a compound of structural parameters, and $\alpha = [\alpha_i, \dots, \alpha_m]$;
- (c) C is aggregate consumption;
- (d) $\mathcal{S}$ identifies sector-specific changes in productivity and consumption;

- (e) $\mathbf{A}$ is the Input-Output matrix, comprising sectoral weight on other sectors, influencing the elements of $\mathbf{X}$, a matrix with intermediate inputs quantities.

Proof: provided in Appendix A.3. ■

PROPOSITION 5 (Direct propagation of variations to sectoral employment) Consider an Input-Output economy defined by a labour market equilibrium as stated in eq. (A.4). Then, the response of sectoral employment to other sectors' employment is a first-order (log-linear) approximation

$$d \log \mathbf{n} = \Theta \left\{ d \log C + d \log \mathbf{S} + \underbrace{\mathbf{A}(\Phi_{\{i,j\}}) \left[d \log \mathbf{z} + \boldsymbol{\alpha} d \log \mathbf{n} \right] + \boldsymbol{\varepsilon} d \log \mathbf{X}}_{d \log \mathbf{V}} \right\} \quad (\text{A.8})$$

where:

- (a) $\mathbf{n}$ identifies sectoral employment levels;
- ...
- (e) $\mathbf{V}$ identifies the production network effect of other sectors' changes in productivities, employment levels, and intermediate inputs usage, where:
 - $\mathbf{A}(\Phi_{\{i,j\}})$ is the Input-Output matrix, comprising the weight of each sector on other sectors, whose entries are set to 0 whenever $i = j$;
 - $\boldsymbol{\varepsilon} = \mathbf{A}(\Phi_{\{i,j\}})' \mathbf{A}(\Phi_{\{j,i\}})$ is a compounded network effect, made of the inner product of the Input-Output matrix, influencing the elements of $\mathbf{X}$.

Proof: provided in Appendix A.3. ■

PROPOSITION 6 (Direct propagation under factor input demand distance) Consider an Input-Output economy defined by a labour market equilibrium as stated in eq. (A.5). Then, the response of sectoral employment to other sectors' employment is a first-order (log-linear) approximation

$$d \log \mathbf{n} = \Theta \left\{ d \log C + d \log \mathbf{S} + d \log \mathbf{V} + \underbrace{\mathbf{D}^{fd} \left[d \log \mathbf{n} - (\mathbf{I}'_{\Phi} d \log \mathbf{n}) \mathbf{1} \right]}_{d \log \mathbf{D}(fd)} \right\} \quad (\text{A.9})$$

where:

- (a) $\mathbf{n}$ identifies sectoral employment levels;
- ...
- (f) $\mathbf{D}^{fd}$ identifies all the demand-based distances across any pair of sectors;
- (g.i) $\mathbf{D}(fd)$ identifies the production network distance effect of other sectors' variations in employment levels.

Proof: provided in Appendix A.3. ■

PROPOSITION 7 (Direct propagation under factor input supply distance) Consider an Input-Output economy defined by a labour market equilibrium as stated in eq. (A.6). Then, the response of sectoral employment to other sectors' employment is a first-order (log-linear) approximation

$$d \log \mathbf{n} = \Theta \left\{ d \log \mathbf{C} + d \log \mathbf{S} + d \log \mathbf{V} + \underbrace{\mathcal{D}^{fs} [d \log \mathbf{p} - (\mathcal{I}'_{\Phi} d \log \mathbf{p}) \mathbf{1}]}_{d \log \mathcal{D}(fs)} \right\} \quad (\text{A.10})$$

where:

- (a) $\mathbf{n}$ identifies sectoral employment levels;
- ...
- (f) $\mathcal{D}^{fs}$ identifies all the supply-based distances across any pair of sectors;
- (g.ii) $\mathcal{D}(fs)$ identifies the production network distance effect of other sectors' variations in employment levels.

Proof: provided in Appendix A.3. ■

A.3. Proofs for model derivation

(Derive labour market equilibrium of eqs. (A.5) and (A.6)) From the sector-level equilibrium definition of the labour market, $n_i = \left(\alpha_i \frac{\mathbf{C}}{c_i} \omega_i y_i \right)^{1/(1+\phi)}$, including the condition $x_{ij} = \vartheta_{ij} y_j$ would imply eq. (A.4), that is

$$n_i = \left[\alpha_i \frac{\mathbf{C}}{c_i} \omega_i z_i n_i^{\alpha_i} \prod_{j \neq i} \left(\vartheta_{ij} y_j \right)^{\alpha_{ij}} \varkappa_i^q \right]^{\frac{1}{1+\phi}} \quad (\text{A.11})$$

where it should be inserted the production function of eq. (A.1) for sector- j , and $\varkappa_i^q \neq 1$, with $q = \{fd, fs\}$, for the purpose of this derivation. The resulting equation can be simplified by imposing an infinite inverse Frisch elasticity of labour, $\phi \rightarrow \infty$. In fact, in the model of Section 3, only the extensive margin of employment (i.e., total number of workers) is important, while it is not the intensive margin (i.e., total hours worked), which is ruled out under $\phi \rightarrow \infty$; refer to Rogerson (1988). However, for the sake of generality, I will proceed with $\phi \in [0, \infty)$.

Factor input demand.— Assume the non-common (constant) element in the production function to be given by^{II}

^{II} In eq. (A.12), the component $\tau_i^{fd} = \prod_{j \in \Phi} \left[\prod_{j \in \Phi} (x_{ji}/x_{ii})^{\alpha_{ji}/\alpha_{ii}} \right]^{1/\alpha_{ij}}$ ensures that the distance effects are not weighted by the Leontief inverse. Specifically, incorporating the condition $x_{ij} = \vartheta_{ij} y_j$ in the labour market equilibrium of eq. (A.11) would result in a double counting of the values from the Input-Output matrix: one of them is used to capture sectoral distances, while the other would have been multiplied by $\mathcal{D}^{fd}$ when analysing the effect of sectoral propagation in Proposition 6. Within this context, it is essential to highlight that, starting from the baseline labour market equilibrium in eq. (A.3), and deriving the results presented in this section and in Proposition 6 without including the exogenous condition $x_{ij} = \vartheta_{ij} y_j$ would produce an outcome reflecting the effect of network distances *without* the influence of the Input-Output Leontief inverse weights, as discussed in the Theorem 3; imposing τ_i^{fd} due to the condition $x_{ij} = \vartheta_{ij} y_j$ allows to avoid that such concept is violated.

$$\mathcal{X}_i^{fd} = \tau_i^{fd} \prod_{j \in \Phi} \left(\frac{\prod_{i \in \Phi} x_{ji}^{\frac{1}{\alpha_{ii}} + \alpha_{ji}}}{\prod_{i \in \Phi} x_{ii}^{\alpha_{ji}}} \right)^{\frac{\alpha_{ij}}{1+\phi}} \quad (\text{A.12})$$

Then, multiply and divide both sides of the above labour market equilibrium, including the production function, by $\prod_{j \in \Phi} \left[\left(\prod_{i \in \Phi} x_{ji}^{\frac{1}{\alpha_{ii}}} \right) / \left(\prod_{i \in \Phi} x_{ii}^{\frac{\alpha_{ji}}{\alpha_{ii}}} \right) \right]^{\frac{\alpha_{ij}}{1+\phi}}$. Finally, making all the required adjustments once including $\mathcal{X}_i^{fd}$, the resulting condition will deliver eq. (A.5). All the multiplications are over Φ .

Factor input supply.— Assume now the non-common (constant) element in the production function to be given by

$$\mathcal{X}_i^{fs} = \tau_i^{fs} \prod_{j \in \Phi} \left(\frac{\prod_{i \in \Phi} x_{ji}^{\frac{1}{\alpha_{ii}}}}{\prod_{j \in \Phi} x_{jj}^{\frac{\alpha_{ji}}{\alpha_{ii}} - \alpha_{jj}}} \right)^{\alpha_{ij}} \quad (\text{A.13})$$

Multiply and divide both sides of the above labour market equilibrium, including the production function, by $\prod_{j \in \Phi} \left[\left(\prod_{i \in \Phi} x_{ji}^{\frac{1}{\alpha_{ii}}} \right) / \left(\prod_{j \in \Phi} x_{jj}^{\frac{\alpha_{ji}}{\alpha_{ii}}} \right) \right]^{\frac{\alpha_{ij}}{1+\phi}}$. Making all the required adjustments once including $\mathcal{X}_i^{fs}$, the resulting condition will deliver eq. (A.6). All the multiplications are over Φ .

□

(Proof of Proposition 3) The proof directly follows Propositions 6 and 7.

□

(Proof of Proposition 4) The interest is in characterizing the way in which sectoral employment changes in response to variations in sector-specific intermediate inputs usage. In order to inspect these changes, it is necessary to express the equilibrium conditions in terms of log-linear conditions. Starting from the sector- i labour market equilibrium of eq. (A.3) – i.e., $n_i = \left(\alpha_i \frac{C}{c_i} \omega_i y_i \right)^{1/(1+\phi)}$ –, the associated log-linear form is $\tilde{n}_i = \frac{1}{1+\phi} \left\{ \tilde{y}_i + \tilde{C} - \tilde{c}_i \right\}$, with “tilded” variables, $\tilde{\cdot}$, expressing the deviation from their respective steady state. Such log-linearized equation contains the log-linearized output, obtained by log-differentiating the production function in eq. (A.1): $\tilde{y}_i = \tilde{z}_i + \alpha_i \tilde{n}_i + \sum_{j \in \Phi} \alpha_{ij} \tilde{x}_{ij}$. Substituting out these two equations, one easily gets that log-linearized sectoral employment is a function of

$$\tilde{n}_i = \frac{1}{1+\phi} \left\{ \tilde{z}_i + \alpha_i \tilde{n}_i + \sum_{j \in \Phi} \alpha_{ij} \tilde{x}_{ij} + \tilde{C} - \tilde{c}_i \right\}$$

expressing the elements whose variations induce changes in employment level of sector- s . In vectorial notation, one can rewrite the inner summation as $\mathbf{a}_i \tilde{\mathbf{x}}_i$ which, when stacked over all sectors, can be written in matrix form:

$$\tilde{\mathbf{n}} = \frac{1}{1+\phi} \left\{ \tilde{\mathbf{C}} - \tilde{\mathbf{c}} + \tilde{\mathbf{z}} + \boldsymbol{\alpha} \tilde{\mathbf{n}} + \mathbf{A}' \tilde{\mathbf{X}} \right\}$$

where $\tilde{\mathbf{n}}$ is an $m \times 1$ vector of changes in sector-specific employment levels, identifies changes in aggregate consumption, $\tilde{\mathbf{c}}$ is an $m \times 1$ vector of changes in sector-specific final consumption by households, $\tilde{\mathbf{z}}$ is an $m \times 1$ vector of changes in sectoral productivities, $\mathbf{A}$ is an $m \times m$ squared Input-Output matrix identifying the structure of inter-sectoral trade, and $\tilde{\mathbf{X}}$ is an $m \times 1$ vector of changes in sector-specific set of intermediate inputs bought within the production network. Finally, ϕ is a scalar for aggregate inverse Frisch elasticity of labour supply to wage level, and $\boldsymbol{\alpha} = [\alpha_i, \alpha_j, \dots, \alpha_m]$ comprises sector-specific labour force as a share of its intermediate output.

Bringing the vector of sectoral employments, $\tilde{\mathbf{n}}$, on the left-hand side, rewriting $\tilde{\cdot} = d \log [\cdot]$, and rearranging terms, then one obtains eq. (A.7) in Proposition 4. $\square$

(Proof of Proposition 5) The interest is in characterizing the way in which sectoral employment changes in response to variations in other sectors' employment levels. In order to inspect these changes, it is necessary to express the equilibrium conditions in terms of log-linear conditions. Starting from the sector- i labour market equilibrium of eq. (A.3) – i.e., $n_i = \left(\alpha_i \frac{C}{c_i} \omega_i y_i \right)^{1/(1+\phi)}$ –, the associated log-linear form is $\tilde{n}_i = \frac{1}{1+\phi} \left\{ \tilde{y}_i + \tilde{C} - \tilde{c}_i \right\}$, with “tilded” variables, $\tilde{\cdot}$, expressing the deviation from their respective steady state. Use the condition $x_{ij} = \vartheta_{ij} y_j$, which states that each intermediate input is just a given share ϑ_{ij} of other sector's output. Inserting its log-linearized form $\tilde{x}_{ij} = \tilde{y}_j$, along the log-differentiation of the production function in eq. (A.1), $\tilde{y}_i = \tilde{z}_i + \alpha_i \tilde{n}_i + \sum_{j \in \Phi} \alpha_{ij} \tilde{x}_{ij}$, into the log-linear labour market equilibrium, one gets

$$\tilde{n}_i = \frac{1}{1+\phi} \left\{ \begin{aligned} &\tilde{C} - \tilde{c}_i + \tilde{z}_i + \alpha_i \tilde{n}_i + \\ &+ \sum_{j \in \Phi_i} \alpha_{ij} \left[\tilde{z}_j + \alpha_j \tilde{n}_j + \sum_{i \in \Phi_j} \alpha_{ji} \tilde{x}_{ji} \right] \end{aligned} \right\}$$

expressing the elements whose variations induce changes in employment level of sector- i . Note that using the above condition $x_{ij} = \vartheta_{ij} y_j$ implies that summations are not over the whole set of sectors, but rather it should be excluded the sector whose one is summing for. To this aim, denote Φ the set of all sectors; then, both Φ_i and Φ_j denotes improper subsets of Φ , since they are excluding sector- i and sector- j , respectively. In other words, all the elements in $\{\Phi_i, \Phi_j\}$ are contained in Φ but $\{\Phi_i, \Phi_j\}$ and Φ are not equal, $\Phi_i \subseteq \Phi$ with $\Phi_i \subsetneq \Phi$, and $\Phi_j \subseteq \Phi$ with $\Phi_j \subsetneq \Phi$. When stacking the above condition over all sectors (i.e., matrix notation), the Input-Output matrix from α_{ij} and α_{ji} are not full but rather are set to zero whenever $i = j$, i.e., using the $\Phi_{\{i,j\}}$ as a matrix indicator.

Using such notation, and expressing the above log-linearized labour market equation in vectorial form and then in matrix form (as in the Proof of Proposition 4), and solving for $d \log n_i$, one obtains eq. (A.8) of Proposition 5. $\square$

(Proof of Proposition 6) *The interest is in characterizing the way in which sectoral employment changes in response to variations in other sectors' employment levels according to their factor input demand network distance relationships. Taking the logarithm of eq. (A.5) would result in*

$$\log n_i = \frac{1}{1+\phi} \left\{ \begin{aligned} & \log \alpha_i + \log C + \log z_i + \alpha_i \log n_i - \log c_i + \\ & + \sum_{j \in \Phi} \alpha_{ij} \left[\log \vartheta_{ij} + \log z_j + \alpha_j \log n_j \right] + \\ & + \sum_{j \in \Phi} \alpha_{ij} \left\{ \sum_{i \in \Phi} \alpha_{ji} \log x_{ji} \right\} + \log \omega_i + \\ & + \sum_{j \in \Phi} \left\{ \frac{\sum_{i \in \Phi} \alpha_{ji}}{\sum_{i \in \Phi} \alpha_{ii}} \log \left[\frac{\sum_{i \in \Phi} x_{ji}}{\sum_{i \in \Phi} x_{ii}} \right] \right\} \end{aligned} \right\}$$

Inserting the condition in Lemma 1 in the $\log [\cdot]$ component in the last row, compute the associate total differentiation, following closely the steps to derive Proposition 5, and using the notation associated to network distance matrix $\mathbf{D}^{fd}$, then one obtains eq. (A.9) characterizing Proposition 6.

□

(Proof of Proposition 7) *The interest is in characterizing the way in which sectoral employment changes in response to variations in other sectors' employment levels according to their factor input supply network distance relationships. Taking the logarithm of eq. (A.6) would result in*

$$\log n_i = \frac{1}{1+\phi} \left\{ \begin{aligned} & \log \alpha_i + \log C + \log z_i + \alpha_i \log n_i - \log c_i + \\ & + \sum_{j \in \Phi} \alpha_{ij} \left[\log \vartheta_{ij} + \log z_j + \alpha_j \log n_j \right] + \\ & + \sum_{j \in \Phi} \alpha_{ij} \left\{ \sum_{j \in \Phi} \alpha_{jj} \log x_{jj} \right\} + \log \omega_i + \\ & + \sum_{i \in \Phi} \left\{ \frac{\sum_{j \in \Phi} \alpha_{ji}}{\sum_{j \in \Phi} \alpha_{jj}} \log \left[\frac{\sum_{j \in \Phi} x_{ji}}{\sum_{j \in \Phi} x_{jj}} \right] \right\} \end{aligned} \right\}$$

Inserting the condition in Lemma 1 in the $\log [\cdot]$ component in the last row, compute the associate total differentiation, following closely the steps to derive Proposition 5, and using the notation associated to network distance matrix $\mathbf{D}^{fs}$, then one obtains eq. (A.10) characterizing Proposition 7.

□

B. DATA, FIGURES AND TABLES

(Data details) *In order to build the main dataset with a panel of 65 private 3-digit U.S. 2017 North American Industry Classification System (NAICS) sectors used in Sections 4-5, I rely on different data sources from the Bureau of Economic Analysis (BEA), otherwise stated.^I For years from 1998 to 2022, these information are:*

- (a) employment (Persons Engaged in Production by Industry, in thousands. Last revised on: September 29, 2023);*
- (b) value-added (Value-Added by Industry, in billions of U.S. dollars. Last revised on: December 19, 2024);*
- (c) aggregate Total Factor Productivity (from Fernald 2014) and Government Consumption Expenditures and Gross Investment (e.g., Ramey and Zubairy 2018);^{II}*
- (d) inter-sectoral trade and Input-Output linkages, in particular:*
 - sectoral share of own production, and thus the sectoral share of intermediate inputs from other sectors (Industry-by-Commodity Market Share Matrix, After Redefinitions – Summary, in producers’ prices);*
 - total inputs required (directly and indirectly) in order to deliver one dollar of output to final users (Commodity-by-Commodity Total Requirements, After Redefinitions – Summary, in producers’ prices);*
 - sector-specific imports and exports of goods and services, government (total, defence, non-defence) expenditure shares, as well as changes in private inventories (The Use of Commodities by Industries, Before Redefinitions – Summary, in producers’ prices and in millions of U.S. dollars).*

Note that “Summary” defines the level of sectoral disaggregation considered in the analysis (3-digit). The BEA also provides information on Input-Output matrices to a more detailed level (6-digit), but these are on a five-yearly basis (thus it would impossible to run the analysis in Subsection 5.1), and data on other variables are not collected. Moreover, the sample begins in 1998 so to have all sectors classified given the NAICS system, and ends in 2022, the last year available when I started this project.^{III} Last time I accessed these public network data was in March 2025.

Turn to the manipulation of Input-Output matrices. To recover the production share of intermediate inputs of a given sector coming from the production of other sectors, each cell

^I Note that I refer to 3-digit but, according to the BEA classification, four sectors are at 2-digit (“construction”, “management of companies and enterprises”, “educational services”, and “other services, except government”), while five sectors related to finance and insurance are at 4-digit level.

^{II} Here is the website for the TFP series. The U.S. government expenditure series is downloaded from the Federal Reserve Economic Data (FRED) database, with reference name *GCEA*; such series, net of federal defence expenditure (named *FDEFX*), represents the non-defence expenditure. Last revised on: June 15, 2026.

^{III} The 3-digit classification is the most granular if one wants to keep track of the evolution in the set of intermediate inputs, and 1998 is the second year in which the NAICS system has been adopted. Before 1997, classification follows the U.S. Standard Industrial Classification (SIC) system.

of the yearly squared matrix contains the intermediate output share from a given sector to another and, on the main diagonal, the share of intermediate output of a given sector directly produced by that sector. Henceforth, labelling by $\mathbf{X} = [x_{ij}]$, with $x_{ij} \geq 0$, this Input-Output matrix I compute, for each sector, $x_i = 1 - x_{ii}$. In other words, to extract the total share of intermediate inputs that sector- i buys from other sectors, I compute $\mathbf{x} = \mathbf{1} - \text{diag}(\mathbf{X})$, where $\mathbf{1}$ is an $m \times 1$ vector of ones; the resulting $m \times 1$ vector $\mathbf{x}$ reports, for every considered year, the total production of a sectoral good which is due to intermediate inputs bought through Input-Output relationships. For what concerns the production network structure, I rely on the main Input-Output matrix (Commodity-by-Commodity Total Requirements, After Redefinitions) which, consistently with the main text, I label as $\mathbf{L} = [\ell_{ij} \geq 0]$.^{IV} Such Leontief inverse matrix investigates its deep characteristics and, following the literature (e.g., Acemoglu et al. 2012; Carvalho 2014) I disregard small transactions across sectors: this allows me to account for a narrower network amplification due to the removal of bias from meaningless inter-sectoral trade intensity. Disregarding small transactions implies that $\ell_{ij} = \frac{\ell_{ij}}{\ell_i}$ is set to 0 if $\ell_{ij} \leq 0.01$, where $\ell_i = \sum_{j \in \text{row}} \ell_{ij}$ is sector- i 's total input purchases (i.e., row sum of matrix $\mathbf{L}$), or $\ell_i = \sum_{j \in \text{column}} \ell_{ij}$ is sector- i 's total input sales (i.e., column sum of matrix $\mathbf{L}$).^V The resulting matrix would therefore display each cell to be greater than 0.01 – i.e., larger than 1%. Leontief inverse matrix, upstream and downstream sectors, and the weight associated to each of them (as in Subsections 5.1-5.2) are identified through it.

To compute network distances between sectors, I rely on the promptly-downloadable BEA Input-Output table (i.e., the Commodity-by-Commodity Total Requirements – After Redefinitions), which reports the total output of each commodity required to satisfy one unit of final demand for any given commodity. By incorporating both direct and indirect production requirements, this table captures Input-Output linkages extending throughout the production network and along the entire supply chain. This representation is particularly suited to my purpose, as it allows me to compare sectors according to common Input-Output structures not only at the local level, through their direct relationships, but also across the full length of their respective supply chains, through the indirect linkages connecting them to further upstream and downstream stages of production. In this way, similarities between sectors reflect commonalities in their production network exposure at different depths of the supply chain, rather than being driven solely by their immediate bilateral connections. Accordingly, the Euclidean distances (eqs. B.1-B.2) between sectors are computed from differences in their total-requirements profiles, capturing the extent to which sectors share similar buyers/suppliers across the production network. From these distances, I compute the intensive margin values for both network “economic” distance metrics.

(Shortest path algorithm, technical but suitable for coding) Given element $\mathbf{G}$ being an $m \times m$ adjacency matrix with non-negative discrete values, the idea is to compute a distance matrix

^{IV} Row sectors identify (upstream) suppliers, while column sectors the (downstream) buyers: row sector is the origin and column sector is the destination of the circulating intermediate input.

^V The “sales-based” Leontief inverse (in the main text) is a normalization of the baseline Input-Output matrix by sectoral total sales, so that each entry represents the share of a sector’s inputs relative to its output.

$\mathcal{D}$ where each cell $d[i, j]$ is the length of the shortest path between node- i and node- j . The numerical algorithm is as follows:

- (a) from the Input-Output matrix $\mathbf{A}$, build an associated identity matrix, $\mathbf{I}$;
- (b) set the initial length of the path, $k = 1$, and build a length matrix $\mathcal{G}_k = \mathcal{G}$;
- (c) build a Boolean matrix, $\mathbf{B}$, whose elements are $b_{ij} = 1$ if $\mathcal{G}_k \neq 0$;
- (d) as long as matrix $\mathbf{B}$ contains a value different from zero (i.e., there are connections among nodes not considered):
 - update identity matrix $\mathbf{I}$ by adding k whenever the condition on $\mathbf{B}$ is true, that is, $I_{ij} = I_{ij} + k$ if $b_{ij} = 1$;
 - update lengths, $k = k + 1$;
 - algebraic matrix calculation: $\mathcal{G}_k = \mathcal{G}'_k \mathcal{G}$;
 - update $\mathbf{B}$ with new paths: set $b_{ij} = 1$ if $\mathcal{G}_{ij,k} \neq 0$ and $d[i, j] = 0$;
- (e) once this loop concludes, set to infinite all the disconnected nodes, that is $d_{ij} = \infty$ if $d[i, j] = 0$, to check consistency;
- (g) set to the minimum distance the elements on the main diagonal, $\mathcal{D} = \mathcal{D} - \mathbf{I}$.

Outside this routine set to zero all the disconnected nodes (those at ∞). Matrix $\mathcal{D}_{\text{ext}}$, whose values are at the extensive margin and identify network distances among any pair $\{i, j\}$, is built. The generic element identifying the existence of a link between node- i and node- j in the distance matrix $\mathcal{D}$ is then $d_{\text{ext}}[i, j] = \{1, 2, \dots, d_{\text{max}}\}$.

(Characteristics of the distance networks) . To illustrate the constructed distance matrices Figure B.1 displays, for each extensive margin value, the horizontal linkages among sectors from both common demand and supply network ties. Importantly, each connection does not necessarily represent inter-sectoral trade between two sectors, but rather it reflects an horizontal linkage based on demand- or supply-driven network distance relationships.

A few concatenated points are then worth noting. First, the distribution of horizontal complementarities is uneven across distance levels. At very short range ($d = 1$), the demand-based network appears slightly denser than the supply-based one, suggesting that sectors are more frequently linked through common upstream suppliers than through shared downstream buyers. However, this asymmetry vanishes as the distance increases ($d = 2$), where both types of horizontal ties spread more diffusely across the network. Second, sectors with the highest number of Input-Output linkages tend to concentrate at the very core of the production system when considering only close connections ($d = 1$). As shown in Panels B.1a and B.1b, both the largest hubs (top 10% by connectivity) and the mid-sized hubs (top 20%) are similarly central, occupying short distances from the network's nucleus. Third, these same highly connected sectors tend to become relatively more peripheral once longer distances ($d = 2$) are considered. Panels B.1c and B.1d show that, at this range, their positions are more evenly distributed across the network, reflecting the fact that their trade structures – both upstream and downstream – are

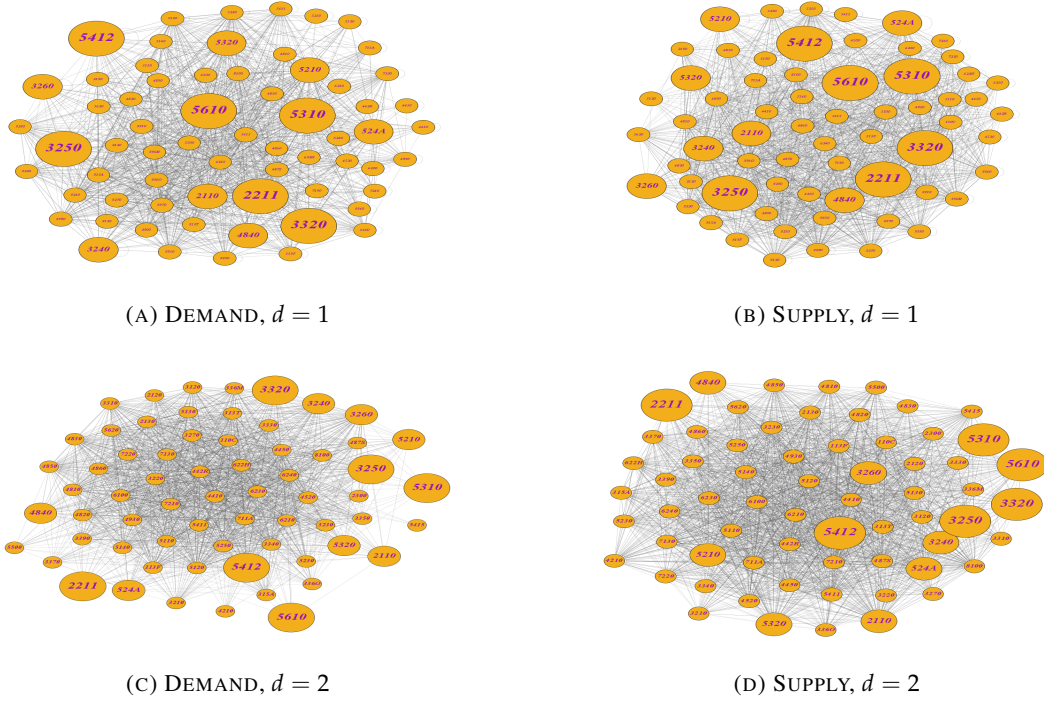


FIGURE B.1: VISUALIZATION OF PRODUCTION NETWORK DISTANCES

Note: each panel of the figure represents the distance-based inter-sectoral production network corresponding to the (*commodity-by-commodity total direct requirements*) U.S. Input-Output matrix in the year 2007. *Factor input demand* and *factor input supply* network distances are computed implementing the *shortest path algorithm* on distance matrices whose values are at the intensive margin, as well derived from the *directed* configuration of the production network. Panels B.1a and B.1c correspond to the demand-based network distances, with $d = 1$ for low distance (closest sectors) and $d = 2$ for major distance (further sectors), among any pair of sectors; analogously it is for the supply-based network distances in Panels B.1b and B.1d. There are 65 3-digit U.S. 2017 NAICS private sectors, and each orange node corresponds to one of them. Two sectors connected under $d = 1$ cannot be linked under $d = 2$, and otherwise; constructed distance matrices are full (Theorem 2). Biggest nodes correspond to top 10% of mostly connected sectors (*i.e.*, major number of inter-sectoral connections), while intermediate ones the additional top 20%. Figure drawn with the software package *Gephi*, version 0.10, exploiting the *ForceAtlas2* layout algorithm. *Source:* BEA and own calculations.

highly similar to those of other sectors. In other words, most connected sectors are less unique at longer distances, as commonalities in demand and supply linkages become more widespread.

Altogether, these observations highlight that the structure of horizontal complementarities depend crucially on the notion of distance. When short, horizontal linkages cluster around the key sectors whereas, at longer distances, such complementarities are more evenly distributed. This pattern confirms that sectors typically share both upstream suppliers and downstream buyers at smaller but not at a greater distance, and the same vertical production system generates distinct horizontal geometries (Theorem 2).

TABLE B.1: LIST OF CONSIDERED SECTORS AND THEIR CODES

<i>denomination</i>	<i>codes</i>	
	<i>NAICS 2017</i>	<i>BEA</i>
<i>Farms</i>	111, 112	111CA
<i>Forestry, Fishing and Related Activities</i>	113, 114, 115	113FF
<i>Oil and Gas Extraction</i>	211	211
<i>Mining, except Oil and Gas</i>	212	212
<i>Support Activities for Mining</i>	213	213
<i>Utilities</i>	22	22
<i>Construction</i>	23	23
<i>Wood Products</i>	321	321
<i>Nonmetallic Mineral Products</i>	327	327
<i>Primary Metals</i>	331	331
<i>Fabricated Metal Products</i>	332	332
<i>Machinery</i>	333	333
<i>Computer and Electronic Products</i>	334	334
<i>Electrical Equipment, Appliances, and Components</i>	335	335
<i>Motor Vehicle, Bodies and Trailers, and Parts</i>	3361-3	3361MV
<i>Other Transportation Equipment</i>	3364-9	3364OT
<i>Furniture and Related Products</i>	337	337
<i>Miscellaneous Manufacturing</i>	339	339
<i>Food and Beverage and Tobacco Products</i>	311, 312	311FT
<i>Textile Mills and Textile Product Mills</i>	313, 314	313TT
<i>Apparel and Leather and Allied Products</i>	315, 316	315AL
<i>Paper Products</i>	322	322
<i>Printing and Related Support Activities</i>	323	323
<i>Petroleum and Coal Products</i>	324	324
<i>Chemical Products</i>	325	325
<i>Plastics and Rubber Products</i>	326	326
<i>Wholesale Trade</i>	42	42
<i>Motor Vehicle and Parts Dealers</i>	441	441
<i>Food and Beverage Stores</i>	445	445
<i>General Merchandise Stores</i>	452	452
<i>Other Retail</i>	442-4, 446-8, 45 ex.	452 4AO
<i>Air Transportation</i>	481	481
<i>Rail Transportation</i>	482	482
<i>Water Transportation</i>	483	483
<i>Truck Transportation</i>	484	484
<i>Transit and Ground Passenger Transportation</i>	485	485
...		

Sectors are defined according to the nomenclature used by the U.S. Bureau of Economic Analysis (BEA).

TABLE B.1: LIST OF CONSIDERED SECTORS AND THEIR CODES (*continued*)

<i>denomination</i>	<i>codes</i>	
	<i>NAICS 2017</i>	<i>BEA</i>
...		
<i>Pipeline Transportation</i>	486	486
<i>Other Transportation and Support Activities</i>	487, 488, 492	487OS
<i>Warehousing and Storage</i>	493	493
<i>Publishing Industries, except Internet (includes Software)</i>	511	511
<i>Motion Picture and Sound Recording industries</i>	512	512
<i>Broadcasting and Telecommunications</i>	515, 517	513
<i>Data Processing, Internet Publishing, and Other Information Service</i>	518, 519	514
<i>Federal Reserve Banks, Credit Intermediation, and Related Activities</i>	521, 5221, 5222, 5223	521CI
<i>Securities, Commodity Contracts, and Investments</i>	523	523
<i>Insurance Carriers and Related Activities</i>	5241, 5242	524
<i>Funds, Trusts, and Other Financial Vehicles</i>	525	525
<i>Housing</i> (not considered in the analysis)	.	HS
<i>Other Real Estate</i>	531	ORE
<i>Rental and Leasing Services and Lessors of Intangible Assets</i>	532, 533	532RL
<i>Legal Services</i>	5411	5411
<i>Computer Systems Design and Related Services</i>	5415	5415
<i>Miscellaneous Professional, Scientific, and Technical Services</i>	541 ex. 5411, 5415	5412OP
<i>Management of Companies and Enterprises</i>	55	55
<i>Administrative and Support Services</i>	561	561
<i>Waste Management and Remediation Services</i>	562	562
<i>Educational Services</i>	61	61
<i>Ambulatory Health Care Services</i>	621	621
<i>Hospitals</i>	622	622
<i>Nursing and Residential Care Facilities</i>	623	623
<i>Social Assistance</i>	624	624
<i>Performing Arts, Spectator Sports, Museums, and Related Activities</i>	711, 712	711AS
<i>Amusement, Gambling, and Recreation industries</i>	713	713
<i>Accommodation</i>	721	721
<i>Food Services and Drinking Places</i>	722	722
<i>Other Services, except Government</i>	81	81

Sectors are defined according to the nomenclature used by the U.S. Bureau of Economic Analysis (BEA).

(Network distances at the intensive margin) Section 4 of the main text characterizes distance matrices at their “extensive margin”, $\mathbf{D}_{ext}^{fd}$ and $\mathbf{D}_{ext}^{fs}$, whose values are $d_{ext}^q[i, j] = \{1, 2, \dots, d_{max}\}$ for $q = \{fd, fs\}$, exploiting the shortest path algorithm on the “intensive margin” distances from the Leontief inverse production network, $\mathbf{L}$, that accounts for direct and indirect sectoral connections.

To characterize the “intensive margin” matrices $\mathbf{D}^{fd}$ and $\mathbf{D}^{fs}$ I exploit the system of Cartesian coordinates typical of any production network (e.g., Conley and Dupor 2003). Given $\alpha_{ij}^{fd} = \frac{\alpha_{ij}}{\sum_{j \in row} \alpha_{ij}}$, the generic element of factor input demand distance matrix is computed as

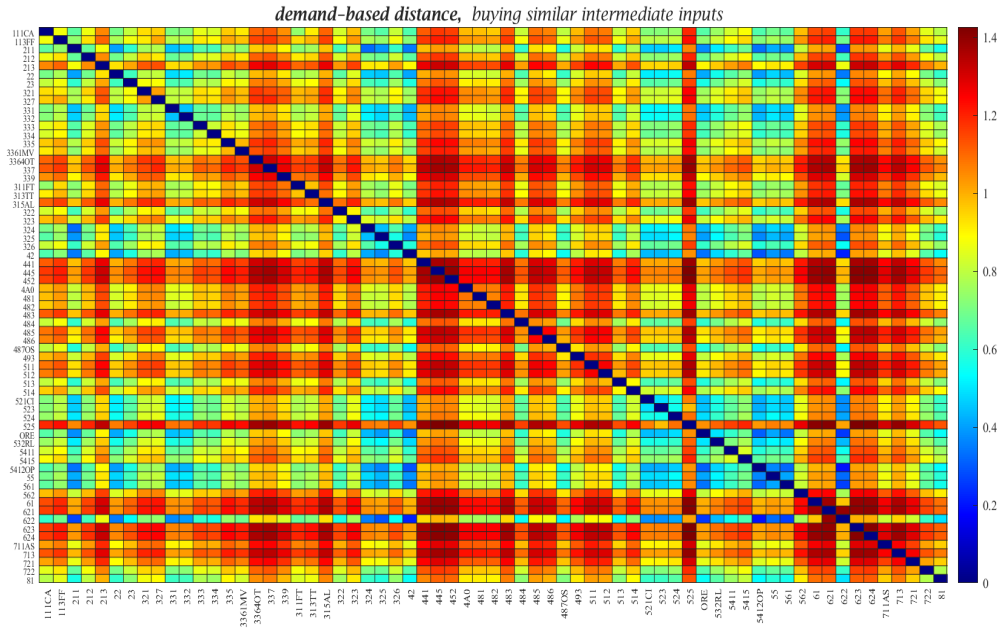
$$d^{fd}[i, j] = \left\{ \sum_{s \in \Phi} [\alpha_{is}^{fd} - \alpha_{js}^{fd}]^2 \right\}^{\frac{1}{2}} \quad (\text{B.1})$$

which defines the length of the line segment (i.e., the Euclidean distance) connecting sectors i and j when both are buying from the same other sector- s , $\forall s \in \Phi$, so that $\mathbf{D}^{fd} = [d^{fd}[i, j]]$ with generic element $d^{fd}[i, j] > 0$. Demand distance is constructed by dividing each cell by its row sum. Why? A row sum represents the total sales of the sector in question, so dividing each entry by its associated row sum yields the share of that (row) sector’s output purchased by each of the column sectors. This normalization allows for a meaningful comparison across dyads of sectors: by examining these shares, one can assess the relative distance between two (column) sectors in terms of their purchases from common upstream suppliers, thereby pointing to the strength of the upstream pass-through (share of total sales of the common sector).

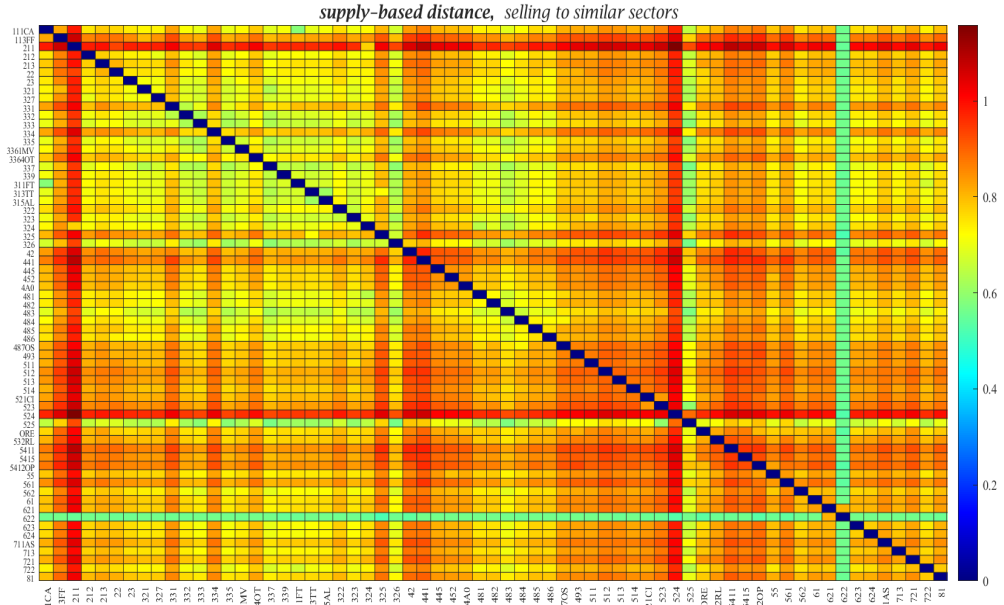
In an analogous way, given $\alpha_{ij}^{fs} = \frac{\alpha_{ij}}{\sum_{j \in column} \alpha_{ij}}$, the generic element of factor input supply distance matrix is shaped by

$$d^{fs}[i, j] = \left\{ \sum_{s \in \Phi} [\alpha_{si}^{fs} - \alpha_{sj}^{fs}]^2 \right\}^{\frac{1}{2}} \quad (\text{B.2})$$

tracing a Euclidean distance between sectors $\{i, j\}$ when both are selling to the same sector- s , $\forall s \in \Phi$, so that $\mathbf{D}^{fs} = [d^{fs}[i, j]]$ with $d^{fs}[i, j] > 0$. Supply distance is constructed by dividing each cell by its column sum. Why? A column sum represents the total purchases of the sector in question, so dividing each entry by its associated column sum yields the share of that (column) sector’s input purchased from each of the row sectors. This normalization allows for a meaningful comparison across dyads of sectors: by examining these shares, one can assess the relative distance between two (row) sectors in terms of their traded quantities to common downstream buyers, thereby pointing to the strength of the downstream pass-through (share of total purchases of the common sector).



(A) FACTOR INPUT DEMAND



(B) FACTOR INPUT SUPPLY

FIGURE B.2: INTENSIVE MARGIN MATRICES FOR NETWORK ECONOMIC DISTANCE

Note: heatmaps of sectoral distances characterizing the U.S. production network in year 2007 for 3-digit U.S. 2017 NAICS sectors. Entry (i, j) represents the intensive marginal value, $d^q[i, j] \geq 0$ for $q = \{fd, fs\}$, of the horizontal relationship between sector- i and sector- j under *factor input demand* network distance (i.e., demand linkages across sectors given their common upstream sellers, Panel B.2a) and *factor input supply* network distance (i.e., supply linkages across sectors given their common downstream buyers, Panel B.2b). Diagonal sparsity (i.e., lots of 0s), is a direct consequence of the fact that the difference of inter-sectoral trade intensities between a sector and itself is zero. Distance matrices are at the *intensive margin*, as in eqs. (B.1)-(B.2), computed from the *directed* sectoral production network. *Source:* BEA and own calculations.

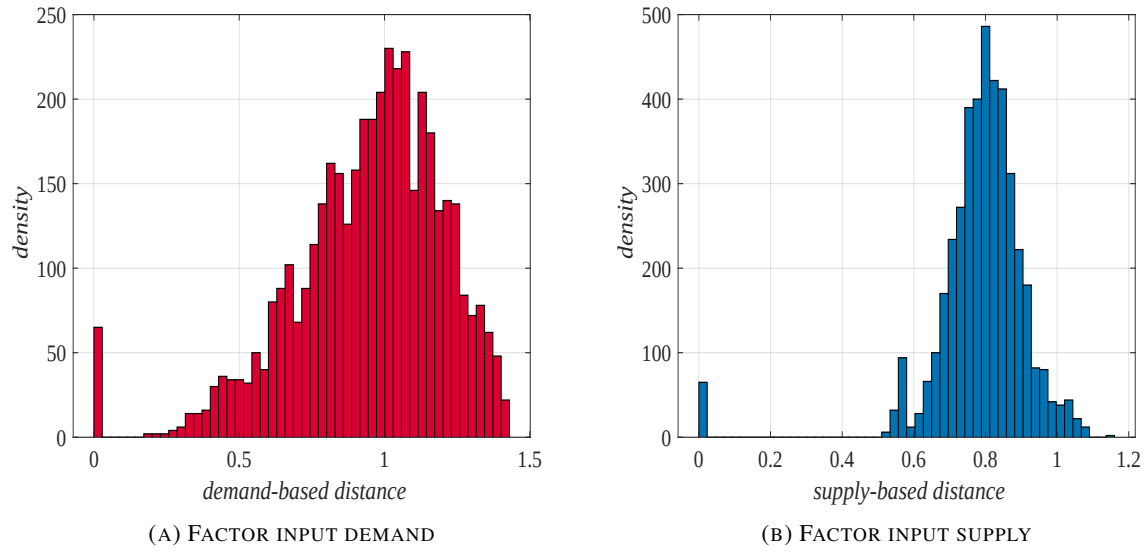


FIGURE B.3: DISTRIBUTION OF NETWORK ECONOMIC DISTANCES

Note: these figures depict the distributions for *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers, Panel B.3a) and *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, Panel B.3b). Distance values on the *x*-axis are at the *intensive margin*, as in eqs. (B.1)-(B.2), computed from the *directed* sectoral production network *Source:* BEA and own calculations.

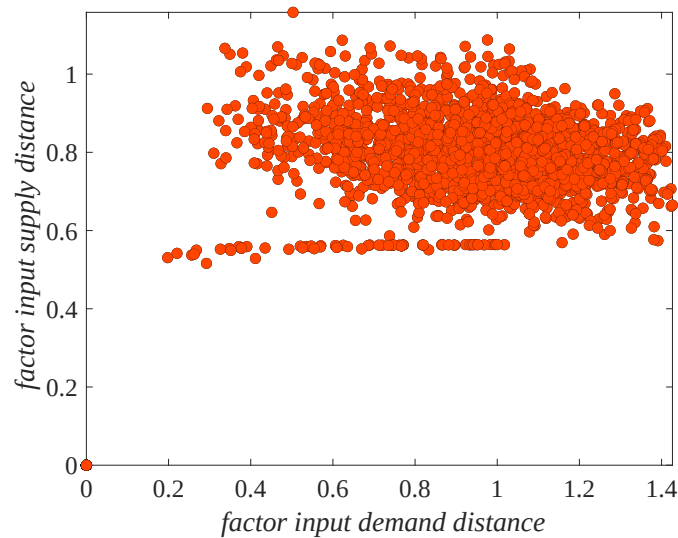
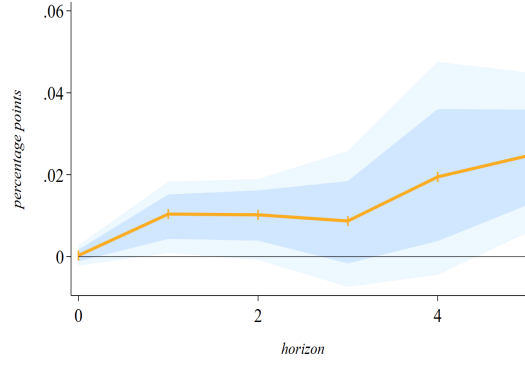


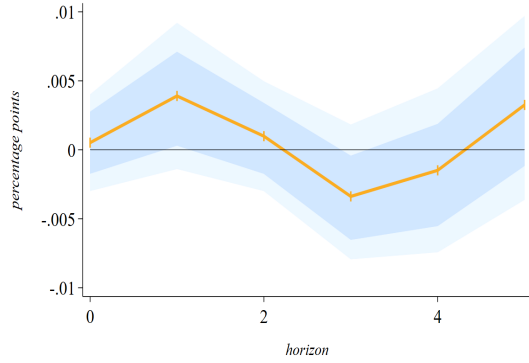
FIGURE B.4: CORRELATION OF NETWORK ECONOMIC DISTANCES

Note: the figure shows the correlation between the measures for *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers, *x*-axis) and *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, *y*-axis). Distances are at the *intensive margin*, as in eqs. (B.1)-(B.2), computed from the *directed* sectoral production network *Source:* BEA and own calculations.

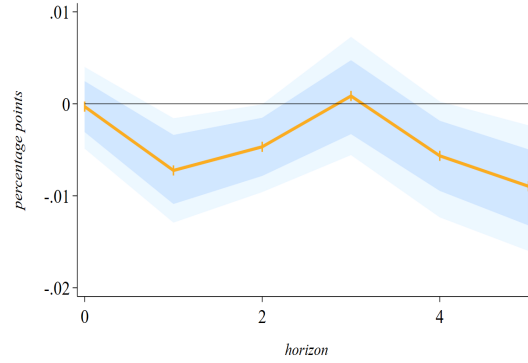
C. COMPLEMENTARY RESULTS



(A) SECTOR-SPECIFIC



(B) $d = 1$



(C) $d = 2$

FIGURE C.1: SECTORAL EMPLOYMENT RESPONSE TO INTERMEDIATE INPUTS

Note: given the *factor input demand* network distance (i.e., demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (i.e., sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the sector-specific set of intermediate inputs as a share of its value-added for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. In particular, Panel C.1a represents the sector-level employment response to changes in its intermediate inputs, while Panel C.1b-C.1c to changes in the set of intermediates in closer (distance equal to 1) and further (distance equal to 2) sectors. The solid-orange line corresponds to the average response of employment across sectors, while shadow-blue and shadow-light blue areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eqs. (7)-(8). *Source:* BEA and own calculations.

(Plotting separated results for highly interlinked sectors) Unpacking all the combined plots, below I report the estimated Local Projection (LP) dynamics for top 5%, top 10%, top 20%, and top 30% of more connected sectors in the production network (i.e., major number of Input-Output linkages) separately.¹ The purpose is to show the significance of the responses appearing in Panels 11a-11b of Figure 11 in Subsection 5.2 of the main text, and in Panels 9a-9b of Figure 9 in Appendix C.

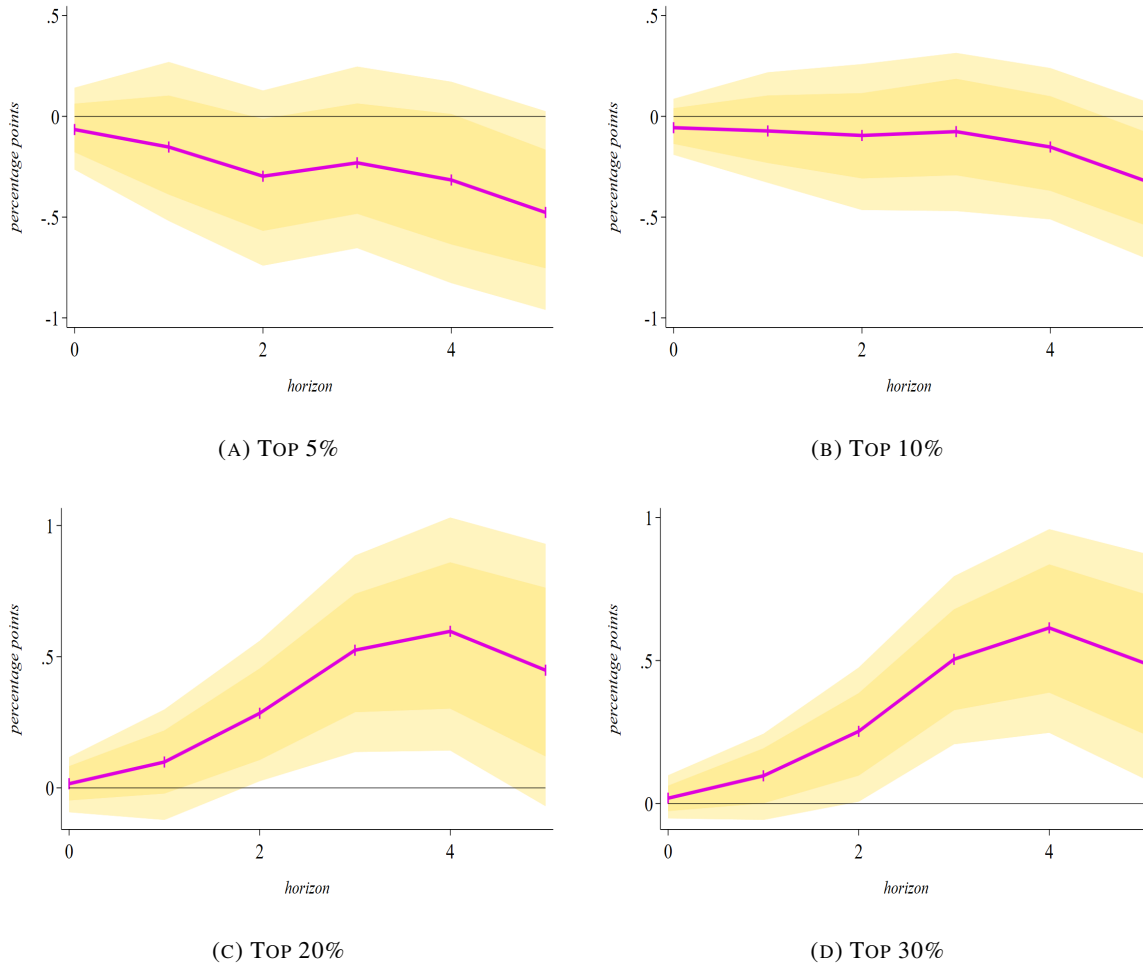


FIGURE C.2: DEMAND ($d = 1$) EFFECTS FOR INTERLINKED SECTORS

Note: given the factor input demand network distance (i.e., demand linkages across sectors given their common upstream sellers) and the Leontief inverse transmission (i.e., sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 9a of Figure 9, each panel of this figure shows the employment response of top 5% (Panel C.2a), 10% (Panel C.2b), 20% (Panel C.2c), and 30% (Panel C.2d) more connected sectors in the production network (i.e., major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (10). *Source:* BEA and own calculations.

¹ Besides my network-based effects outlined in the main text, further theories beyond employment comovement are in Rogerson (1987) and Boldrin et al. (2001).

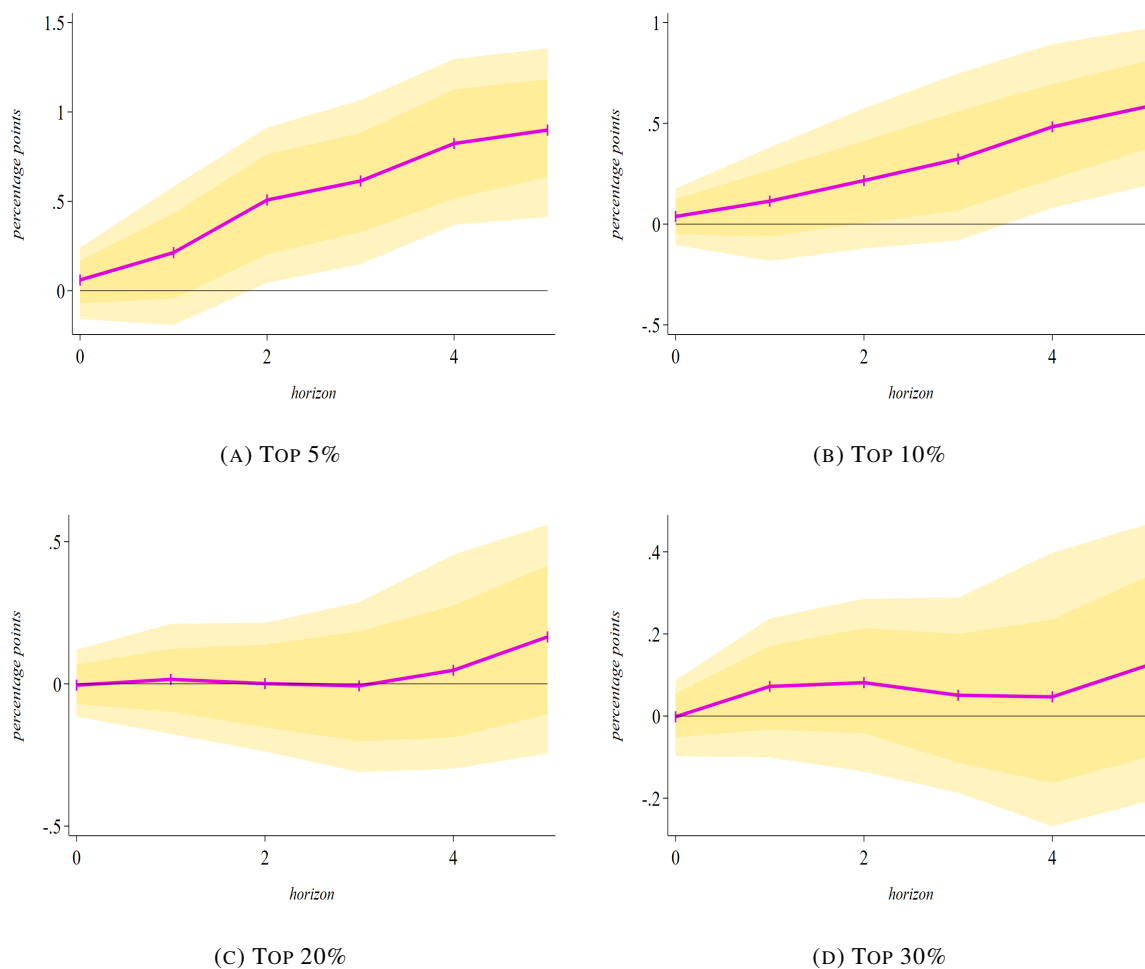


FIGURE C.3: DEMAND ($d = 2$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 9b of Figure 9, each panel of this figure shows the employment response of top 20% (Panel C.3c) and 30% (Panel C.3d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) to employment changes in their further (distance equal to 2) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (10). *Source:* BEA and own calculations.

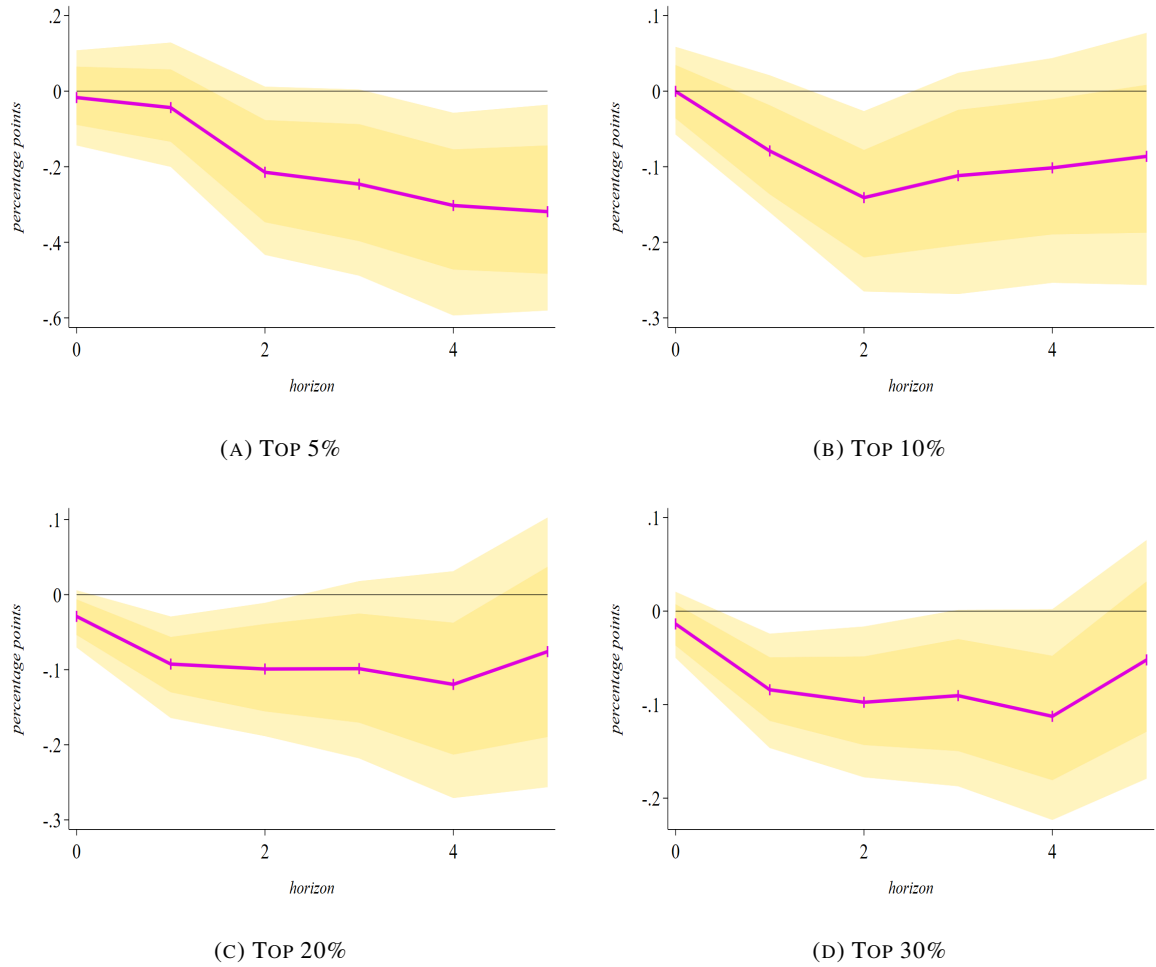


FIGURE C.4: SUPPLY ($d = 1$) EFFECTS FOR INTERLINKED SECTORS

Note: given the factor input supply network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the Leontief inverse transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 11a of Figure 11, each panel of this figure shows the employment response of top 5% (Panel C.4a), 10% (Panel C.4b), 20% (Panel C.4c), and 30% (Panel C.4d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (10). *Source:* BEA and own calculations.

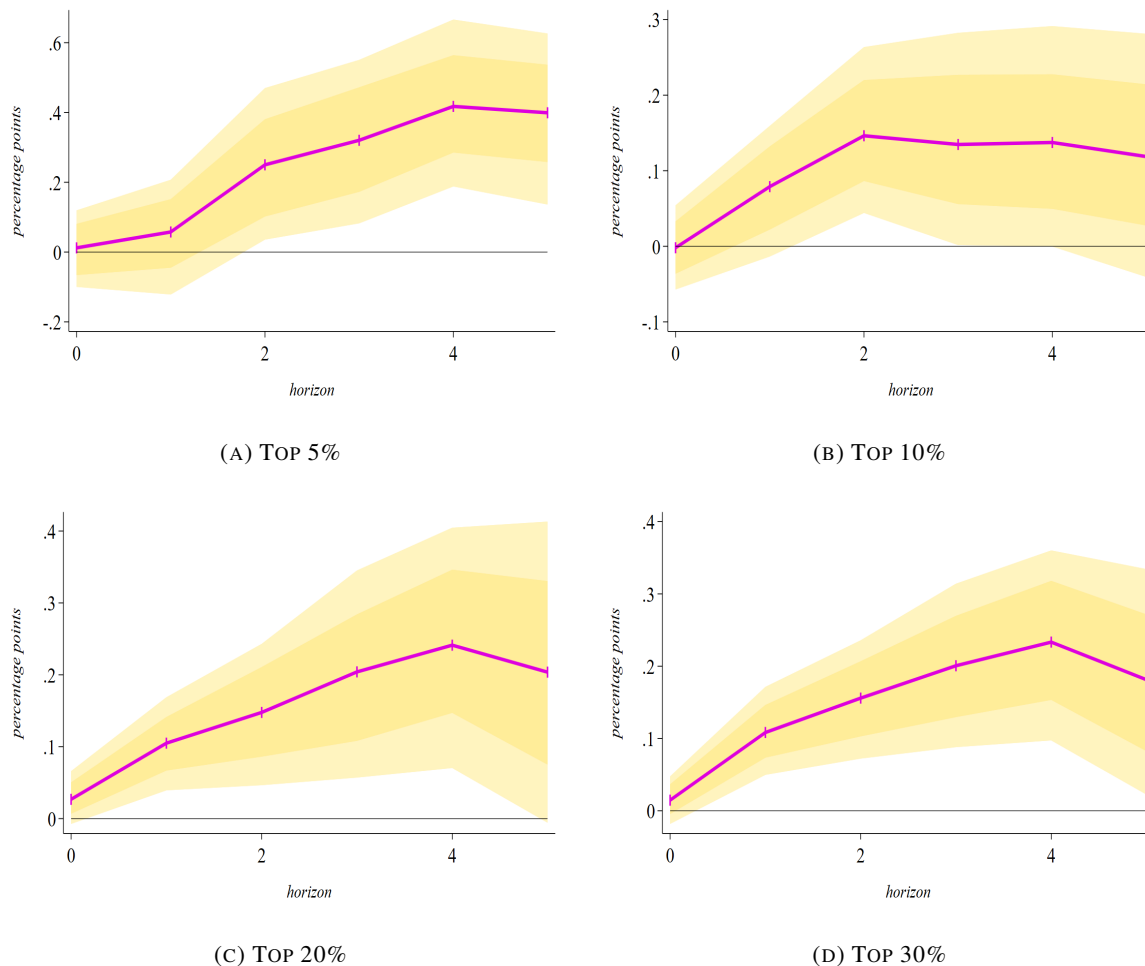


FIGURE C.5: SUPPLY ($d = 2$) EFFECTS FOR INTERLINKED SECTORS

Note: given the factor input supply network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the Leontief inverse transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 11b of Figure 11, each panel of this figure shows the employment response of top 5% (Panel C.5a), 10% (Panel C.5b), 20% (Panel C.5c), and 30% (Panel C.5d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their further (distance equal to 2) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (10). *Source:* BEA and own calculations.

(Robustness with centrality measures) To identify the centrality degree of each sector, I follow the approach outlined in Carvalho (2014) to measure the sector-level Bonacich-Katz centrality.^{II} Using the Leontief-inverse matrix $\mathcal{L}$ for the U.S. production network as of 2007, the Bonacich-Katz centrality $m \times 1$ vector, $\mathbf{bk}$, is defined as:

$$\mathbf{bk} = \sum_{k \geq 0} \zeta^k \mathcal{L}^k \mathbf{1} = \left(\mathbf{I} - \zeta \mathcal{L} \right)^{-1} \mathbf{1} \quad (\text{C.1})$$

where index- k refers to the length of the paths in the production network – that is, the number of steps (or network “hops”) connecting one sector to another – and, given a finite set of sectors $i, j = \{1, \dots, m\} \in \Phi$:

- $\mathbf{1}$ is an $m \times 1$ vector of ones, reflecting the base influence;
- $\zeta \in \left(0, \frac{1}{v_{\max}}\right)$ is a dampening scalar parameter, with $v_{\max}$ being the largest eigenvalue of matrix $\mathcal{L}$;
- $\mathbf{I}$ is the $m \times m$ identity matrix.

The value of ζ determines the weight assigned to indirect linkages: smaller values place more emphasis on direct connections, while larger values account for deeper propagation within the network. Following Carvalho (2014), a value of $\zeta = 0.5$ is typically used, which lies safely below the inverse of the largest eigenvalue of most empirically observed Input-Output matrices, ensuring convergence of the series. This measure reflects the idea that a sector is central not only if it is directly connected to many others, but also if it is connected to sectors that are themselves highly central.

This formulation ensures that the centrality score of each sector captures recursive amplification along both its direct connections and its indirect influence through other sectors due to the use of matrix $\mathcal{L} = (\mathbf{I} - \mathbf{A})^{-1} = [\ell_{ij} \geq 0]$, where $\mathbf{A} = [\alpha_{ij} \geq 0]$ is the directed production network matrix (i.e., that in the BEA Input-Output tables considering direct and indirect connections but not amplification), defining the intensity of good produced by sector- j in the total intermediate inputs used by sector- i , with $\alpha_{ij} = 0$ indicating that sector- i does not make use of the good produced by sector- j in producing its own intermediate good.

^{II} Outlined by Bonacich (1987), it is a measure of a sector’s overall importance within an Input-Output system, capturing not only its direct connections to other sectors but also the centrality of those it is connected to. In essence, a sector is considered central if it is linked to other central sectors, creating a recursive structure of influence. This measure goes beyond simple counts of linkages by incorporating indirect connections – weighted by a dampening factor – to reflect the diminishing influence of more distant sectors in the network. In the context of production network, a sector with high Bonacich-Katz centrality is one that plays a key role in the propagation of shocks, as its position allows it to influence (or be influenced by) large portions of the economy through both direct and indirect channels. Compared to other centrality measures, Bonacich-Katz is particularly well-suited for economic applications, as it accounts for the intensity of inter-sectoral relationships and the structure of the network in a realistic and nuanced way.

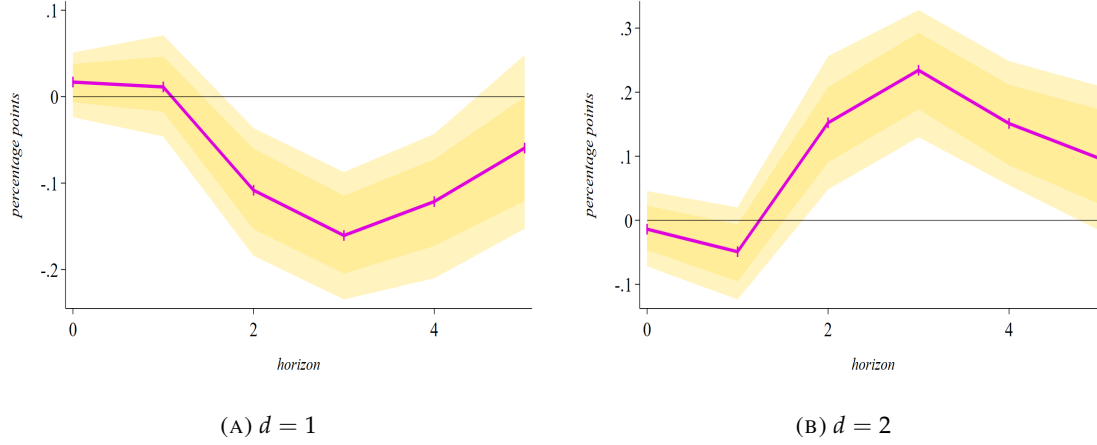


FIGURE C.6: DEMAND LINKAGES AND COMOVEMENT IN PERIPHERAL SECTORS

Note: given the *factor input demand* network distance (i.e., demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (i.e., sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of most central sectors (with *centrality* defined according to sector's relevance in final consumption) to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel C.6a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel C.6b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (10). *Source:* BEA and own calculations.

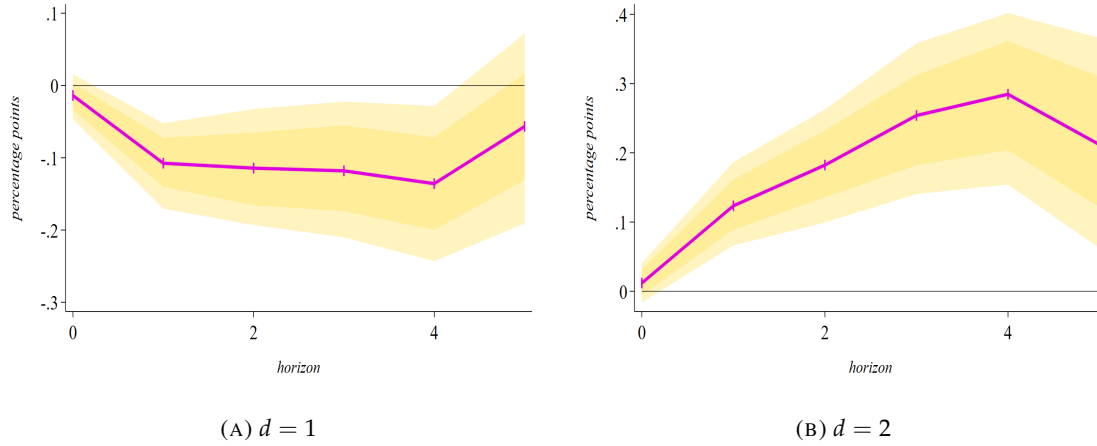


FIGURE C.7: SUPPLY LINKAGES AND COMOVEMENT IN PERIPHERAL SECTORS

Note: given the *factor input supply* network distance (i.e., supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (i.e., sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of most central sectors (with *centrality* defined according to sector's relevance in final consumption) to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel C.7a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel C.7b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (10). *Source:* BEA and own calculations.

(Robustness with private inventories) Responses for comovement in sectoral employment levels when controlling for changes in sectoral inventories, distinguishing further between sectors with inventories and those without. The results are shown in Figure C.8. Note that the plots refer dynamic changes in inventories and the Input-Output matrix held at the baseline year 2007.

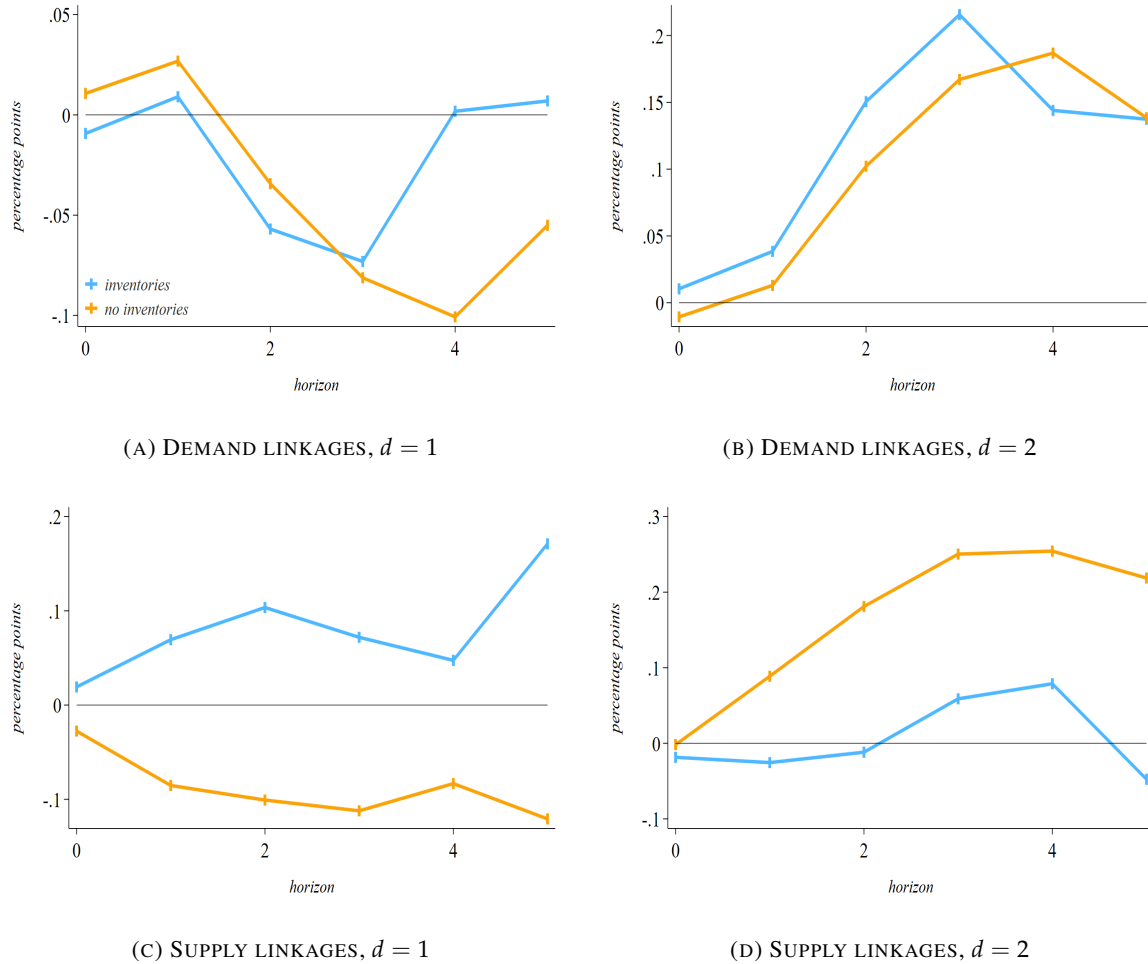


FIGURE C.8: COMOVEMENT UNDER CHANGES IN SECTORAL INVENTORIES

Note: given both the *factor input demand* (i.e., demand linkages across sectors given their common upstream sellers) and the *factor input supply* (i.e., supply linkages across sectors given their common downstream buyers) network distances, and the *Leontief inverse* transmission (i.e., sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors to a 1% increase in the employment level of other sectors when controlling for the changes in private inventories, for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Solid lines corresponds to the average response of employment across sectors, given 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (10). Responses are referred to employment changes in closer (distance equal to 1) and further (distance equal to 2) sectors. Since not all sectors have inventories, the panels are disentangled by subgroups that make use or not of private inventories. *Source:* BEA and own calculations.

(Robustness with production network at different base years) Responses for comovement in sectoral employment levels in the main analysis of Section 5 are centred on the implementation of a sectoral production network represented by an Input-Output matrix in the baseline year 2007. Differently, Figures C.9, C.10 and C.11 perform again all the main analysis by referring, instead, to a production network structure five years before (2002) and five years after (2012) the baseline year.

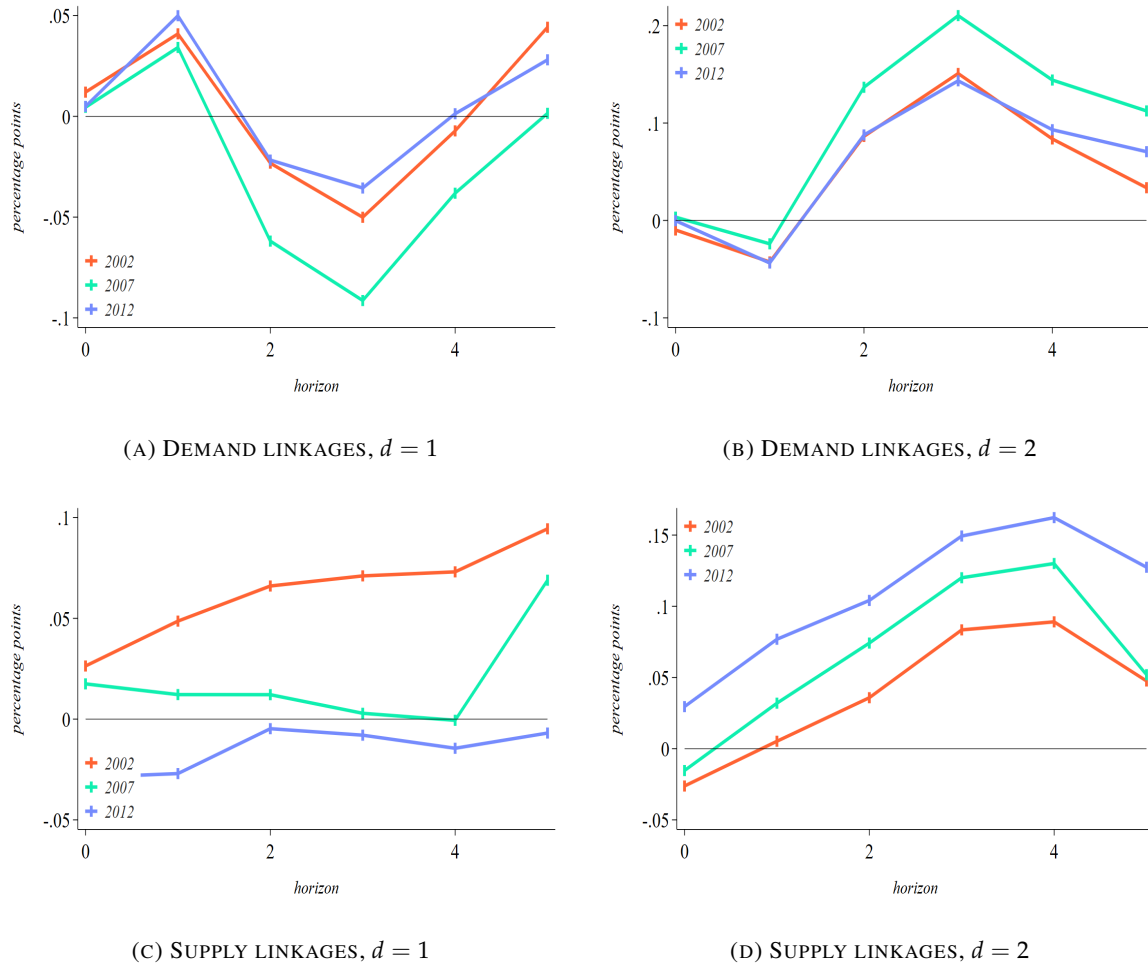


FIGURE C.9: COMOVEMENT OVER DIFFERENT BASELINE YEARS

Note: given both the *factor input demand* (i.e., demand linkages across sectors given their common upstream sellers) and the *factor input supply* (i.e., supply linkages across sectors given their common downstream buyers) network distances, and the *Leontief inverse* transmission (i.e., sectoral direct and indirect network exposure) characterizing the North American production network over different years (2002, 2007, and 2012), the figure shows the response of sectoral employment of sectors to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the results in the main text: Panels C.9a-C.9b expand their relatives of Figure 8, while Panels C.9c-C.9d do so for their relatives in Figure 10. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (10), roughly correspond to the associated figures in the main text. Responses are referred to employment changes in closer (distance equal to 1) and further (distance equal to 2) sectors. *Source:* BEA and own calculations.

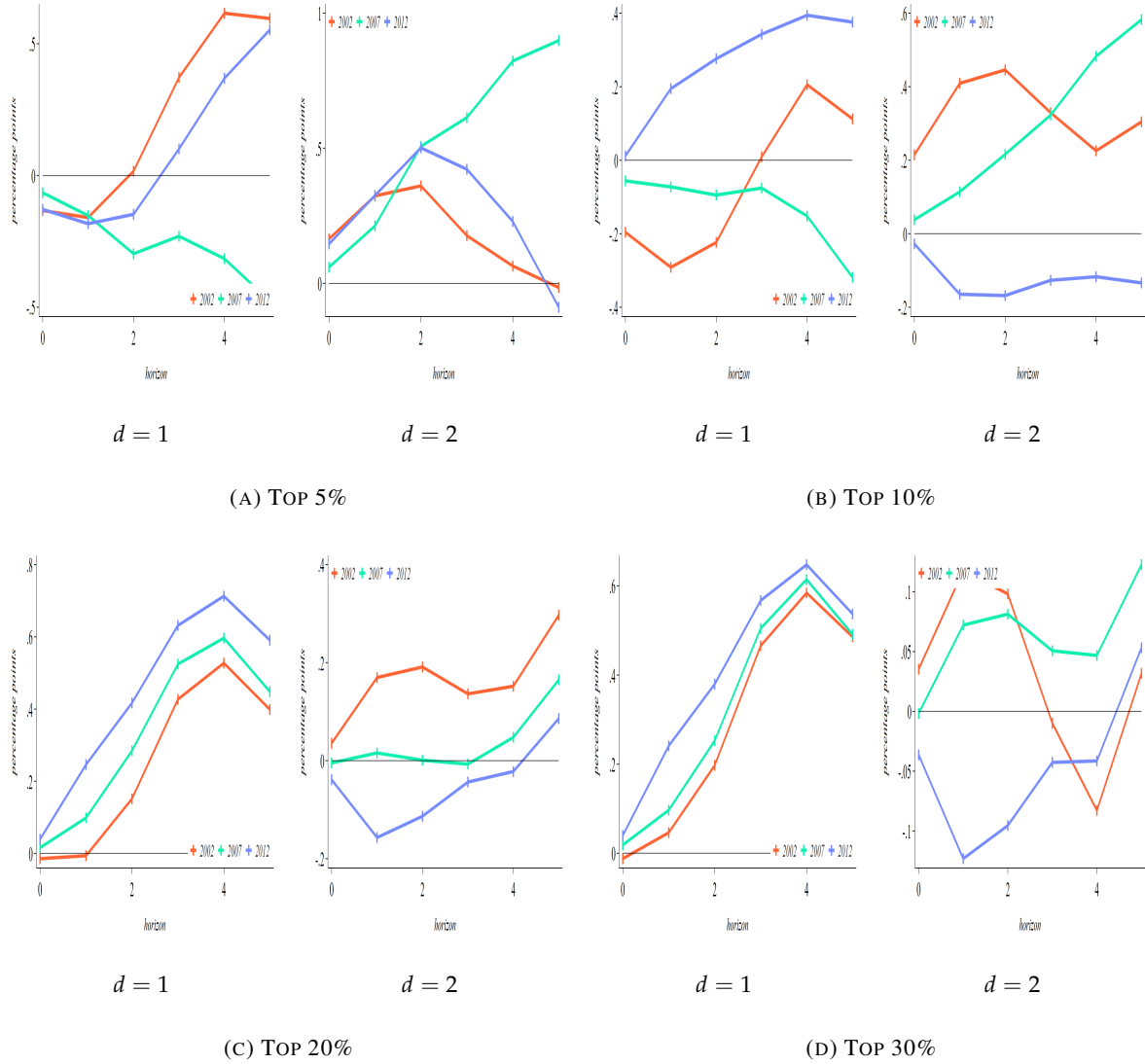


FIGURE C.10: DEMAND LINKAGES AND INTERLINKED SECTORS OVER THE YEARS

Note: given both the *factor input demand* network distance (i.e., demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (i.e., sectoral direct and indirect network exposure) characterizing the North American production network over different years (2002, 2007, and 2012), the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the responses of Panels 9a-9b of Figure 9, each panel of this figure shows the employment response of top 5% (Panel C.10a), 10% (Panel C.10b), 20% (Panel C.10c), and 30% (Panel C.10d) more connected sectors in the production network (i.e., major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) or further (distance equal to 2) sectors. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (10), roughly correspond to the associated figure. *Source:* BEA and own calculations.

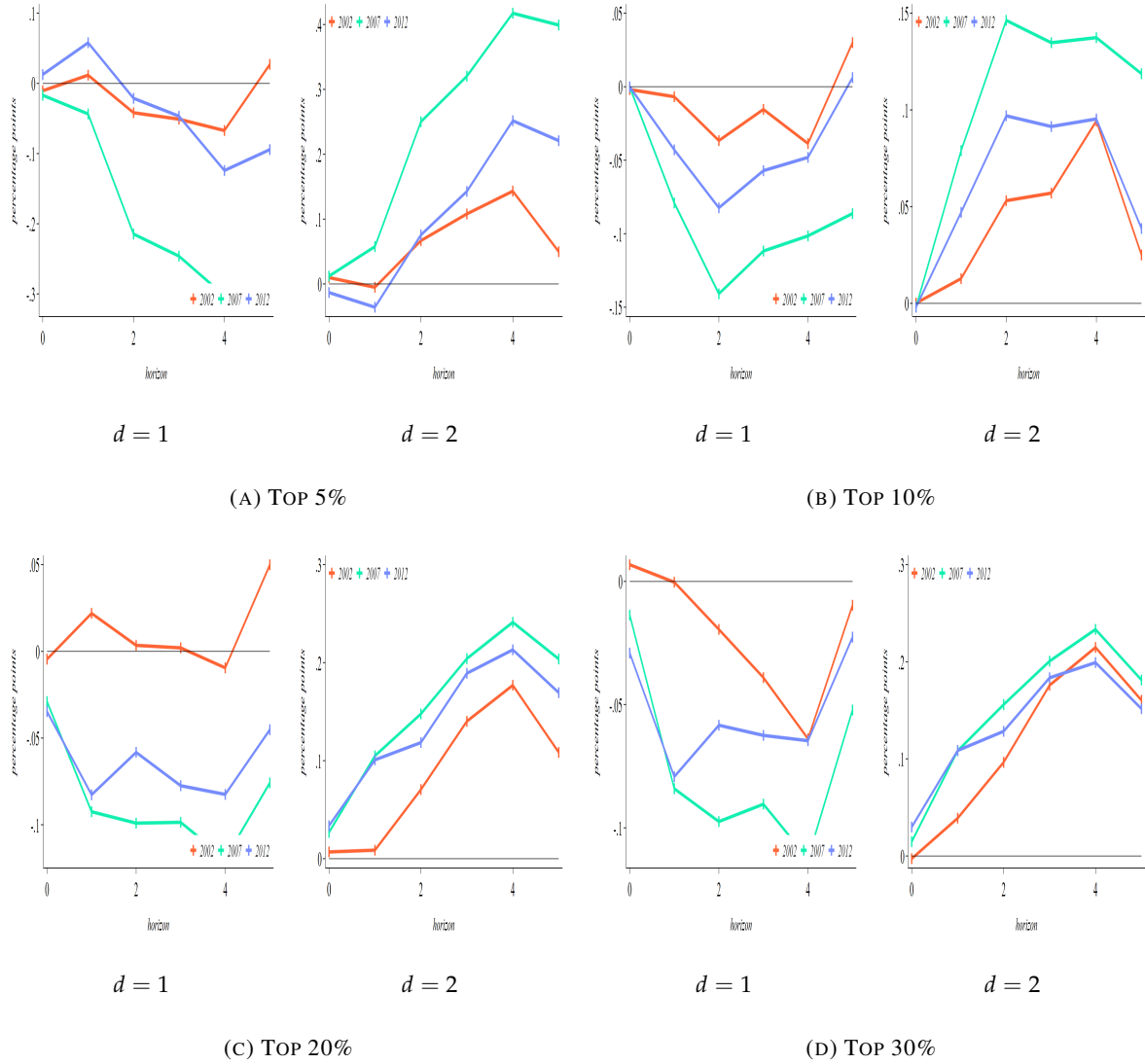


FIGURE C.11: SUPPLY LINKAGES AND INTERLINKED SECTORS OVER THE YEARS

Note: given both the *factor input supply* network distance (i.e., supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (i.e., sectoral direct and indirect network exposure) characterizing the North American production network over different years (2002, 2007, and 2012), the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the responses of Panels 11a-11b of Figure 11, each panel of this figure shows the employment response of top 5% (Panel C.11a), 10% (Panel C.11b), 20% (Panel C.11c), and 30% (Panel C.11d) more connected sectors in the production network (i.e., major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) or further (distance equal to 2) sectors. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (10), roughly correspond to the associated figure. *Source:* BEA and own calculations.

(Illustrative episodes of horizontal transmission) *In conclusion, a set of recent episodes that exemplify the formalized mechanisms of horizontal shock transmission are presented. Firms/sectors not directly linked through bilateral Input-Output transactions nonetheless exhibit simultaneous outcomes because they share a common downstream buyer (supply-based distance) or a common upstream supplier (demand-based distance). These episodes illustrate how network-based horizontal economic distance, rather than observed vertical exposure, can generate comovement.*

Technological shock in Samsung and supply-based spillovers to competing memory suppliers (2026).— *On February 2, 2026, Samsung Electronics announced significant technological progress in the development and production of next-generation High Bandwidth Memory 4 (HBM4) chips. HBM is a vertically stacked dynamic memory technology optimized for high-speed data processing and is a critical input for Artificial Intelligence (AI) accelerators produced by firms such as NVIDIA. The announcement constituted a positive, firm-specific productivity shock for Samsung in the high-performance memory segment. Because Samsung and other memory producers – most notably Micron and SK Hynix – compete to supply advanced memory chips to the same downstream buyers (including NVIDIA), market expectations regarding future procurement patterns were immediately revised. Financial press coverage reported that investors anticipated a potential reallocation of orders toward Samsung, reflecting its improved technological position. Importantly, Samsung and its rival suppliers do not maintain vertical trade relationships with one another. Nevertheless, the positive shock in Samsung generated adverse valuation effects for competing firms, reflecting expected substitution by the common downstream buyer.*^{III}

Supply-based Apple's reallocation of AirPods assembly orders (2022).— *In 2022, Apple reduced and partially suspended assembly orders for AirPods 2 at Goertek, a major Chinese contract manufacturer. The shock was idiosyncratic to Goertek, stemming from operational issues that impaired its ability to meet Apple's standards and delivery schedules. Rather than reducing overall AirPods production proportionally, Apple responded by shifting demand from the negatively shocked supplier to its competitor, reallocating a substantial share of the affected orders to Luxshare Precision Industry, another contract manufacturer producing similar components for Apple. As a result, Luxshare experienced an increase in expected output and revenues, while Goertek faced declining orders and downward revisions in market expectations. Goertek and Luxshare are not vertically linked to each other; however, they are horizontally proximate because they depend on the same downstream buyer (Apple). The negative shock in one supplier therefore translated into a positive demand shift toward its competitor: Luxshare experienced an increase in expected orders and revenues despite having no direct vertical linkage with Goertek.*^{IV}

^{III} Reported by financial outlets such as Barron's, investing.com, and MarketWatch in February 2026.

^{IV} Covered by Yicai Global, Euronew and MacRumors in November 2022.

Supply-based Apple's reallocation for OLED panels (2026).— In January 2026, Apple shifted a significant portion of its iPhone OLED (Organic Light-Emitting Diode) panel orders from BOE Technology Group in China to the American company Samsung Display after persistent production and yield issues affected BOE's ability to meet quality and volume requirements. OLED panels are critical display components used in iPhones, and Apple relies on multiple suppliers to ensure supply stability and bargaining power. The operational difficulties at BOE constituted a negative firm-specific shock. In response, Apple reallocated millions of display panel orders to Samsung Display, which had available capacity and superior production reliability. The reallocation of downstream demand increased expected revenues for Samsung Display while reducing BOE's projected sales and market share. As in the AirPods episode, the two suppliers are not connected through vertical production relationships; rather, they are horizontally linked because they sell to the same downstream buyer. The transmission of the shock occurred through Apple's procurement decision, not through direct trade between BOE and Samsung Display.^V

AI-driven surge in memory chips demand (2025).— In 2025, the rapid expansion of Artificial Intelligence (AI) applications led to an unprecedented increase in global demand for advanced memory chips, particularly DRAM (Dynamic Random Access Memory) and HBM (High Bandwidth Memory). DRAM constitutes the primary volatile memory used in computing systems, while HBM is a high-performance, vertically stacked memory technology optimized for AI accelerators and data-intensive workloads. The surge in AI-related computing, especially for large language models and generative AI systems, generated a sharp increase in procurement by major cloud service providers and AI hardware producers. Because global production capacity for advanced memory is concentrated among a small number of firms (e.g., Samsung, SK Hynix, Micron), the downstream demand shock tightened upstream supply conditions and increased prices. Downstream sectors relying on these same memory inputs – including consumer electronics manufacturers, enterprise hardware producers, and specialized computing firms – faced rising input costs and constrained availability, even though they were not directly linked to each other through trade.^{VI}

Cloud-based services expansion (2023-2024).— Between 2023 and 2024, large-scale investment in cloud computing infrastructure – driven by AI adoption and data center expansion – significantly increased downstream demand for advanced semiconductors, including graphics processing units (GPUs) and specialized AI accelerators. This expansion placed substantial pressure on upstream semiconductor manufacturing capacity. Because leading-edge chip fabrication is highly concentrated – particularly in firms such as Taiwan Semiconductor Manufacturing Company (TSMC) – the increase in cloud-sector demand reduced effective upstream supply available to other industries relying on the same fabrication processes. Automotive manufacturers, consumer electronics firms, and industrial equipment producers faced delays

^V Reported by MacRumors, TechNode, and DisplayDaily in January 2026.

^{VI} Reported in 2025 by investing.com, Business Standard, and Reuters.

and supply constraints despite having no direct trade relationship with cloud service providers. The resulting synchronized production adjustments across otherwise unrelated sectors reflected shared exposure to upstream fabrication capacity rather than vertical trade interdependence.^{VII}

Reallocation of TSMC supply toward AI customers (2026).— In 2023, the rapid expansion of Artificial Intelligence (AI) applications triggered a sharp increase in downstream demand for high-performance computing chips, particularly graphics processing units (GPUs). These chips rely on advanced fabrication nodes supplied by TSMC, the world’s leading contract chip manufacturer; operating as a pure-play foundry, producing chips designed by numerous fab-less semiconductor firms across industries, and whose cutting-edge production capacity is both technologically complex and capacity-constrained. Given the long planning horizon of semiconductor manufacturing – with wafer capacity typically allocated several quarters, and often years, in advance – the AI-driven surge in orders during 2023 induced TSMC to reprioritize production toward high-margin AI customers: in 2024, TSMC reallocated portions of its advanced fabrication capacity toward AI chip designers such as NVIDIA, responding to exceptionally strong demand for AI accelerators. Industry reports indicate that such reallocations affected fabrication schedules extending into 2025–2026, as a growing share of advanced-node capacity was reserved for AI-related chips. Given finite leading-edge fabrication capacity, TSMC’s prioritization of AI-related chip production effectively reduced available output for other clients, including smartphone and consumer electronics chip designers. As a consequence, other downstream clients competing for the same foundry capacity faced relatively tighter access or lower priority in allocation. These experienced longer lead times and constrained access to advanced nodes, despite not interacting directly with the AI hardware sector. The shock therefore originated in heightened demand from one downstream segment (AI hardware) but propagated through TSMC as a shared upstream production node, generating horizontal effects on multiple unrelated industries.^{VIII}

Across all episodes, the defining feature is that idiosyncratic variations are mediated by the horizontal geometry of the production network rather than by bilateral trade, where shared downstream buyers (supply-based distance) or shared upstream suppliers (demand-based distance) determine the strength and the direction of micro-level shock transmission. These cases provide qualitative empirical grounding for the paper’s central theoretical insight: comovement can arise from horizontal network proximity even in the absence of vertical Input-Output linkages.

^{VII} Documented by the CNBC, Reuters, and Financial Times during 2023-2024.

^{VIII} Reported in January-February 2026 by MacRumors, Business Insider, and AppleInsider.

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